

**Design and Implementation of an Interoperable Risk based Food  
Safety Inspection Information Management System (IIMS) for the  
FCAU of Ministry of Health**

**Software Design Specification Annexure 01**

**Prepared By**



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## Key Summary

|                     |                                                                                                                                                         |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Name</b> | Design and Implementation of an Interoperable Risk based Food Safety Inspection Information Management System (IIMS) for the FCAU of Ministry of Health |
| <b>Tender No</b>    | 2026/FASRL/FASRL/135600                                                                                                                                 |
| <b>Contract No</b>  | 304/2026/UNJP/SRL/076/EC                                                                                                                                |
| <b>Sponsor</b>      | UNFAO                                                                                                                                                   |
| <b>Approver</b>     | Food Control and Administration Unit (FCAU) of Ministry of Health (MOH)                                                                                 |
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# 1. System Workflows

System workflows describe the end-to-end operational processes that govern how different components of the system interact to deliver business functionality. Within the context of this Software Design Document (SDD), workflows provide a structured representation of how user actions, system services, and data flows are coordinated across the microservices based architecture.

These workflows serve as a bridge between the functional requirements defined in the Software Requirements Specification (SRS) and the technical implementation outlined in the SDD. They illustrate how discrete modules collaborate to execute business processes in a cohesive and efficient manner.

Each workflow captures:

- The sequence of interactions between users and system components
- The orchestration
- Data exchange patterns across services and databases
- Trigger points for events such as notifications, validations, and escalations

Given the adoption of a microservices architecture, workflows are particularly critical in defining service boundaries, asynchronous communication mechanisms (e.g., event-driven messaging), and inter-service dependencies. They ensure that the system maintains consistency, scalability, and fault tolerance while handling complex real-world scenarios.

The workflows documented in this section will be represented using standardized process flow diagrams, enabling clear visualization of system behavior for both technical and non-technical stakeholders.

Overall, system workflows provide a comprehensive view of how the system operates dynamically, ensuring alignment between business objectives and technical design while supporting maintainability and future scalability.

## 1.1 Premises Registration and Lifecycle Management Module

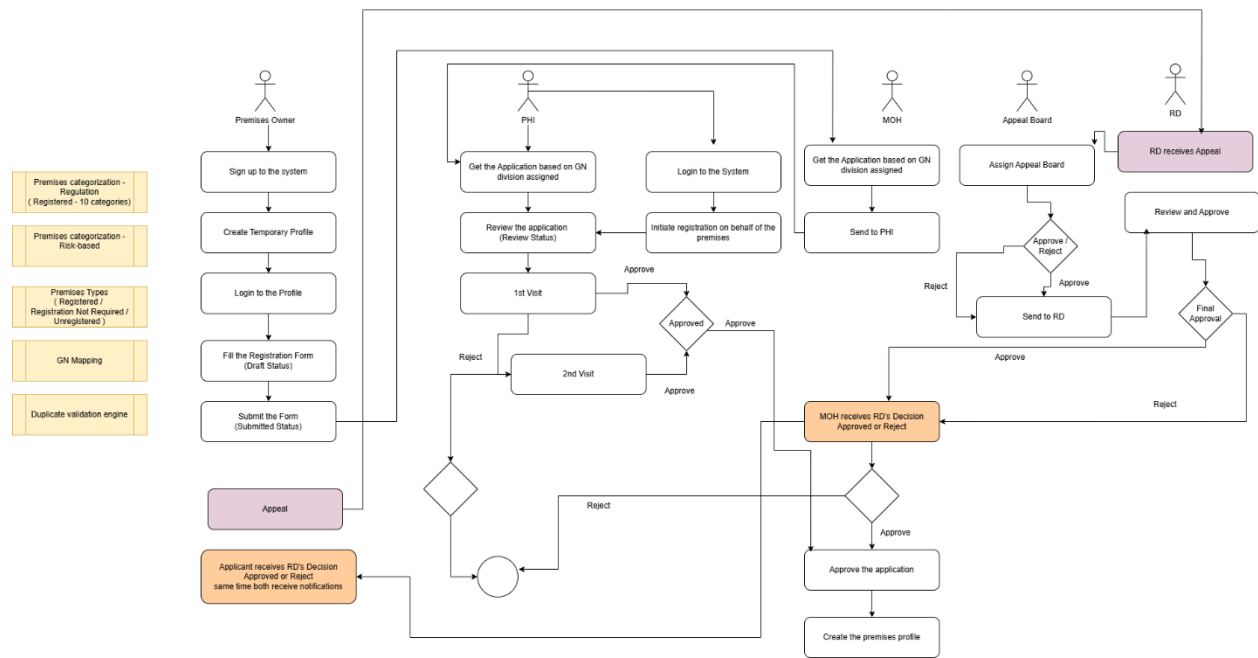


Figure 1.0

## 1.2 Inspection and Grading Module

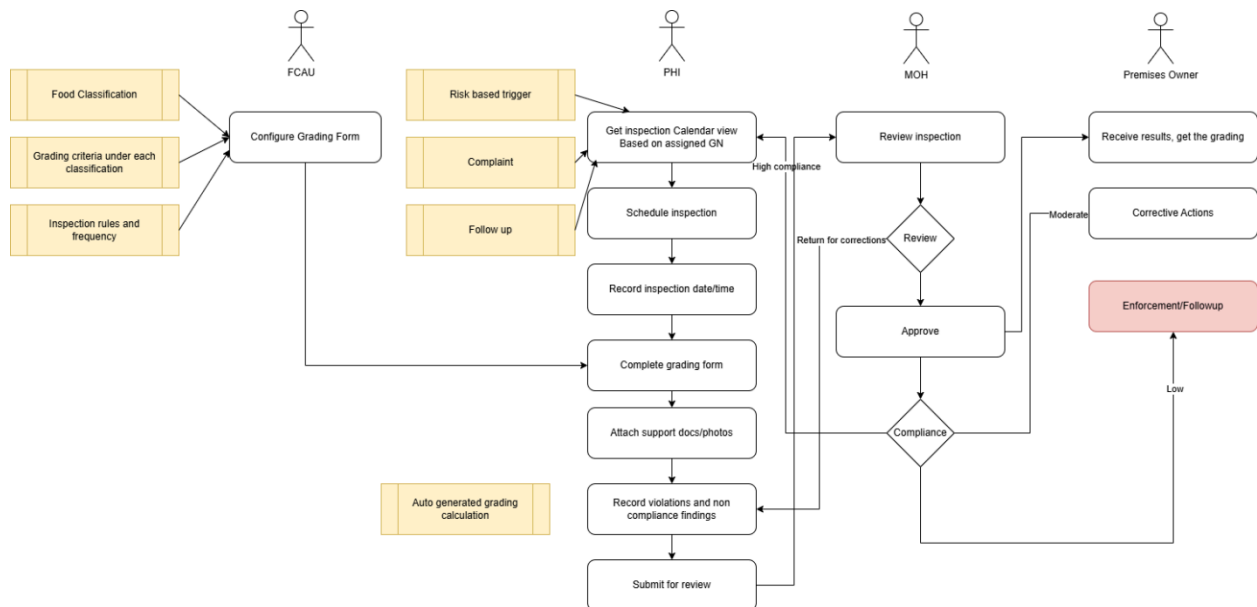


Figure 1.1

### 1.3 Laboratory Coordination Module

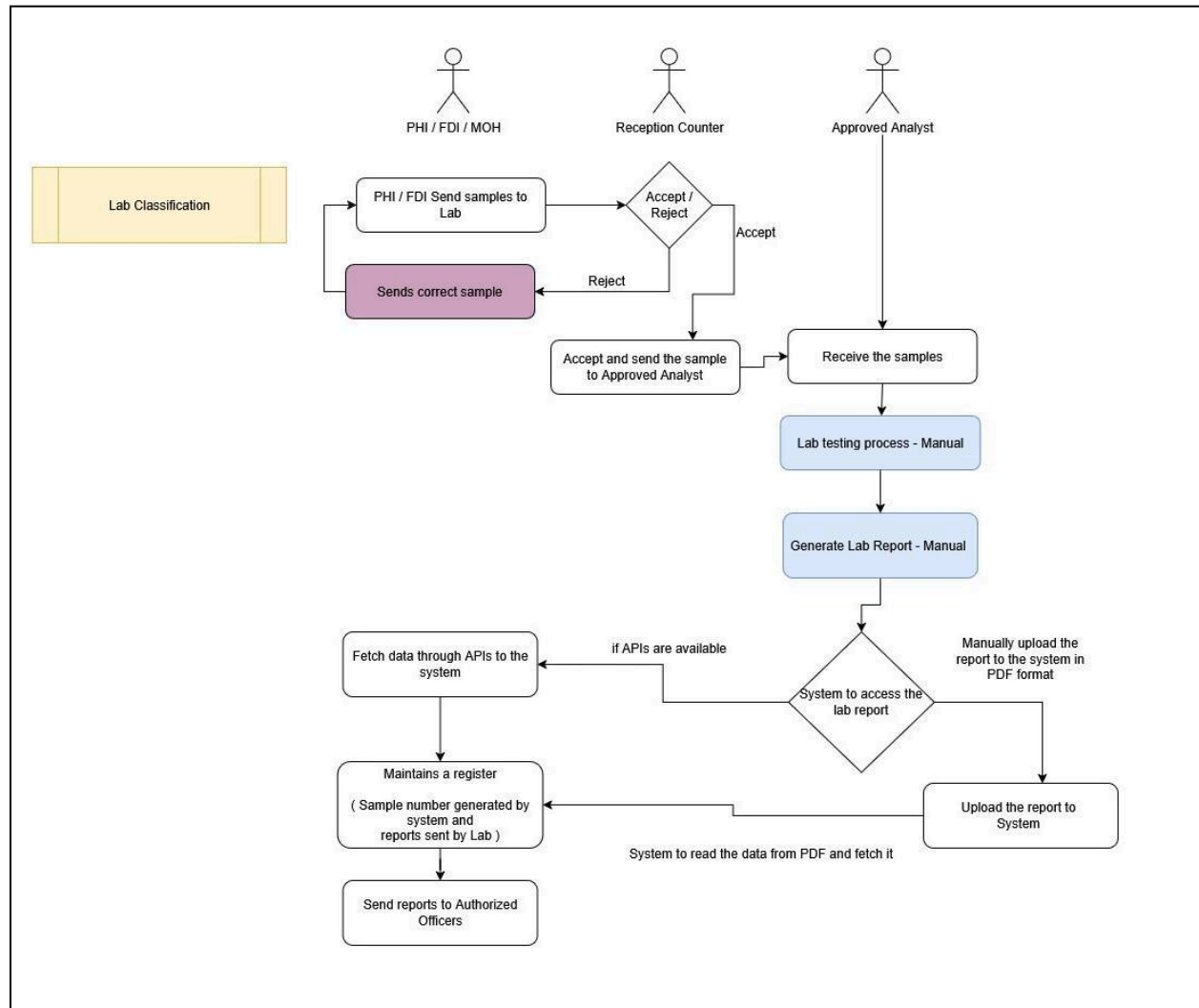


Figure 1.2

## 1.4 Legal, Prosecution and Enforcement Management Module

### 1.4.1 Summarized

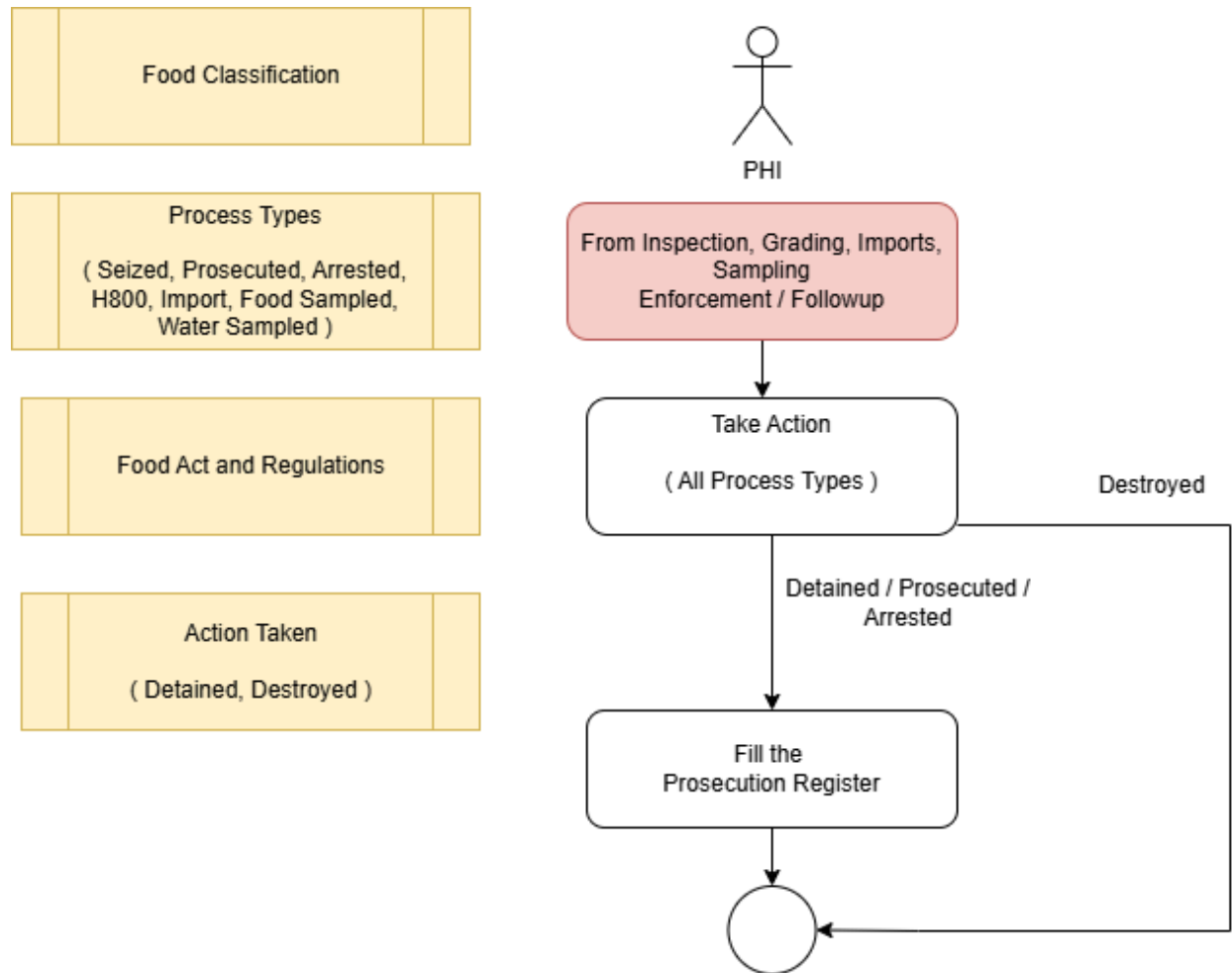


Figure 1.3



## 1.4.2 Seized

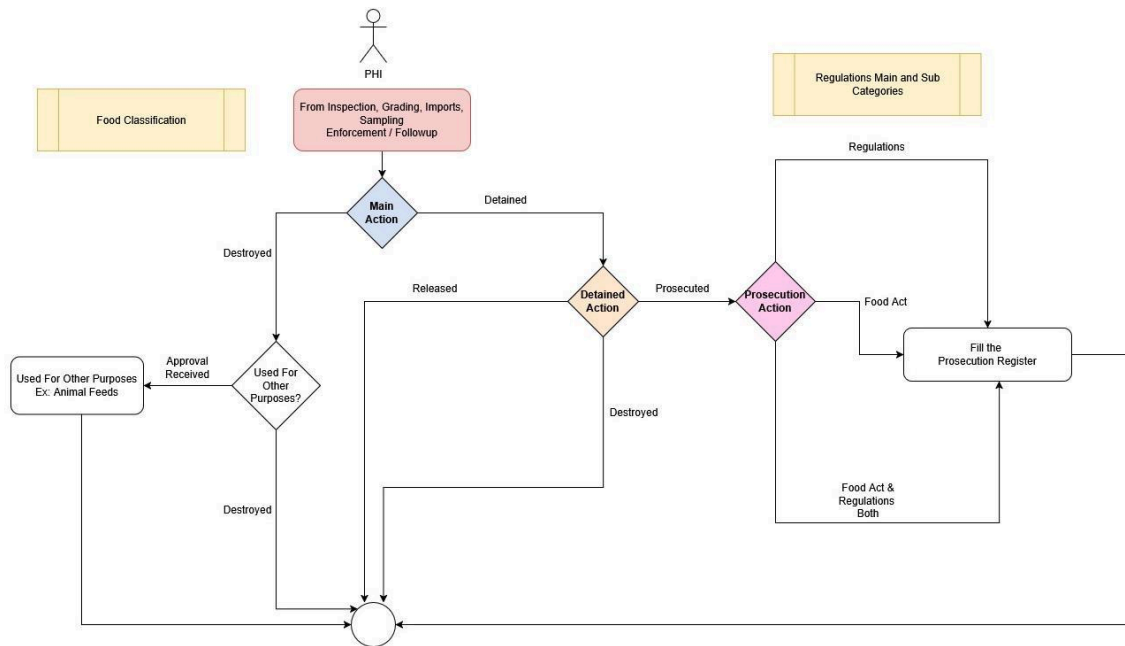


Figure 1.4

### 1.4.3 Prosecution

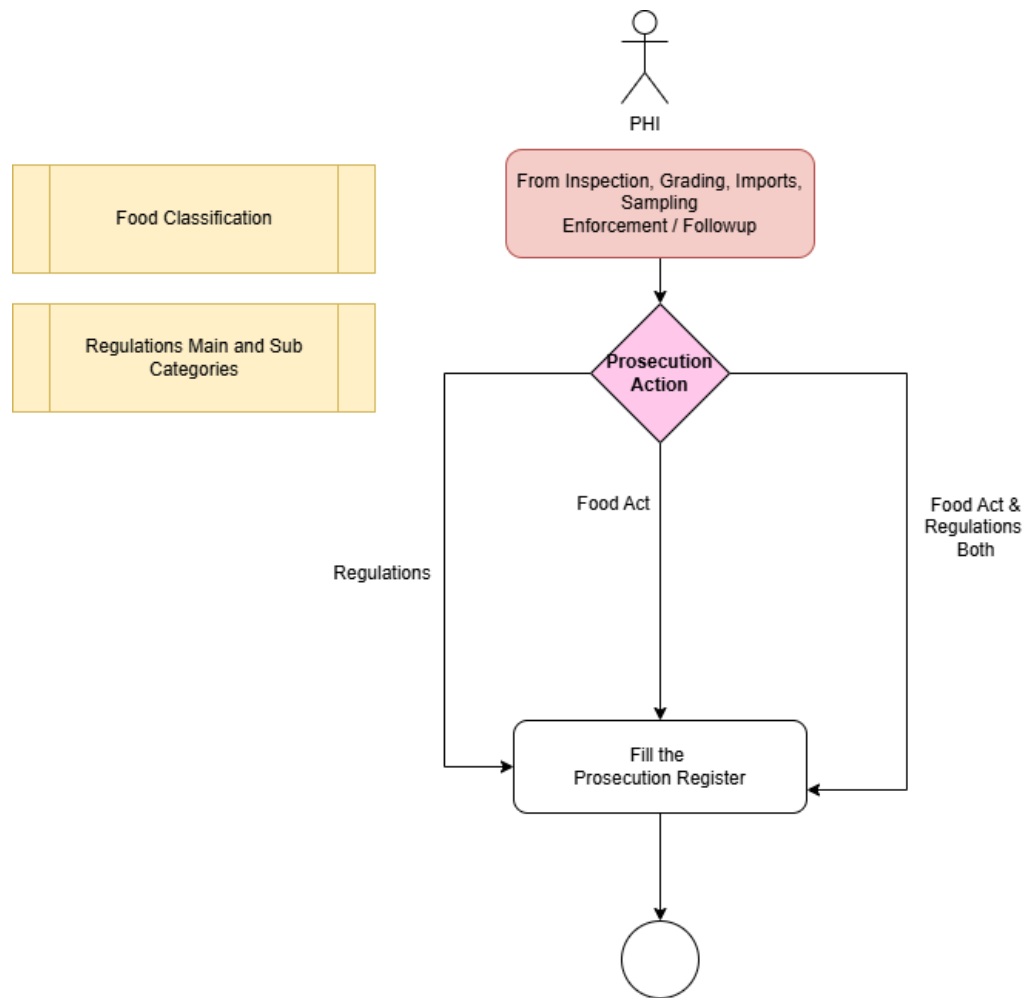


Figure 1.5

### 1.4.4 Arrested

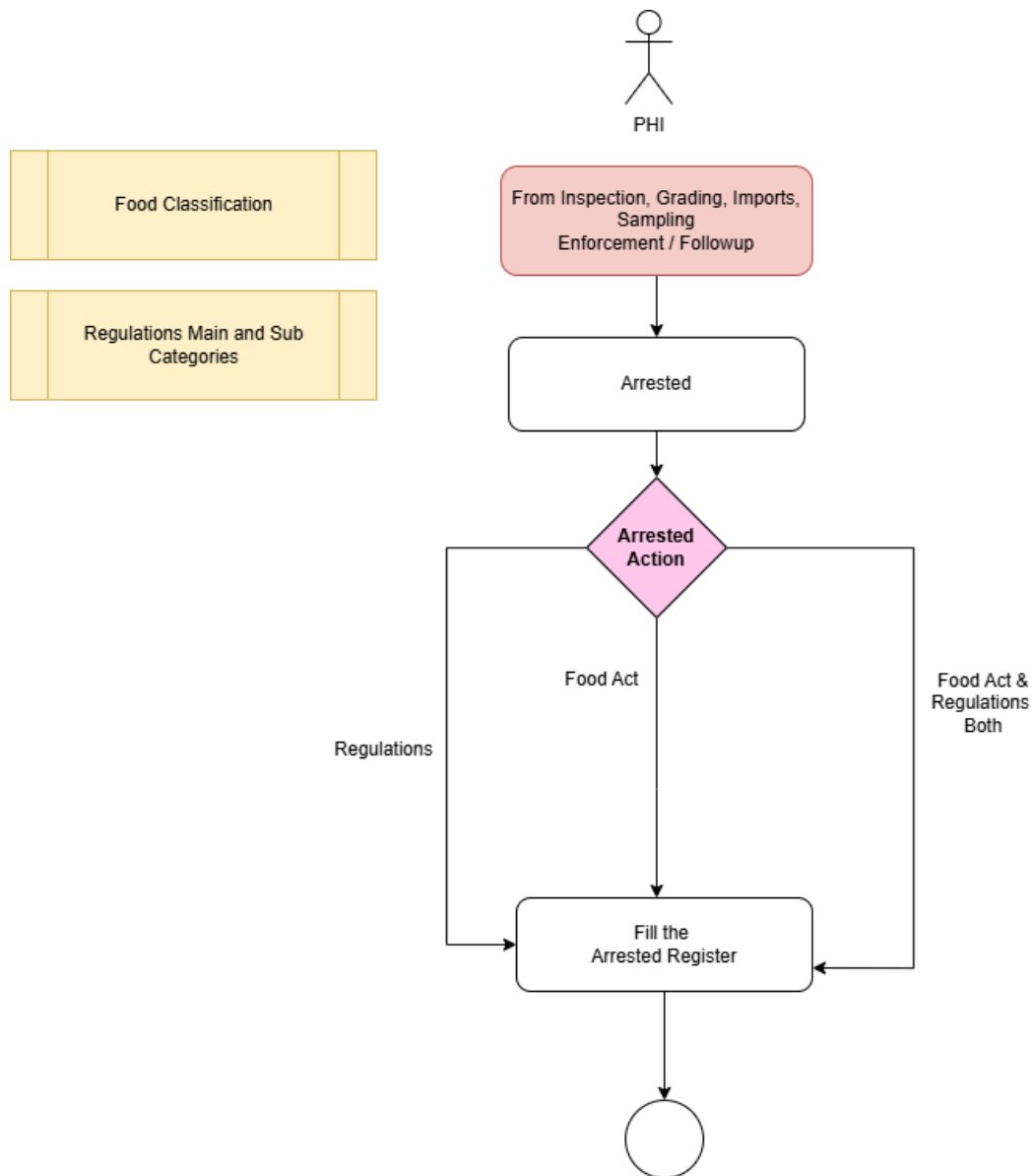


Figure 1.6



## 1.6 Export Control Module

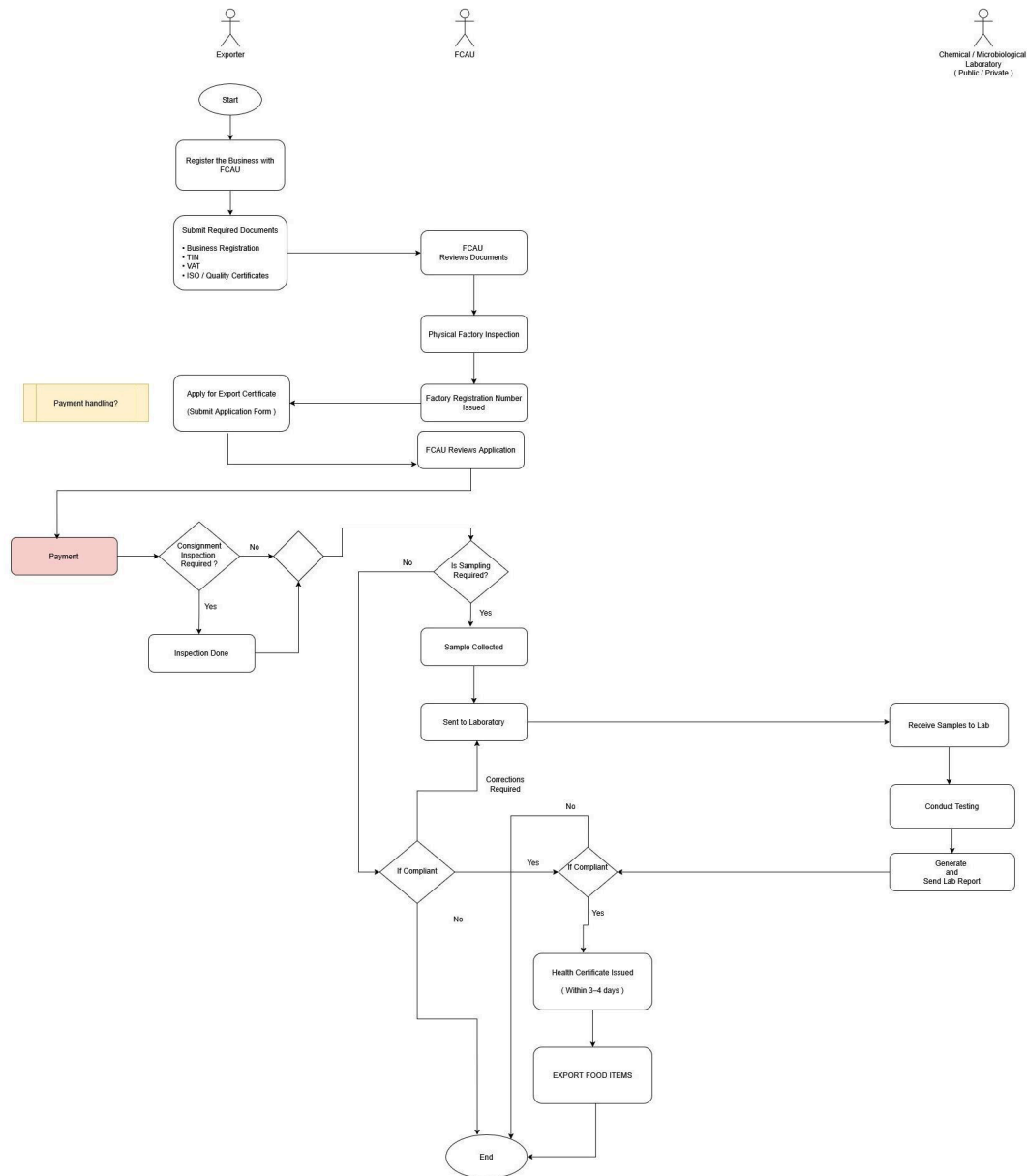


Figure 1.8

## 1.7 Registration of Common Salt Manufacturing and Transports Management Module

### 1.7.1 Salt Factory Registration

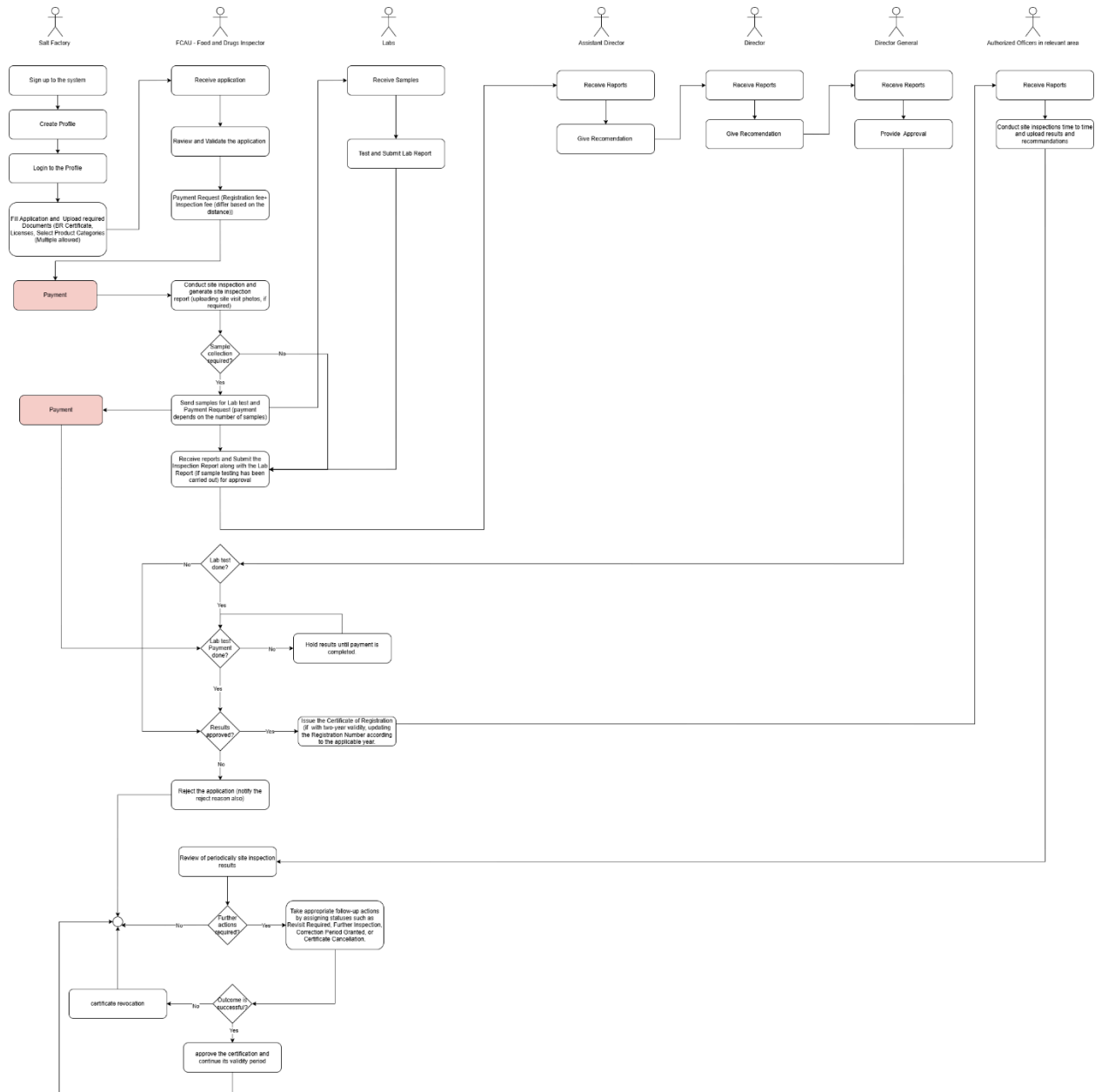


Figure 1.9

## 1.7.2 Salt Factory Annual Renewal

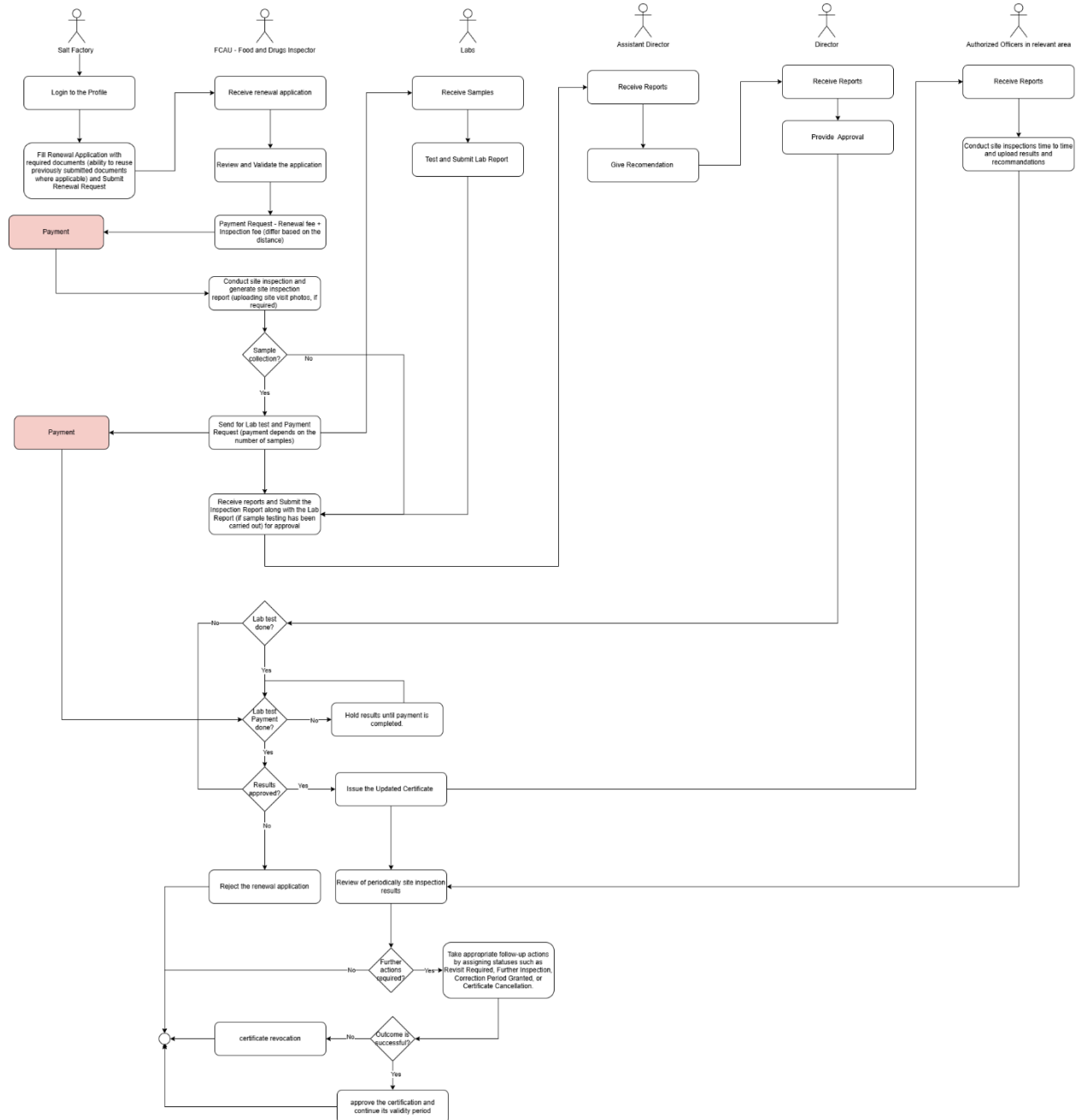


Figure 1.10

### 1.7.3 Saltern Registration

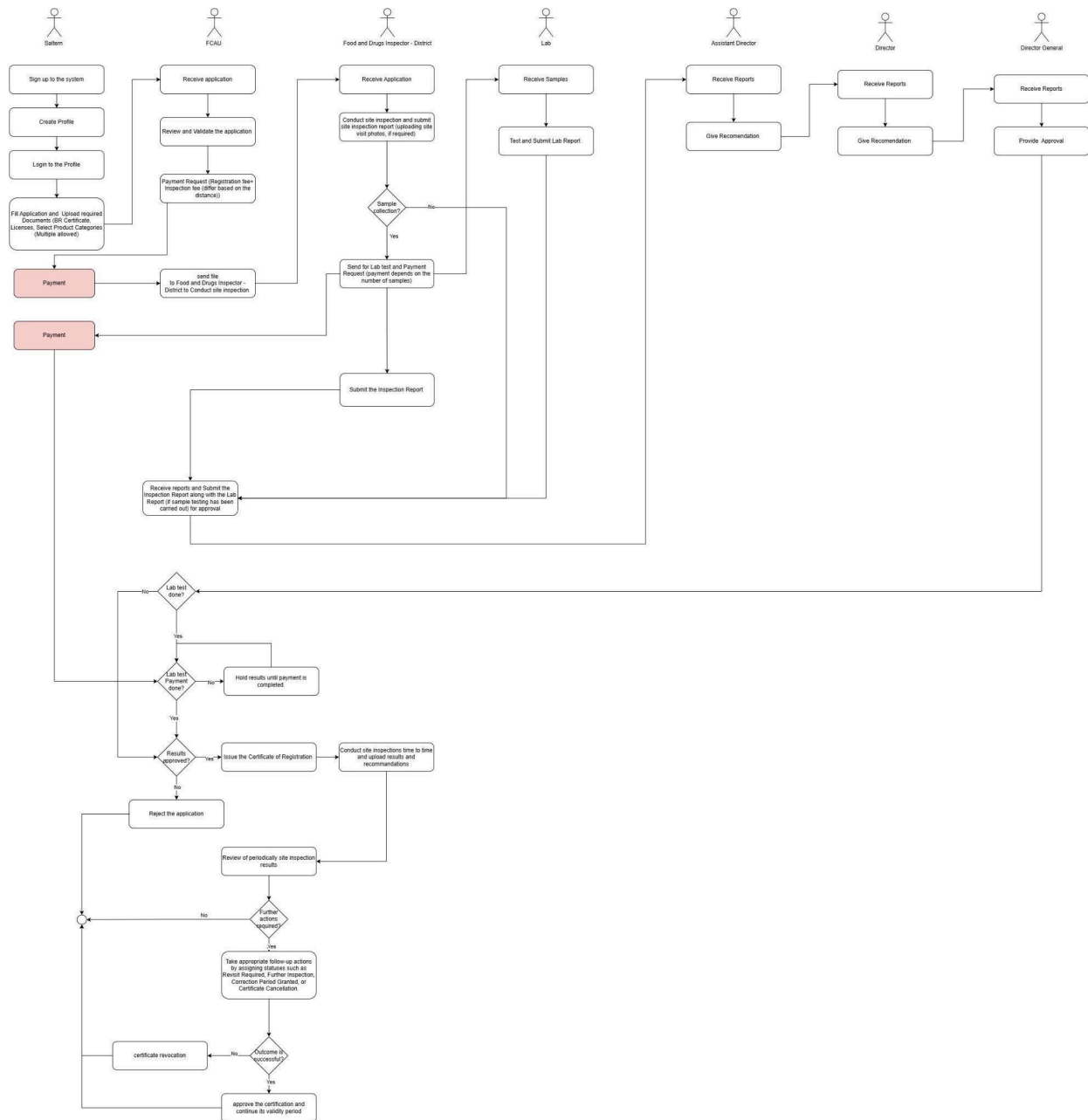


Figure 1.11



### 1.7.4 Saltern Annual Renewal

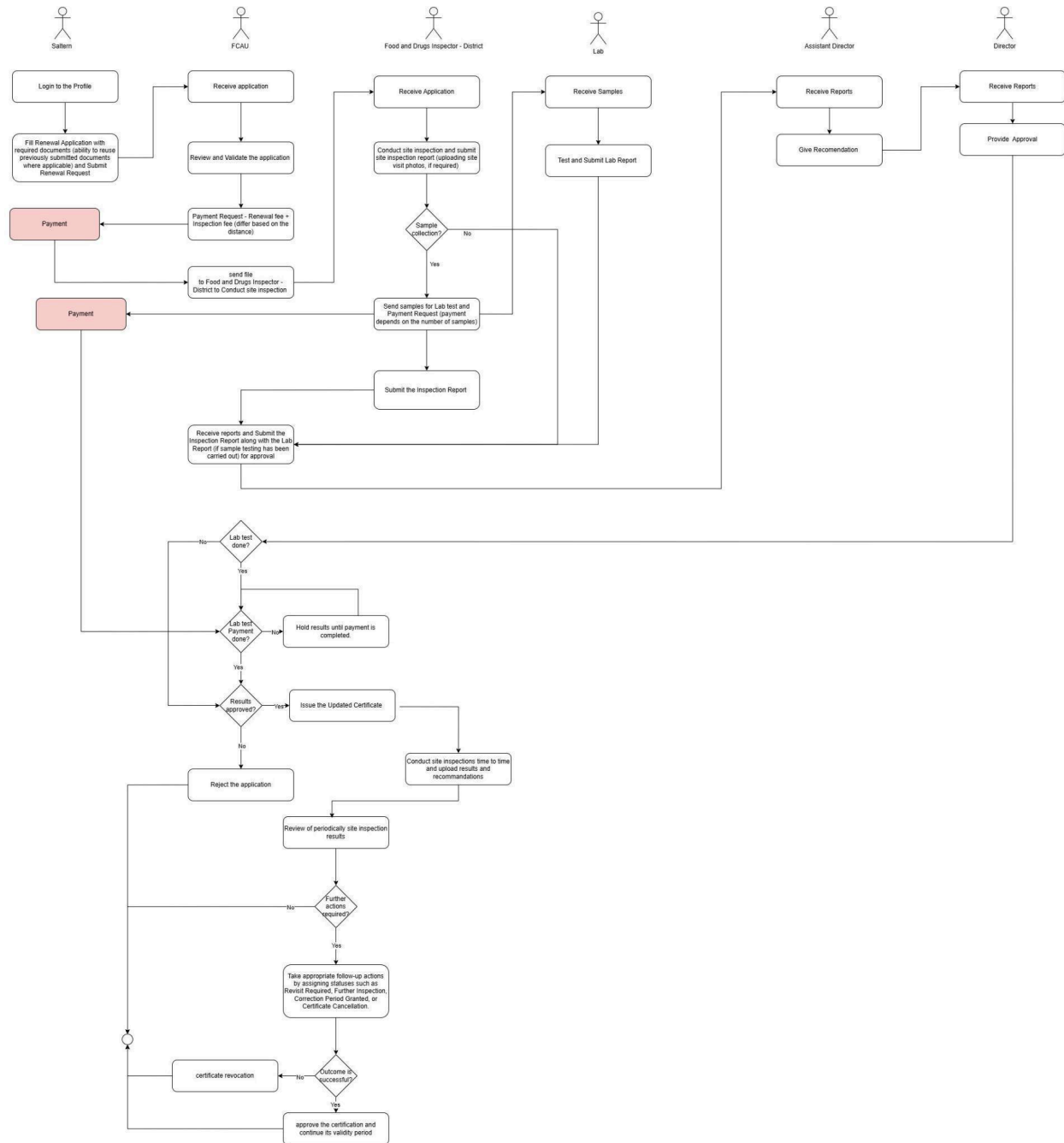


Figure 1.12

### 1.7.5 Salt Factory/Saltern Registration Amendment

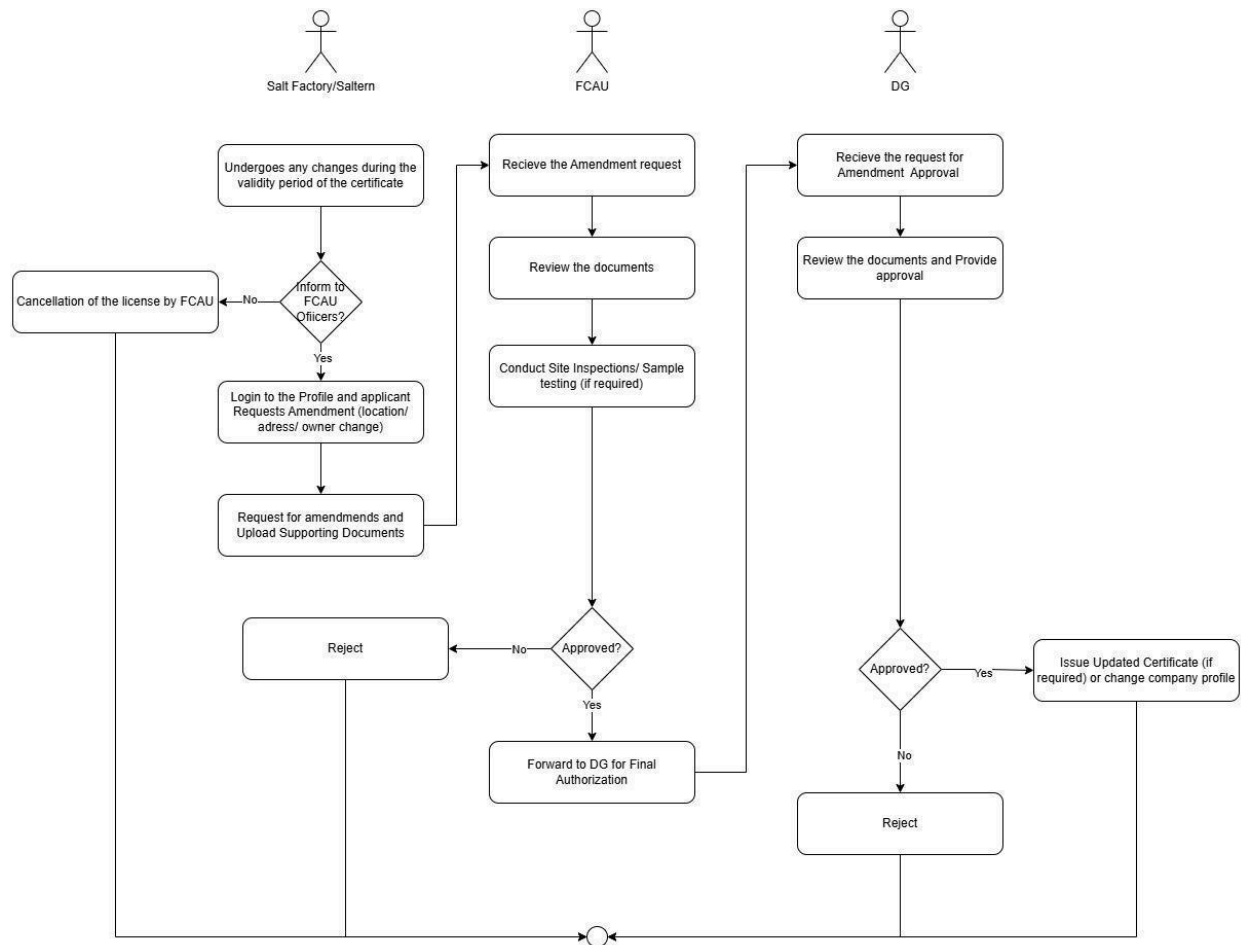


Figure 1.13

### 1.7.6 Salt Factory/Saltern Multiple Brands

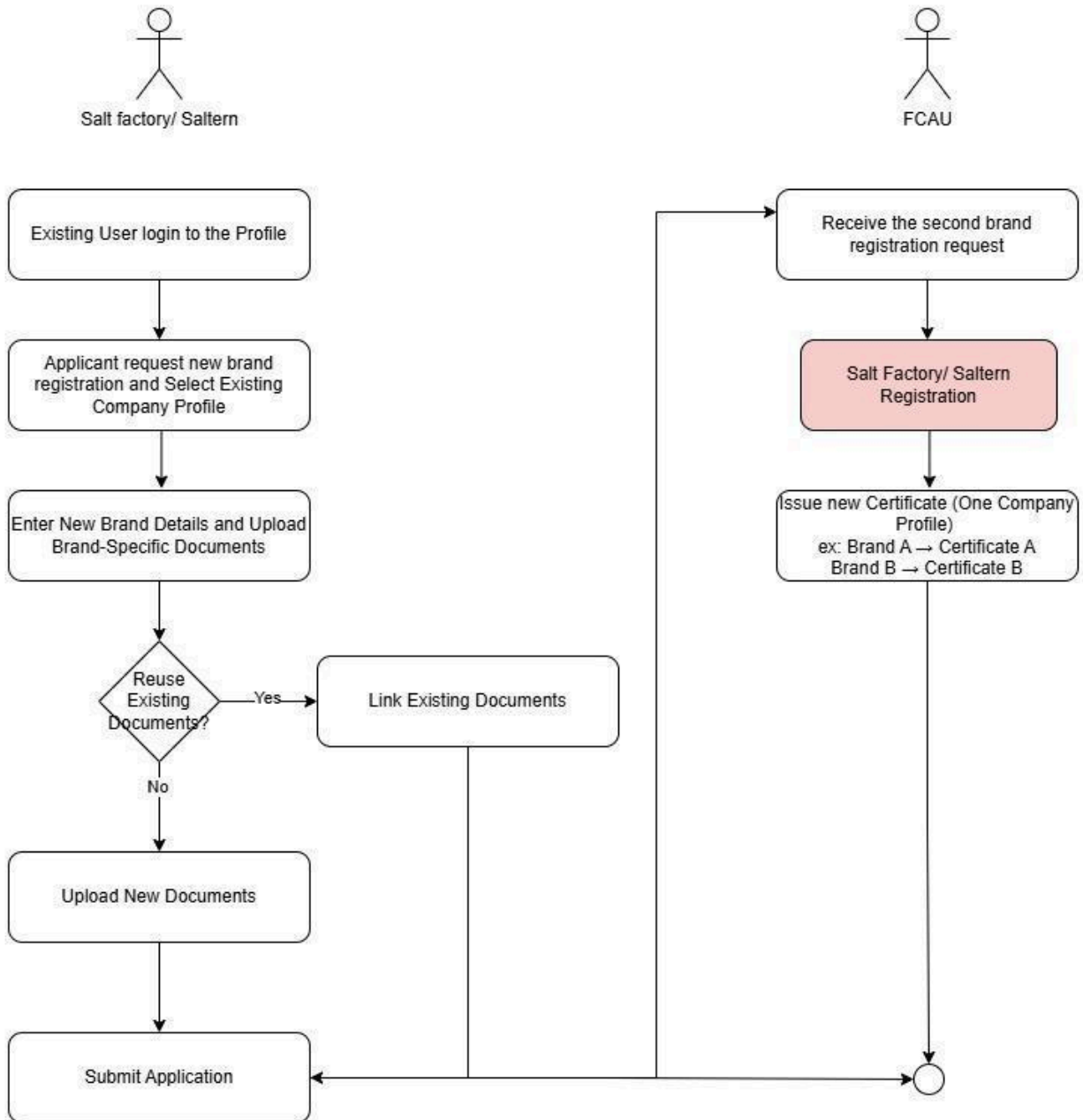


Figure 1.14



### 1.8.2 Bottled Water Registration Renewal

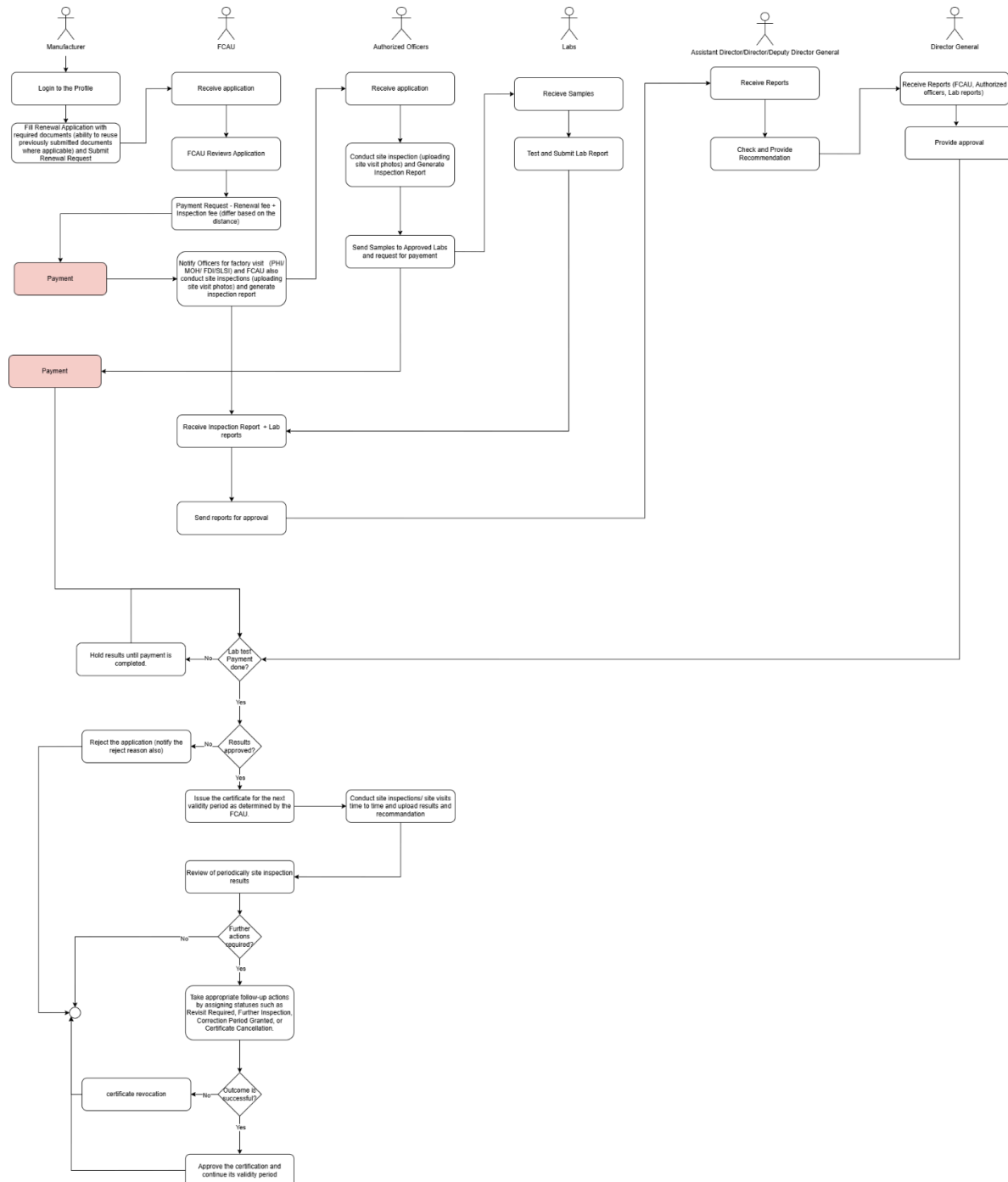


Figure 1.16

### 1.8.3 Bottled Water Registration Amendment

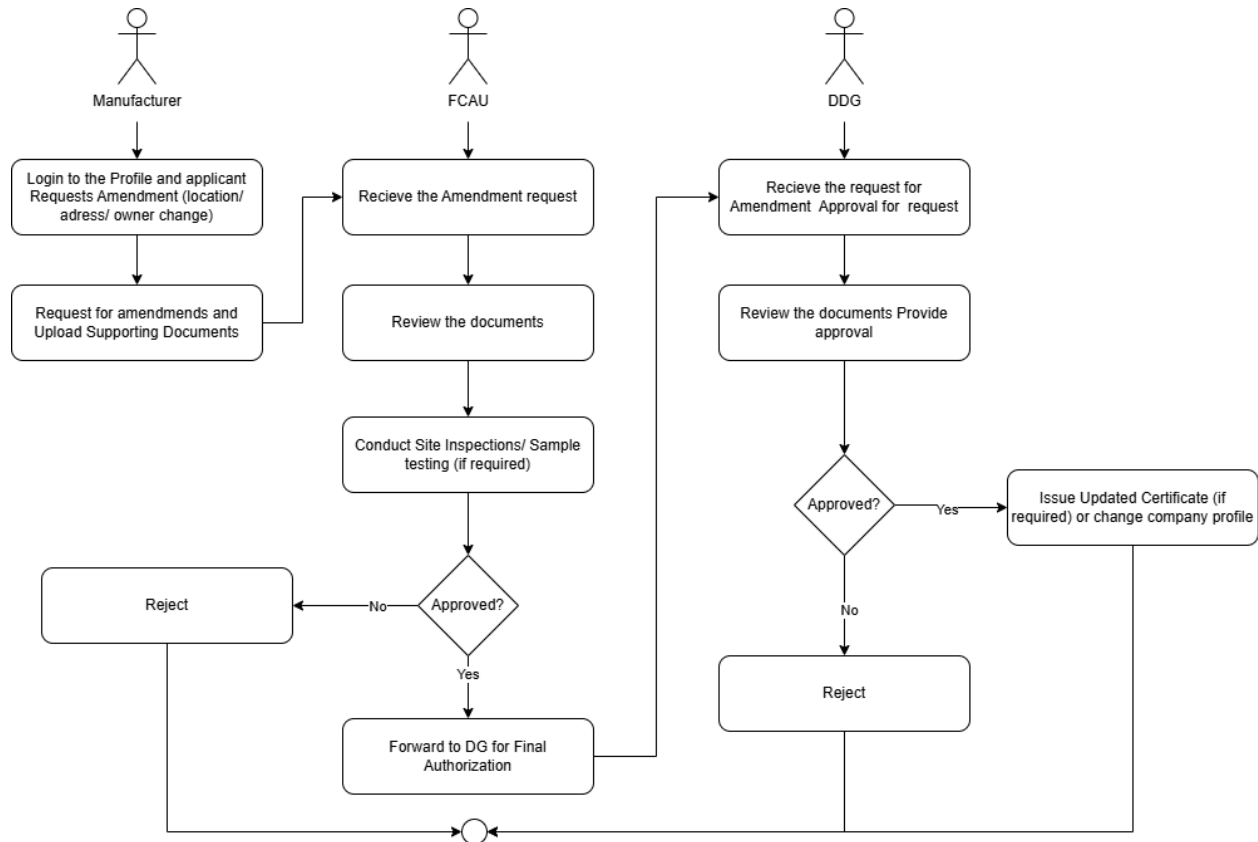


Figure 1.17

### 1.8.4 Bottled Water Multiple Brands

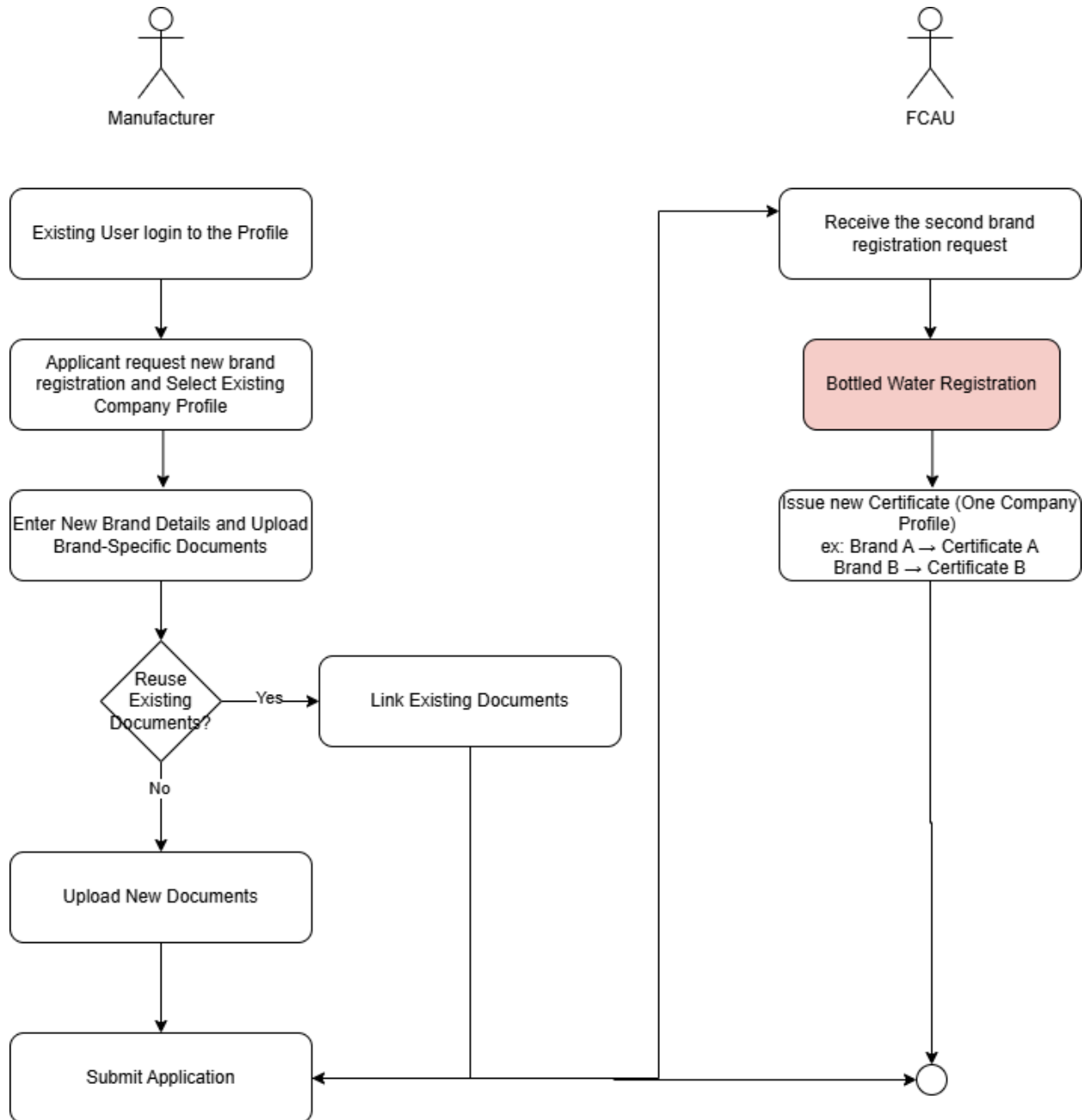


Figure 1.18

## 2. Use Case Modeling

Use case modeling is a fundamental technique used to capture and represent the functional behavior of a system from the perspective of its users and external actors. Within this Software Design Document (SDD), use case modeling provides a clear and structured way to describe how different stakeholders interact with the system to achieve specific goals.

The primary purpose of use case modeling is to translate high-level functional requirements defined in the Software Requirements Specification (SRS) into user centric scenarios. Each use case outlines a sequence of interactions between an actor (such as a system user, administrator, or external system) and the system, demonstrating how a particular function or service is executed.

Use case models typically consist of:

- **Actors** – entities that interact with the system (e.g., public users, system administrators, regulatory officers)
- **Use Cases** – specific functionalities or services provided by the system (e.g., register premises, submit complaints, generate reports)
- **System Boundary** – defines the scope of the system and its interactions with external entities
- **Relationships** – associations such as include, extend, and generalization that show dependencies and reuse among use cases

In the context of a microservices-based architecture, use case modeling plays a critical role in identifying system functionalities and mapping them to corresponding services. It helps ensure that each microservice is aligned with specific business capabilities while maintaining loose coupling and high cohesion.

Use case diagrams and detailed use case descriptions also support effective communication among stakeholders, including developers, designers, project managers, and domain experts. They provide a shared understanding of system behavior, reduce ambiguity in requirements, and serve as a foundation for further design activities such as sequence diagrams, activity diagrams, and API design.

Overall, use case modeling ensures that the system is designed with a strong focus on user interactions and business processes, enabling the development of a solution that is both functionally complete and aligned with stakeholder expectations.



## 2.1 Premises Registration and Lifecycle Management Module



Figure 2.0

## Use Case Narratives

| UC-01: Register Account |                                                                                                                                                                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>   | Food Business Operator (FBO)                                                                                                                                                                                                       |
| <b>Description:</b>     | Allows a new user to create an account in the system.                                                                                                                                                                              |
| <b>Preconditions:</b>   | <ul style="list-style-type: none"> <li>User does not already have an account</li> </ul>                                                                                                                                            |
| <b>Postconditions:</b>  | <ul style="list-style-type: none"> <li>User account is successfully created</li> <li>User can log in to the system</li> </ul>                                                                                                      |
| <b>Main Flow:</b>       | <ul style="list-style-type: none"> <li>User selects “Register Account”</li> <li>System displays registration form</li> <li>User enters required details</li> <li>System validates input</li> <li>System creates account</li> </ul> |
| <b>Alternate Flow:</b>  | <ul style="list-style-type: none"> <li>4a. Invalid input → System shows error message</li> </ul>                                                                                                                                   |

| UC-02: Login           |                                                                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>  | FBO                                                                                                                                           |
| <b>Description:</b>    | Allows registered users to access the system.                                                                                                 |
| <b>Preconditions:</b>  | <ul style="list-style-type: none"> <li>User must have a registered account</li> </ul>                                                         |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>User is authenticated and granted access</li> </ul>                                                    |
| <b>Main Flow:</b>      | <ul style="list-style-type: none"> <li>User enters credentials</li> <li>System validates credentials</li> <li>System grants access</li> </ul> |
| <b>Alternate Flow:</b> | <ul style="list-style-type: none"> <li>2a. Invalid credentials → Error message displayed</li> </ul>                                           |

| UC-03: Fill Registration Form |                                                                                                                                                                                           |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                | FBO                                                                                                                                                                                       |
| Description:                  | User fills details required for premises registration.                                                                                                                                    |
| Preconditions:                | <ul style="list-style-type: none"> <li>User is logged in</li> </ul>                                                                                                                       |
| Postconditions:               | <ul style="list-style-type: none"> <li>Form data is saved (draft or ready for submission)</li> </ul>                                                                                      |
| Main Flow:                    | <ul style="list-style-type: none"> <li>User opens registration form</li> <li>User enters premises details</li> <li>System validates inputs</li> <li>User proceeds to next step</li> </ul> |
| Alternate Flow:               | N/A                                                                                                                                                                                       |

| UC-04: Upload Documents |                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:          | FBO                                                                                                                                               |
| Description:            | Upload supporting documents for registration.                                                                                                     |
| Preconditions:          | <ul style="list-style-type: none"> <li>Registration form initiated</li> </ul>                                                                     |
| Postconditions:         | <ul style="list-style-type: none"> <li>Documents stored successfully</li> </ul>                                                                   |
| Main Flow:              | <ul style="list-style-type: none"> <li>User selects documents</li> <li>System validates file type/size</li> <li>Documents are uploaded</li> </ul> |
| Alternate Flow:         | <ul style="list-style-type: none"> <li>2a. Invalid file → Error displayed</li> </ul>                                                              |

| UC-05: Save Draft Application |                                                                                      |
|-------------------------------|--------------------------------------------------------------------------------------|
| Primary Actor:                | FBO                                                                                  |
| Description:                  | Save incomplete application for later completion.                                    |
| Preconditions:                | <ul style="list-style-type: none"> <li>Registration form partially filled</li> </ul> |
| Postconditions:               | <ul style="list-style-type: none"> <li>Draft stored in system</li> </ul>             |
| Main Flow:                    | <ul style="list-style-type: none"> <li>User clicks "Save Draft"</li> </ul>           |

|                 |                                                                             |
|-----------------|-----------------------------------------------------------------------------|
|                 | <ul style="list-style-type: none"> <li>System saves current data</li> </ul> |
| Alternate Flow: | N/A                                                                         |

| UC-06: Submit Application |                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:            | FBO                                                                                                                                                                                                                                                                                                                                                                                            |
| Description:              | Submit completed application for processing.                                                                                                                                                                                                                                                                                                                                                   |
| Preconditions:            | <ul style="list-style-type: none"> <li>Form completed</li> <li>Documents uploaded</li> </ul>                                                                                                                                                                                                                                                                                                   |
| Postconditions:           | <ul style="list-style-type: none"> <li>Application submitted</li> <li>Reference number generated</li> </ul>                                                                                                                                                                                                                                                                                    |
| Main Flow:                | <ul style="list-style-type: none"> <li>User clicks “Submit”</li> <li>System validates full application</li> <li>System performs: <ul style="list-style-type: none"> <li>Premises categorization &amp; risk profiling</li> <li>GN division mapping</li> <li>Duplicate validation</li> </ul> </li> <li>System generates reference number</li> <li>Application is forwarded to MOH/PHI</li> </ul> |
| Alternate Flow:           | <ul style="list-style-type: none"> <li>2a. Validation fails → Errors displayed</li> </ul>                                                                                                                                                                                                                                                                                                      |

| UC-07: Track Application Status |                                                                               |
|---------------------------------|-------------------------------------------------------------------------------|
| Primary Actor:                  | FBO                                                                           |
| Description:                    | Allows user to track submitted application.                                   |
| Preconditions:                  | <ul style="list-style-type: none"> <li>Application submitted</li> </ul>       |
| Postconditions:                 | <ul style="list-style-type: none"> <li>Status displayed</li> </ul>            |
| Main Flow:                      | <ul style="list-style-type: none"> <li>User selects “Track Status”</li> </ul> |

|                 |                                                                                                                      |
|-----------------|----------------------------------------------------------------------------------------------------------------------|
|                 | <ul style="list-style-type: none"> <li>• System retrieves application details</li> <li>• Status displayed</li> </ul> |
| Alternate Flow: | N/A                                                                                                                  |

| UC-08: Submit Appeal |                                                                                                                                                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:       | FBO                                                                                                                                                                                                                        |
| Description:         | Submit appeal against rejected application.                                                                                                                                                                                |
| Preconditions:       | <ul style="list-style-type: none"> <li>• Application rejected</li> </ul>                                                                                                                                                   |
| Postconditions:      | <ul style="list-style-type: none"> <li>• Appeal submitted to system</li> </ul>                                                                                                                                             |
| Main Flow:           | <ul style="list-style-type: none"> <li>• User selects rejected application</li> <li>• User submits appeal with justification</li> <li>• System records appeal</li> <li>• Appeal forwarded to relevant authority</li> </ul> |
| Alternate Flow:      | N/A                                                                                                                                                                                                                        |

| UC-09: View Assigned Applications |                                                                                                                  |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                    | PHI                                                                                                              |
| Description:                      | View applications assigned for inspection.                                                                       |
| Preconditions:                    | <ul style="list-style-type: none"> <li>• Applications assigned by MOH</li> </ul>                                 |
| Postconditions:                   | <ul style="list-style-type: none"> <li>• List of applications displayed</li> </ul>                               |
| Main Flow:                        | <ul style="list-style-type: none"> <li>• PHI logs in</li> <li>• System displays assigned applications</li> </ul> |
| Alternate Flow:                   | N/A                                                                                                              |

| UC-10: Conduct Inspections |                                  |
|----------------------------|----------------------------------|
| Primary Actor:             | PHI                              |
| Description:               | Perform inspections on premises. |

|                        |                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Preconditions:</b>  | <ul style="list-style-type: none"> <li>Applications assigned by MOH</li> </ul>                                                                                                                                                                                                                                                                                                        |
| <b>Postconditions:</b> |                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Main Flow:</b>      | <p><b>UC-10A: Conduct 1st Inspection</b></p> <ul style="list-style-type: none"> <li>PHI selects application</li> <li>Conducts inspection</li> <li>Records observations</li> </ul> <p><b>UC-10B: Conduct 2nd Inspection (if required)</b></p> <ul style="list-style-type: none"> <li>Triggered based on first inspection outcome</li> <li>PHI performs follow-up inspection</li> </ul> |
| <b>Alternate Flow:</b> | N/A                                                                                                                                                                                                                                                                                                                                                                                   |

| <b>UC-11: Submit Inspection Report</b> |                                                                                                                              |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>                  | PHI                                                                                                                          |
| <b>Description:</b>                    | Submit inspection findings.                                                                                                  |
| <b>Preconditions:</b>                  | <ul style="list-style-type: none"> <li>Inspection completed</li> </ul>                                                       |
| <b>Postconditions:</b>                 | <ul style="list-style-type: none"> <li>Report submitted to MOH</li> </ul>                                                    |
| <b>Main Flow:</b>                      | <ul style="list-style-type: none"> <li>PHI fills report</li> <li>System validates input</li> <li>Report submitted</li> </ul> |
| <b>Alternate Flow:</b>                 | N/A                                                                                                                          |

| <b>UC-12: Review PHI Recommendation</b> |                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>                   | MOH                                                                                     |
| <b>Description:</b>                     | Review inspection report and recommendation.                                            |
| <b>Preconditions:</b>                   | <ul style="list-style-type: none"> <li>Inspection report submitted</li> </ul>           |
| <b>Postconditions:</b>                  | <ul style="list-style-type: none"> <li>Decision ready for approval/rejection</li> </ul> |

|                        |                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Main Flow:</b>      | <ul style="list-style-type: none"> <li>• Review inspection</li> <li>• System validates input</li> <li>• Submitted for approval</li> </ul> |
| <b>Alternate Flow:</b> | N/A                                                                                                                                       |

| UC-13: Approve Application |                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>      | MOH                                                                                                          |
| <b>Description:</b>        | Approve premises registration.                                                                               |
| <b>Preconditions:</b>      | <ul style="list-style-type: none"> <li>• Review inspection</li> </ul>                                        |
| <b>Postconditions:</b>     | <ul style="list-style-type: none"> <li>• Application marked approved</li> <li>• Notification sent</li> </ul> |
| <b>Main Flow:</b>          | <ul style="list-style-type: none"> <li>• Submit approval/rejection</li> </ul>                                |
| <b>Alternate Flow:</b>     | N/A                                                                                                          |

## 2.2 Inspection and Grading Module

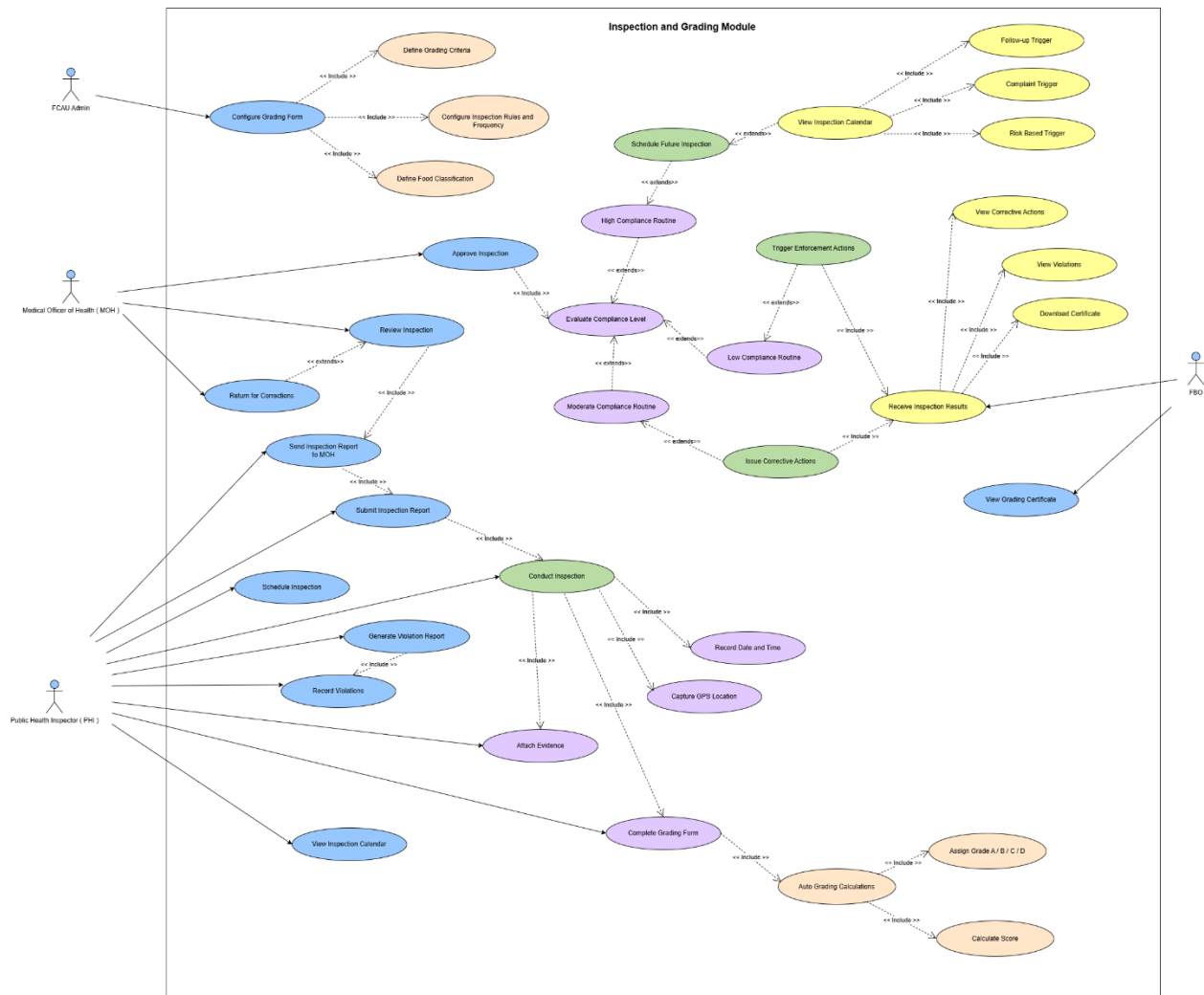


Figure 2.1



## Use Case Narratives

| UC-01: Configure Grading Framework |                                                                                                                                                                                                                                                                   |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                     | FCAU Admin                                                                                                                                                                                                                                                        |
| Description:                       | Configure grading criteria, inspection rules, and food classifications.                                                                                                                                                                                           |
| Preconditions:                     | <ul style="list-style-type: none"> <li>Admin is authenticated</li> </ul>                                                                                                                                                                                          |
| Postconditions:                    | <ul style="list-style-type: none"> <li>Grading framework is configured and available for inspections</li> </ul>                                                                                                                                                   |
| Main Flow:                         | <ul style="list-style-type: none"> <li>Admin selects “Configure Grading Form”</li> <li>Defines grading criteria</li> <li>Configures inspection rules and frequency</li> <li>Defines food classification categories</li> <li>System saves configuration</li> </ul> |
| Alternate Flow:                    | N/A                                                                                                                                                                                                                                                               |

| UC-02: Schedule Inspection |                                                                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:             | PHI                                                                                                                                                                                           |
| Description:               | Schedule inspections for registered premises.                                                                                                                                                 |
| Preconditions:             | <ul style="list-style-type: none"> <li>Premises registered and assigned</li> </ul>                                                                                                            |
| Postconditions:            | <ul style="list-style-type: none"> <li>Inspection scheduled and visible in calendar</li> </ul>                                                                                                |
| Main Flow:                 | <ul style="list-style-type: none"> <li>PHI selects premises</li> <li>Chooses inspection date/time</li> <li>System schedules inspection</li> <li>Entry added to inspection calendar</li> </ul> |
| Alternate Flow:            | N/A                                                                                                                                                                                           |

| UC-03: View Inspection Calendar |                                                                                                                                                                                                       |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                  | PHI, FBO                                                                                                                                                                                              |
| Description:                    | View scheduled inspections and triggers.                                                                                                                                                              |
| Preconditions:                  | <ul style="list-style-type: none"> <li>User logged in</li> </ul>                                                                                                                                      |
| Postconditions:                 | <ul style="list-style-type: none"> <li>Calendar displayed with relevant events</li> </ul>                                                                                                             |
| Main Flow:                      | <ul style="list-style-type: none"> <li>User opens calendar</li> <li>System displays: <ul style="list-style-type: none"> <li>Scheduled inspections</li> <li>Risk-based triggers</li> </ul> </li> </ul> |

|                        |                                                                                                  |
|------------------------|--------------------------------------------------------------------------------------------------|
|                        | <ul style="list-style-type: none"> <li>Complaint triggers</li> <li>Follow-up triggers</li> </ul> |
| <b>Alternate Flow:</b> | N/A                                                                                              |

| UC-04: Conduct Inspection |                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>     | PHI                                                                                                                                                                                                                 |
| <b>Description:</b>       | Perform on-site inspection and collect data.                                                                                                                                                                        |
| <b>Preconditions:</b>     | <ul style="list-style-type: none"> <li>Inspection scheduled</li> </ul>                                                                                                                                              |
| <b>Postconditions:</b>    | <ul style="list-style-type: none"> <li>Inspection data recorded</li> </ul>                                                                                                                                          |
| <b>Main Flow:</b>         | <ul style="list-style-type: none"> <li>PHI starts inspection</li> <li>Records date and time</li> <li>Captures GPS location</li> <li>Completes grading form</li> <li>Attaches evidence (images/documents)</li> </ul> |
| <b>Alternate Flow:</b>    | N/A                                                                                                                                                                                                                 |

| UC-05: Record Violations |                                                                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>    | PHI                                                                                                                                        |
| <b>Description:</b>      | Record any violations observed during inspection.                                                                                          |
| <b>Preconditions:</b>    | <ul style="list-style-type: none"> <li>Inspection in progress or completed</li> </ul>                                                      |
| <b>Postconditions:</b>   | <ul style="list-style-type: none"> <li>Violations stored in system</li> </ul>                                                              |
| <b>Main Flow:</b>        | <ul style="list-style-type: none"> <li>PHI identifies violation</li> <li>Records violation details</li> <li>System saves record</li> </ul> |
| <b>Alternate Flow:</b>   | N/A                                                                                                                                        |

| UC-06: Generate Violation Report |                                                                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>            | PHI                                                                                                                                |
| <b>Description:</b>              | Generate report of violations.                                                                                                     |
| <b>Preconditions:</b>            | <ul style="list-style-type: none"> <li>Violations recorded</li> </ul>                                                              |
| <b>Postconditions:</b>           | <ul style="list-style-type: none"> <li>Violation report generated</li> </ul>                                                       |
| <b>Main Flow:</b>                | <ul style="list-style-type: none"> <li>PHI selects violations</li> <li>System compiles report</li> <li>Report generated</li> </ul> |
| <b>Alternate Flow:</b>           | N/A                                                                                                                                |

| UC-07: Submit Inspection Report |                                                                                                                                             |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                  | PHI                                                                                                                                         |
| Description:                    | Submit completed inspection report to MOH.                                                                                                  |
| Preconditions:                  | <ul style="list-style-type: none"> <li>• Inspection completed</li> </ul>                                                                    |
| Postconditions:                 | <ul style="list-style-type: none"> <li>• Report sent to MOH</li> </ul>                                                                      |
| Main Flow:                      | <ul style="list-style-type: none"> <li>• PHI reviews inspection data</li> <li>• Submits report</li> <li>• System forwards to MOH</li> </ul> |
| Alternate Flow:                 | N/A                                                                                                                                         |

| UC-08: Review Inspection |                                                                                                                                                 |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:           | MOH                                                                                                                                             |
| Description:             | Review submitted inspection report.                                                                                                             |
| Preconditions:           | <ul style="list-style-type: none"> <li>• Report submitted by PHI</li> </ul>                                                                     |
| Postconditions:          | <ul style="list-style-type: none"> <li>• Decision taken (approve/return)</li> </ul>                                                             |
| Main Flow:               | <ul style="list-style-type: none"> <li>• MOH accesses report</li> <li>• Reviews findings and grading</li> <li>• Proceeds to decision</li> </ul> |
| Alternate Flow:          | N/A                                                                                                                                             |

| UC-09: Approve Inspection |                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------|
| Primary Actor:            | MOH                                                                                                         |
| Description:              | Approve inspection results.                                                                                 |
| Preconditions:            |                                                                                                             |
| Postconditions:           | <ul style="list-style-type: none"> <li>• Inspection approved</li> <li>• Results published to FBO</li> </ul> |
| Main Flow:                |                                                                                                             |
| Alternate Flow:           | N/A                                                                                                         |

| UC-10: Return for Corrections |                                                                                  |
|-------------------------------|----------------------------------------------------------------------------------|
| Primary Actor:                | MOH                                                                              |
| Description:                  | Return report to PHI for corrections.                                            |
| Preconditions:                |                                                                                  |
| Postconditions:               | <ul style="list-style-type: none"> <li>• PHI notified for corrections</li> </ul> |

|                        |                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Main Flow:</b>      | <ul style="list-style-type: none"> <li>• MOH identifies issues</li> <li>• Sends report back with comments</li> </ul> |
| <b>Alternate Flow:</b> | N/A                                                                                                                  |

| <b>UC-11: Evaluate Compliance Level</b> |                                                                                                                                                                                                                                               |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>                   | System                                                                                                                                                                                                                                        |
| <b>Description:</b>                     | Determine compliance level based on inspection results.                                                                                                                                                                                       |
| <b>Preconditions:</b>                   | <ul style="list-style-type: none"> <li>• Inspection data available</li> </ul>                                                                                                                                                                 |
| <b>Postconditions:</b>                  | <ul style="list-style-type: none"> <li>• Compliance level assigned</li> </ul>                                                                                                                                                                 |
| <b>Main Flow:</b>                       | <ul style="list-style-type: none"> <li>• System analyzes grading inputs</li> <li>• Categorizes into: <ul style="list-style-type: none"> <li>◦ High compliance</li> <li>◦ Moderate compliance</li> <li>◦ Low compliance</li> </ul> </li> </ul> |
| <b>Alternate Flow:</b>                  | N/A                                                                                                                                                                                                                                           |

| <b>UC-12: Compliance-Based Actions</b> |                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>                  |                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Description:</b>                    |                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Preconditions:</b>                  |                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Postconditions:</b>                 |                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Main Flow:</b>                      | <p><b>UC-12A: High Compliance Routine</b></p> <ul style="list-style-type: none"> <li>• Schedule future inspections</li> </ul> <p><b>UC-12B: Moderate Compliance Routine</b></p> <ul style="list-style-type: none"> <li>• Issue corrective actions</li> </ul> <p><b>UC-12C: Low Compliance Routine</b></p> <ul style="list-style-type: none"> <li>• Trigger enforcement actions</li> </ul> |
| <b>Alternate Flow:</b>                 | N/A                                                                                                                                                                                                                                                                                                                                                                                       |

| <b>UC-13: Trigger Enforcement Actions</b> |                                                                                   |
|-------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Primary Actor:</b>                     | System / MOH                                                                      |
| <b>Description:</b>                       | Initiate enforcement for low compliance.                                          |
| <b>Preconditions:</b>                     |                                                                                   |
| <b>Postconditions:</b>                    | <ul style="list-style-type: none"> <li>• Enforcement actions triggered</li> </ul> |
| <b>Main Flow:</b>                         |                                                                                   |
| <b>Alternate Flow:</b>                    | N/A                                                                               |

| UC-14: Issue Corrective Actions |                                                                  |
|---------------------------------|------------------------------------------------------------------|
| Primary Actor:                  | MOH / System                                                     |
| Description:                    | Issue corrective actions to FBO                                  |
| Preconditions:                  |                                                                  |
| Postconditions:                 | <ul style="list-style-type: none"> <li>● FBO notified</li> </ul> |
| Main Flow:                      |                                                                  |
| Alternate Flow:                 | N/A                                                              |

| UC-15: Receive Inspection Results |                                                                                                                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                    | FBO                                                                                                                                                                                                                                                    |
| Description:                      | View inspection outcome.                                                                                                                                                                                                                               |
| Preconditions:                    | <ul style="list-style-type: none"> <li>● Inspection approved</li> </ul>                                                                                                                                                                                |
| Postconditions:                   | <ul style="list-style-type: none"> <li>● Results visible to FBO</li> </ul>                                                                                                                                                                             |
| Main Flow:                        | <ul style="list-style-type: none"> <li>● FBO logs in</li> <li>● Views inspection results</li> <li>● Accesses: <ul style="list-style-type: none"> <li>◦ Violations</li> <li>◦ Corrective actions</li> <li>◦ Certification status</li> </ul> </li> </ul> |
| Alternate Flow:                   | N/A                                                                                                                                                                                                                                                    |

| UC-16: View Violations |                                                                                                                   |
|------------------------|-------------------------------------------------------------------------------------------------------------------|
| Primary Actor:         | FBO                                                                                                               |
| Description:           | View violations recorded.                                                                                         |
| Preconditions:         | PHI to submit violations                                                                                          |
| Postconditions:        |                                                                                                                   |
| Main Flow:             | <ul style="list-style-type: none"> <li>● PHI submit violation record</li> <li>● FBO to view the record</li> </ul> |
| Alternate Flow:        | N/A                                                                                                               |

| UC-17: View Corrective Actions |                                   |
|--------------------------------|-----------------------------------|
| Primary Actor:                 | FBO                               |
| Description:                   | View required corrective actions. |
| Preconditions:                 | Get corrective actions from PHI   |

|                        |                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Postconditions:</b> | Update the system after corrective actions implemented                                                             |
| <b>Main Flow:</b>      | <ul style="list-style-type: none"> <li>• Receive actions</li> <li>• Update the system after implemented</li> </ul> |
| <b>Alternate Flow:</b> | N/A                                                                                                                |

|                                    |                                                                            |
|------------------------------------|----------------------------------------------------------------------------|
| <b>UC-18: Download Certificate</b> |                                                                            |
| <b>Primary Actor:</b>              | FBO                                                                        |
| <b>Description:</b>                | Download inspection certificate.                                           |
| <b>Preconditions:</b>              | <ul style="list-style-type: none"> <li>• Inspection approved</li> </ul>    |
| <b>Postconditions:</b>             | <ul style="list-style-type: none"> <li>• Certificate downloaded</li> </ul> |
| <b>Main Flow:</b>                  |                                                                            |
| <b>Alternate Flow:</b>             | N/A                                                                        |

|                                        |                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>UC-19: View Grading Certificate</b> |                                                                                                           |
| <b>Primary Actor:</b>                  | FBO                                                                                                       |
| <b>Description:</b>                    | View grading classification (A/B/C/D)                                                                     |
| <b>Preconditions:</b>                  | Final grading should submit                                                                               |
| <b>Postconditions:</b>                 |                                                                                                           |
| <b>Main Flow:</b>                      | <ul style="list-style-type: none"> <li>• Authority to submit final grade</li> <li>• View grade</li> </ul> |
| <b>Alternate Flow:</b>                 | N/A                                                                                                       |

|                                        |                                                                                                                |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>UC-20: Auto Grading Calculation</b> |                                                                                                                |
| <b>Primary Actor:</b>                  | System                                                                                                         |
| <b>Description:</b>                    | Automatically calculate inspection score and grade.                                                            |
| <b>Preconditions:</b>                  | <ul style="list-style-type: none"> <li>• Grading form completed</li> </ul>                                     |
| <b>Postconditions:</b>                 | <ul style="list-style-type: none"> <li>• Grade assigned</li> </ul>                                             |
| <b>Main Flow:</b>                      | <ul style="list-style-type: none"> <li>• System calculates score</li> <li>• Assigns grade (A/B/C/D)</li> </ul> |
| <b>Alternate Flow:</b>                 | N/A                                                                                                            |

## 2.3 Laboratory Coordination Module

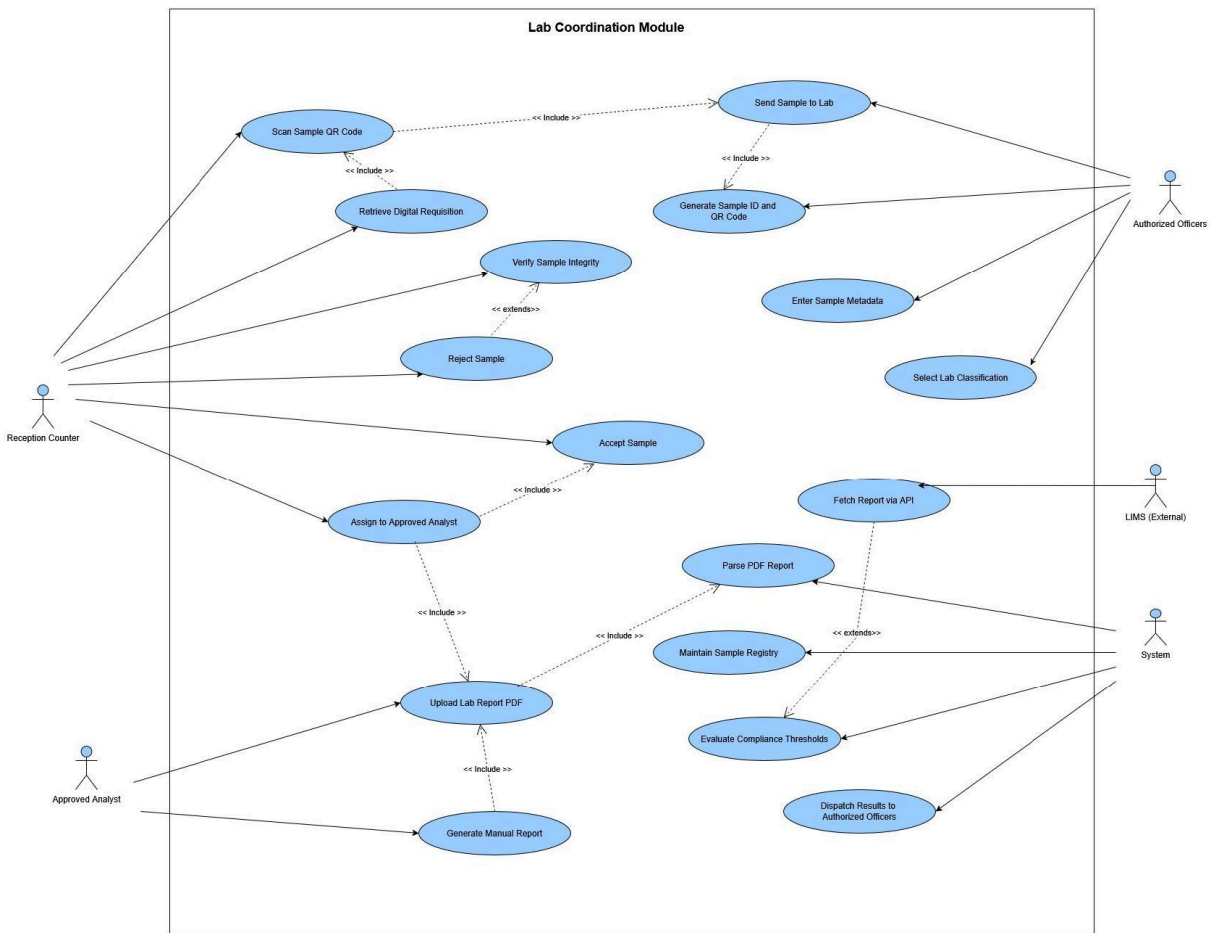


Figure 2.2

## Use Case Narratives

| UC-01: Generate Sample ID and QR Code |                                                                                                                                                                                                               |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                        | PHI / FDI                                                                                                                                                                                                     |
| Description:                          | Generate a unique identifier and QR code for each collected sample.                                                                                                                                           |
| Preconditions:                        | <ul style="list-style-type: none"> <li>Sample collection initiated</li> </ul>                                                                                                                                 |
| Postconditions:                       | <ul style="list-style-type: none"> <li>Unique sample ID and QR code generated</li> </ul>                                                                                                                      |
| Main Flow:                            | <ul style="list-style-type: none"> <li>User initiates sample registration</li> <li>System generates unique sample ID</li> <li>System generates QR code</li> <li>QR code is linked to sample record</li> </ul> |
| Alternate Flow:                       | N/A                                                                                                                                                                                                           |

| UC-02: Enter Sample Metadata |                                                                                                                                                                   |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:               | PHI / FDI                                                                                                                                                         |
| Description:                 | Enter details about the collected sample.                                                                                                                         |
| Preconditions:               | <ul style="list-style-type: none"> <li>Sample ID generated</li> </ul>                                                                                             |
| Postconditions:              | <ul style="list-style-type: none"> <li>Sample metadata stored</li> </ul>                                                                                          |
| Main Flow:                   | <ul style="list-style-type: none"> <li>User inputs sample details (type, location, date, etc.)</li> <li>System validates inputs</li> <li>Data is saved</li> </ul> |
| Alternate Flow:              | N/A                                                                                                                                                               |

| UC-03: Select Lab Classification |                                                                                                                          |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                   | PHI / FDI                                                                                                                |
| Description:                     | Select appropriate lab category/type for testing.                                                                        |
| Preconditions:                   | <ul style="list-style-type: none"> <li>Sample registered</li> </ul>                                                      |
| Postconditions:                  | <ul style="list-style-type: none"> <li>Lab classification assigned</li> </ul>                                            |
| Main Flow:                       | <ul style="list-style-type: none"> <li>User selects lab classification</li> <li>System records classification</li> </ul> |
| Alternate Flow:                  | N/A                                                                                                                      |

| UC-04: Send Sample to Lab |                                                                       |
|---------------------------|-----------------------------------------------------------------------|
| Primary Actor:            | PHI / FDI                                                             |
| Description:              | Dispatch sample to laboratory.                                        |
| Preconditions:            | <ul style="list-style-type: none"> <li>Sample ID generated</li> </ul> |



|                        |                                                                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <ul style="list-style-type: none"> <li>● Metadata completed</li> </ul>                                                                     |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>● Sample marked as sent</li> </ul>                                                                  |
| <b>Main Flow:</b>      | <ul style="list-style-type: none"> <li>● User confirms dispatch</li> <li>● System updates sample status</li> <li>● Lab notified</li> </ul> |
| <b>Alternate Flow:</b> | N/A                                                                                                                                        |

| UC-05: Scan Sample QR Code |                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>      | Reception Counter                                                                                                                             |
| <b>Description:</b>        | Scan QR code to retrieve sample information.                                                                                                  |
| <b>Preconditions:</b>      | <ul style="list-style-type: none"> <li>● Sample received at lab</li> </ul>                                                                    |
| <b>Postconditions:</b>     | <ul style="list-style-type: none"> <li>● Sample details displayed</li> </ul>                                                                  |
| <b>Main Flow:</b>          | <ul style="list-style-type: none"> <li>● Staff scans QR code</li> <li>● System retrieves sample record</li> <li>● Displays details</li> </ul> |
| <b>Alternate Flow:</b>     | N/A                                                                                                                                           |

| UC-06: Retrieve Digital Requisition |                                                                           |
|-------------------------------------|---------------------------------------------------------------------------|
| <b>Primary Actor:</b>               | Reception Counter                                                         |
| <b>Description:</b>                 | Retrieve associated digital requisition form.                             |
| <b>Preconditions:</b>               | <ul style="list-style-type: none"> <li>● QR code scanned</li> </ul>       |
| <b>Postconditions:</b>              | <ul style="list-style-type: none"> <li>● Requisition displayed</li> </ul> |
| <b>Main Flow:</b>                   |                                                                           |
| <b>Alternate Flow:</b>              | N/A                                                                       |

| UC-07: Verify Sample Integrity |                                                                                                                                                |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>          | Reception Counter                                                                                                                              |
| <b>Description:</b>            | Check condition of received sample.                                                                                                            |
| <b>Preconditions:</b>          | <ul style="list-style-type: none"> <li>● Sample received</li> </ul>                                                                            |
| <b>Postconditions:</b>         | <ul style="list-style-type: none"> <li>● Sample marked as accepted or rejected</li> </ul>                                                      |
| <b>Main Flow:</b>              | <ul style="list-style-type: none"> <li>● Staff inspects sample</li> <li>● Confirms integrity</li> <li>● System records verification</li> </ul> |
| <b>Alternate Flow:</b>         | <ul style="list-style-type: none"> <li>● 2a. Sample compromised → proceed to reject sample</li> </ul>                                          |

| UC-08: Accept Sample |                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------|
| Primary Actor:       | Reception Counter                                                                         |
| Description:         | Accept valid sample for analysis.                                                         |
| Preconditions:       | <ul style="list-style-type: none"> <li>Sample integrity verified</li> </ul>               |
| Postconditions:      | <ul style="list-style-type: none"> <li>Sample accepted and queued for analysis</li> </ul> |
| Main Flow:           |                                                                                           |
| Alternate Flow:      | N/A                                                                                       |

| UC-09: Reject Sample |                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------|
| Primary Actor:       | Reception Counter                                                                                 |
| Description:         | Reject invalid sample.                                                                            |
| Preconditions:       |                                                                                                   |
| Postconditions:      | <ul style="list-style-type: none"> <li>Sample marked rejected</li> <li>Reason recorded</li> </ul> |
| Main Flow:           |                                                                                                   |
| Alternate Flow:      | N/A                                                                                               |

| UC-10: Assign to Approved Analyst |                                                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                    | Reception Counter                                                                                                         |
| Description:                      | Assign accepted sample to certified analyst.                                                                              |
| Preconditions:                    | <ul style="list-style-type: none"> <li>Sample accepted</li> </ul>                                                         |
| Postconditions:                   | <ul style="list-style-type: none"> <li>Analyst assigned</li> </ul>                                                        |
| Main Flow:                        | <ul style="list-style-type: none"> <li>Select analyst</li> <li>System assigns sample</li> <li>Analyst notified</li> </ul> |
| Alternate Flow:                   | N/A                                                                                                                       |

| UC-11: Upload Lab Report PDF |                                                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:               | Approved Analyst                                                                                                              |
| Description:                 | Upload finalized lab report.                                                                                                  |
| Preconditions:               | <ul style="list-style-type: none"> <li>Sample analysis completed</li> </ul>                                                   |
| Postconditions:              | <ul style="list-style-type: none"> <li>Report stored in system</li> </ul>                                                     |
| Main Flow:                   | <ul style="list-style-type: none"> <li>Analyst uploads report</li> <li>System validates file</li> <li>Report saved</li> </ul> |
| Alternate Flow:              | N/A                                                                                                                           |

| UC-12: Generate Manual Report |                                                                                 |
|-------------------------------|---------------------------------------------------------------------------------|
| Primary Actor:                | Approved Analyst                                                                |
| Description:                  | Manually create lab report if needed.                                           |
| Preconditions:                | <ul style="list-style-type: none"> <li>Analysis completed</li> </ul>            |
| Postconditions:               | <ul style="list-style-type: none"> <li>Report generated and uploaded</li> </ul> |
| Main Flow:                    |                                                                                 |
| Alternate Flow:               | N/A                                                                             |

| UC-13: Fetch Report via API |                                                                             |
|-----------------------------|-----------------------------------------------------------------------------|
| Primary Actor:              | LIMS (External System)                                                      |
| Description:                | Retrieve lab report through integration.                                    |
| Preconditions:              | <ul style="list-style-type: none"> <li>Report available in LIMS</li> </ul>  |
| Postconditions:             | <ul style="list-style-type: none"> <li>Report imported to system</li> </ul> |
| Main Flow:                  |                                                                             |
| Alternate Flow:             | N/A                                                                         |

| UC-14: Parse PDF Report |                                                                                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:          | System                                                                                                                                    |
| Description:            | Extract structured data from uploaded report.                                                                                             |
| Preconditions:          | <ul style="list-style-type: none"> <li>Report uploaded</li> </ul>                                                                         |
| Postconditions:         | <ul style="list-style-type: none"> <li>Data extracted and stored</li> </ul>                                                               |
| Main Flow:              | <ul style="list-style-type: none"> <li>System reads PDF</li> <li>Extracts relevant data fields</li> <li>Stores structured data</li> </ul> |
| Alternate Flow:         | N/A                                                                                                                                       |

| UC-15: Maintain Sample Registry |                                                                         |
|---------------------------------|-------------------------------------------------------------------------|
| Primary Actor:                  | System                                                                  |
| Description:                    | Maintain lifecycle records of samples.                                  |
| Preconditions:                  | <ul style="list-style-type: none"> <li>Sample data available</li> </ul> |
| Postconditions:                 | <ul style="list-style-type: none"> <li>Registry updated</li> </ul>      |
| Main Flow:                      |                                                                         |
| Alternate Flow:                 | N/A                                                                     |

| UC-16: Evaluate Compliance Thresholds |        |
|---------------------------------------|--------|
| Primary Actor:                        | System |

|                        |                                                                                                                                     |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description:</b>    | Evaluate lab results against regulatory thresholds.                                                                                 |
| <b>Preconditions:</b>  | <ul style="list-style-type: none"> <li>● Parsed report data available</li> </ul>                                                    |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>● Compliance status determined</li> </ul>                                                    |
| <b>Main Flow:</b>      | <ul style="list-style-type: none"> <li>● System compares results with thresholds</li> <li>● Determines compliance status</li> </ul> |
| <b>Alternate Flow:</b> | N/A                                                                                                                                 |

| UC-17: Dispatch Results to Authorized Officers |                                                                                                                                         |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>                          | System                                                                                                                                  |
| <b>Description:</b>                            | Send analyzed results to the Authorized Officers                                                                                        |
| <b>Preconditions:</b>                          | <ul style="list-style-type: none"> <li>● Compliance evaluation completed</li> </ul>                                                     |
| <b>Postconditions:</b>                         | <ul style="list-style-type: none"> <li>● MOH notified with results</li> </ul>                                                           |
| <b>Main Flow:</b>                              | <ul style="list-style-type: none"> <li>● System compiles results</li> <li>● Sends notification/report to Authorized Officers</li> </ul> |
| <b>Alternate Flow:</b>                         | N/A                                                                                                                                     |

## 2.4 Legal, Prosecution and Enforcement Management Module

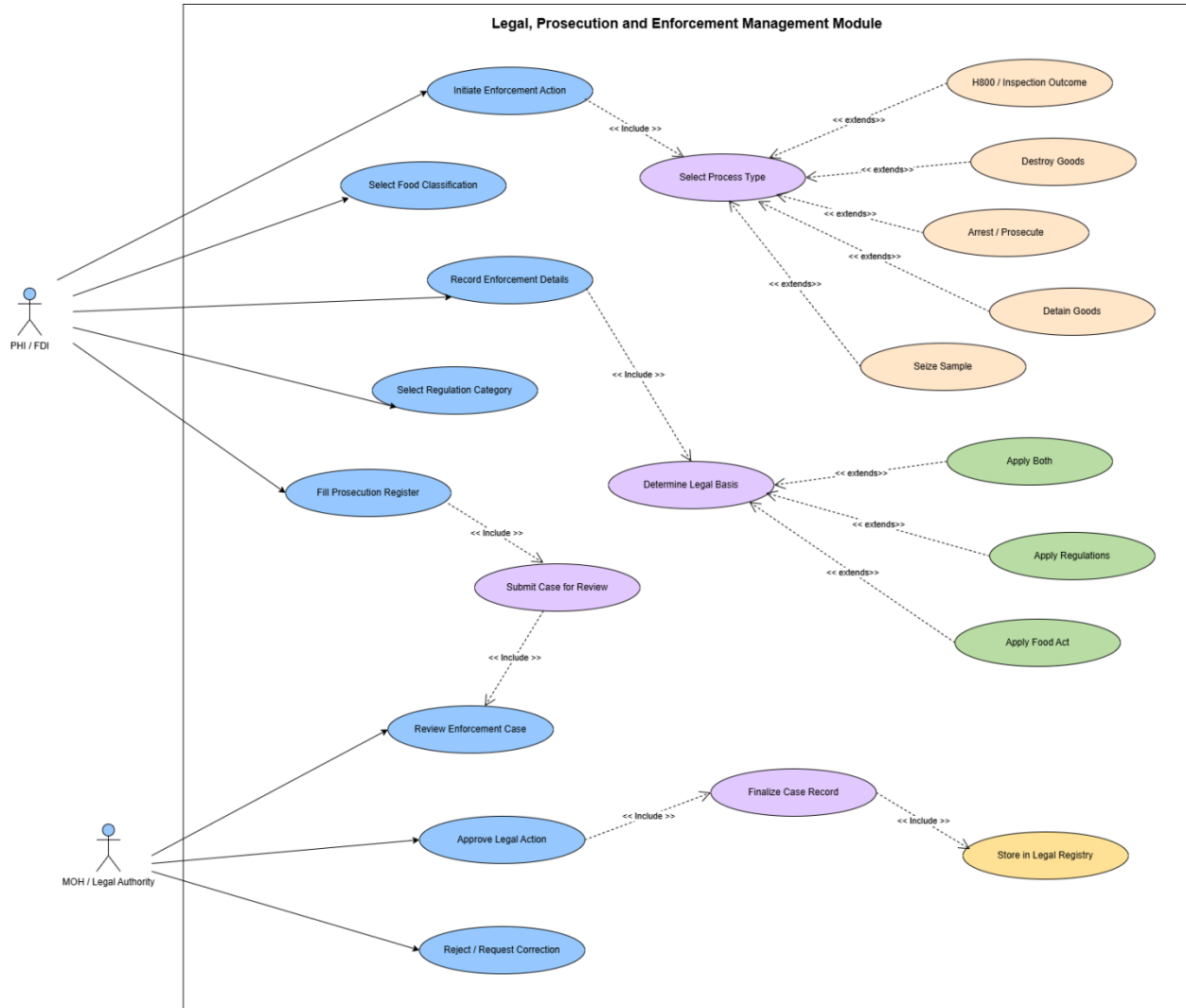


Figure 2.3

## Use Case Narratives

| UC-01: Initiate Enforcement Action |                                                                                                                                                                                                |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                     | PHI / FDI                                                                                                                                                                                      |
| Description:                       | Initiate enforcement process based on inspection findings or violations.                                                                                                                       |
| Preconditions:                     | <ul style="list-style-type: none"> <li>● Violation identified from inspection or lab results</li> </ul>                                                                                        |
| Postconditions:                    | <ul style="list-style-type: none"> <li>● Enforcement case initiated</li> </ul>                                                                                                                 |
| Main Flow:                         | <ul style="list-style-type: none"> <li>● Actor selects “Initiate Enforcement Action”</li> <li>● System creates new enforcement case</li> <li>● User proceeds to select process type</li> </ul> |
| Alternate Flow:                    | N/A                                                                                                                                                                                            |

| UC-02: Select Process Type |                                                                                                                                                                                                                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:             | PHI / FDI                                                                                                                                                                                                                                                                                                    |
| Description:               | Choose the type of enforcement action to be taken.                                                                                                                                                                                                                                                           |
| Preconditions:             | <ul style="list-style-type: none"> <li>● Enforcement action initiated</li> </ul>                                                                                                                                                                                                                             |
| Postconditions:            | <ul style="list-style-type: none"> <li>● Process type selected</li> </ul>                                                                                                                                                                                                                                    |
| Main Flow:                 | <ul style="list-style-type: none"> <li>● User selects enforcement type: <ul style="list-style-type: none"> <li>◦ H800 / Inspection Outcome</li> <li>◦ Destroy Goods</li> <li>◦ Arrest / Prosecute</li> <li>◦ Detain Goods</li> <li>◦ Seize Sample</li> </ul> </li> <li>● System records selection</li> </ul> |
| Alternate Flow:            | N/A                                                                                                                                                                                                                                                                                                          |

| UC-03: Select Food Classification |                                                                                  |
|-----------------------------------|----------------------------------------------------------------------------------|
| Primary Actor:                    | PHI / FDI                                                                        |
| Description:                      | Define food category relevant to enforcement.                                    |
| Preconditions:                    | <ul style="list-style-type: none"> <li>● Enforcement initiated</li> </ul>        |
| Postconditions:                   | <ul style="list-style-type: none"> <li>● Food classification recorded</li> </ul> |
| Main Flow:                        |                                                                                  |

|                        |     |
|------------------------|-----|
| <b>Alternate Flow:</b> | N/A |
|------------------------|-----|

| UC-04: Select Regulation Category |                                                                                |
|-----------------------------------|--------------------------------------------------------------------------------|
| <b>Primary Actor:</b>             | PHI / FDI                                                                      |
| <b>Description:</b>               | Identify applicable regulatory category.                                       |
| <b>Preconditions:</b>             |                                                                                |
| <b>Postconditions:</b>            | <ul style="list-style-type: none"> <li>Regulation category assigned</li> </ul> |
| <b>Main Flow:</b>                 |                                                                                |
| <b>Alternate Flow:</b>            | N/A                                                                            |

| UC-05: Record Enforcement Details |                                                                                                                                                             |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>             | PHI / FDI                                                                                                                                                   |
| <b>Description:</b>               | Capture all enforcement-related details.                                                                                                                    |
| <b>Preconditions:</b>             | <ul style="list-style-type: none"> <li>Enforcement process started</li> </ul>                                                                               |
| <b>Postconditions:</b>            | <ul style="list-style-type: none"> <li>Case details recorded</li> </ul>                                                                                     |
| <b>Main Flow:</b>                 | <ul style="list-style-type: none"> <li>User enters violation details</li> <li>Adds supporting evidence</li> <li>System validates and stores data</li> </ul> |
| <b>Alternate Flow:</b>            | N/A                                                                                                                                                         |

| UC-06: Determine Legal Basis |                                                                                                                                                                                                                |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>        | System / PHI                                                                                                                                                                                                   |
| <b>Description:</b>          | Identify applicable legal framework.                                                                                                                                                                           |
| <b>Preconditions:</b>        | <ul style="list-style-type: none"> <li>Enforcement details recorded</li> </ul>                                                                                                                                 |
| <b>Postconditions:</b>       | <ul style="list-style-type: none"> <li>Legal basis determined</li> </ul>                                                                                                                                       |
| <b>Main Flow:</b>            | <ul style="list-style-type: none"> <li>System evaluates case details</li> <li>Determines applicable: <ul style="list-style-type: none"> <li>Food Act</li> <li>Regulations</li> <li>Both</li> </ul> </li> </ul> |
| <b>Alternate Flow:</b>       | N/A                                                                                                                                                                                                            |

| UC-07: Fill Prosecution Register |  |
|----------------------------------|--|
|----------------------------------|--|

|                        |                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>  | PHI / FDI                                                                                                                                 |
| <b>Description:</b>    | Enter prosecution details for legal processing.                                                                                           |
| <b>Preconditions:</b>  | <ul style="list-style-type: none"> <li>• Legal basis determined</li> </ul>                                                                |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>• Prosecution record created</li> </ul>                                                            |
| <b>Main Flow:</b>      | <ul style="list-style-type: none"> <li>• User fills prosecution form</li> <li>• System validates entries</li> <li>• Data saved</li> </ul> |
| <b>Alternate Flow:</b> | N/A                                                                                                                                       |

|                                      |                                                                                                               |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>UC-08: Submit Case for Review</b> |                                                                                                               |
| <b>Primary Actor:</b>                | PHI / FDI                                                                                                     |
| <b>Description:</b>                  | Submit enforcement case for authority review.                                                                 |
| <b>Preconditions:</b>                | <ul style="list-style-type: none"> <li>• All required details completed</li> </ul>                            |
| <b>Postconditions:</b>               | <ul style="list-style-type: none"> <li>• Case forwarded to MOH / Legal Authority</li> </ul>                   |
| <b>Main Flow:</b>                    | <ul style="list-style-type: none"> <li>• User submits case</li> <li>• System forwards to authority</li> </ul> |
| <b>Alternate Flow:</b>               | N/A                                                                                                           |

|                                       |                                                                                                                                 |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>UC-09: Review Enforcement Case</b> |                                                                                                                                 |
| <b>Primary Actor:</b>                 | MOH / Legal Authority                                                                                                           |
| <b>Description:</b>                   | Review submitted enforcement case.                                                                                              |
| <b>Preconditions:</b>                 | <ul style="list-style-type: none"> <li>• Case submitted</li> </ul>                                                              |
| <b>Postconditions:</b>                | <ul style="list-style-type: none"> <li>• Decision prepared</li> </ul>                                                           |
| <b>Main Flow:</b>                     | <ul style="list-style-type: none"> <li>• Authority reviews case details</li> <li>• Verifies evidence and legal basis</li> </ul> |
| <b>Alternate Flow:</b>                | N/A                                                                                                                             |

|                                    |                                                                                                                        |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>UC-10: Approve Legal Action</b> |                                                                                                                        |
| <b>Primary Actor:</b>              | MOH / Legal Authority                                                                                                  |
| <b>Description:</b>                | Approve enforcement/legal action.                                                                                      |
| <b>Preconditions:</b>              |                                                                                                                        |
| <b>Postconditions:</b>             | <ul style="list-style-type: none"> <li>• Case approved for execution</li> </ul>                                        |
| <b>Main Flow:</b>                  | <ul style="list-style-type: none"> <li>• Authority approves case</li> <li>• System proceeds to finalization</li> </ul> |
| <b>Alternate Flow:</b>             | N/A                                                                                                                    |

|                                           |  |
|-------------------------------------------|--|
| <b>UC-11: Reject / Request Correction</b> |  |
|-------------------------------------------|--|



|                        |                                                                                                                                    |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>  | MOH / Legal Authority                                                                                                              |
| <b>Description:</b>    | Reject case or request corrections.                                                                                                |
| <b>Preconditions:</b>  |                                                                                                                                    |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>• Case returned to PHI for updates</li> </ul>                                               |
| <b>Main Flow:</b>      | <ul style="list-style-type: none"> <li>• Authority identifies issues</li> <li>• Sends feedback</li> <li>• Case returned</li> </ul> |
| <b>Alternate Flow:</b> | N/A                                                                                                                                |

|                                    |                                                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>UC-12: Finalize Case Record</b> |                                                                                                                      |
| <b>Primary Actor:</b>              | System                                                                                                               |
| <b>Description:</b>                | Finalize approved enforcement case.                                                                                  |
| <b>Preconditions:</b>              | <ul style="list-style-type: none"> <li>• Case approved</li> </ul>                                                    |
| <b>Postconditions:</b>             | <ul style="list-style-type: none"> <li>• Case closed and finalized</li> </ul>                                        |
| <b>Main Flow:</b>                  | <ul style="list-style-type: none"> <li>• System consolidates case data</li> <li>• Marks case as finalized</li> </ul> |
| <b>Alternate Flow:</b>             | N/A                                                                                                                  |

|                                       |                                                                                     |
|---------------------------------------|-------------------------------------------------------------------------------------|
| <b>UC-13: Store in Legal Registry</b> |                                                                                     |
| <b>Primary Actor:</b>                 | System                                                                              |
| <b>Description:</b>                   | Store finalized case in legal registry.                                             |
| <b>Preconditions:</b>                 | <ul style="list-style-type: none"> <li>• Case finalized</li> </ul>                  |
| <b>Postconditions:</b>                | <ul style="list-style-type: none"> <li>• Case archived for recordkeeping</li> </ul> |
| <b>Main Flow:</b>                     |                                                                                     |
| <b>Alternate Flow:</b>                | N/A                                                                                 |

## 2.5 Import Control Module

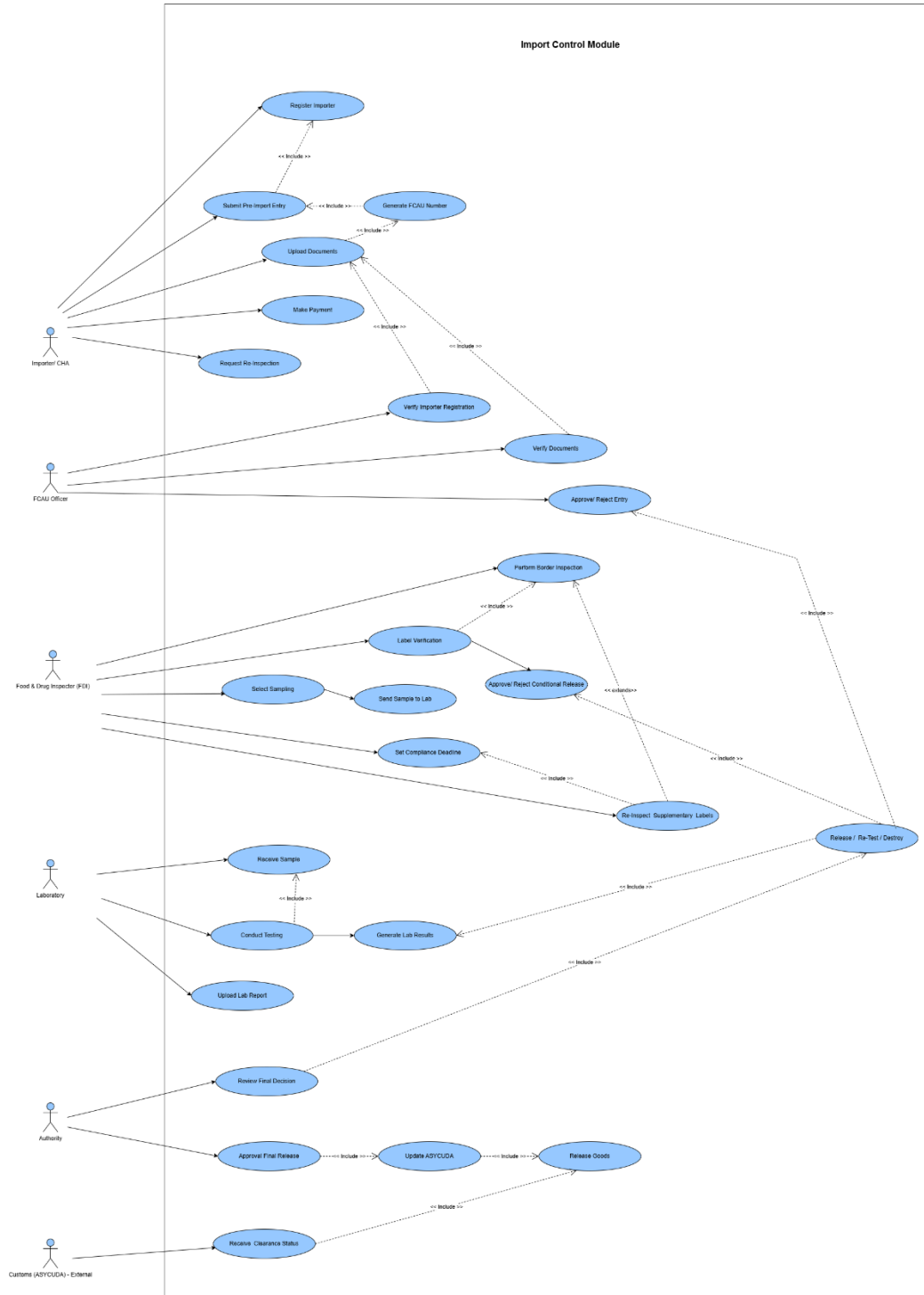


Figure 2.4

## Use Case Narratives

| UC-01: Register Importer |                                                                                                                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:           | Importer / CHA                                                                                                                                                                                                       |
| Description:             | Register a new importer in the system.                                                                                                                                                                               |
| Preconditions:           | <ul style="list-style-type: none"> <li>Importer not previously registered</li> </ul>                                                                                                                                 |
| Postconditions:          | <ul style="list-style-type: none"> <li>Importer account created and verified</li> </ul>                                                                                                                              |
| Main Flow:               | <ul style="list-style-type: none"> <li>Importer submits registration details</li> <li>System validates information</li> <li>FCAU Officer verifies registration</li> <li>System activates importer account</li> </ul> |
| Alternate Flow:          | N/A                                                                                                                                                                                                                  |

| UC-02: Submit Pre-Import Entry |                                                                                                                                                                                                  |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                 | Importer / CHA                                                                                                                                                                                   |
| Description:                   | Submit details of goods prior to import.                                                                                                                                                         |
| Preconditions:                 | <ul style="list-style-type: none"> <li>Importer registered</li> </ul>                                                                                                                            |
| Postconditions:                | <ul style="list-style-type: none"> <li>Pre-import entry created with reference number</li> </ul>                                                                                                 |
| Main Flow:                     | <ul style="list-style-type: none"> <li>Importer enters shipment details</li> <li>Uploads required documents</li> <li>System generates FCAU number</li> <li>Entry submitted for review</li> </ul> |
| Alternate Flow:                | N/A                                                                                                                                                                                              |

| UC-03: Upload Documents |                                                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:          | Importer / CHA                                                                                                                       |
| Description:            | Upload supporting documents.                                                                                                         |
| Preconditions:          | <ul style="list-style-type: none"> <li>Pre-import entry initiated</li> </ul>                                                         |
| Postconditions:         | <ul style="list-style-type: none"> <li>Documents stored</li> </ul>                                                                   |
| Main Flow:              | <ul style="list-style-type: none"> <li>User uploads documents</li> <li>System validates file format</li> <li>Files stored</li> </ul> |
| Alternate Flow:         | N/A                                                                                                                                  |

| UC-04: Make Payment |                                                                              |
|---------------------|------------------------------------------------------------------------------|
| Primary Actor:      | Importer / CHA                                                               |
| Description:        | Pay applicable inspection or processing fees.                                |
| Preconditions:      | <ul style="list-style-type: none"> <li>Pre-import entry submitted</li> </ul> |

|                        |                                                                      |
|------------------------|----------------------------------------------------------------------|
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>• Payment recorded</li> </ul> |
| <b>Main Flow:</b>      |                                                                      |
| <b>Alternate Flow:</b> | N/A                                                                  |

| UC-05: Verify Importer Registration |                                                                                   |
|-------------------------------------|-----------------------------------------------------------------------------------|
| <b>Primary Actor:</b>               | FCAU Officer                                                                      |
| <b>Description:</b>                 | Verify importer registration details.                                             |
| <b>Preconditions:</b>               | <ul style="list-style-type: none"> <li>• Importer registered</li> </ul>           |
| <b>Postconditions:</b>              | <ul style="list-style-type: none"> <li>• Importer verified or rejected</li> </ul> |
| <b>Main Flow:</b>                   |                                                                                   |
| <b>Alternate Flow:</b>              | N/A                                                                               |

| UC-06: Verify Documents |                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------|
| <b>Primary Actor:</b>   | FCAU Officer                                                                      |
| <b>Description:</b>     | Validate submitted documents.                                                     |
| <b>Preconditions:</b>   | <ul style="list-style-type: none"> <li>• Documents uploaded</li> </ul>            |
| <b>Postconditions:</b>  | <ul style="list-style-type: none"> <li>• Documents approved or flagged</li> </ul> |
| <b>Main Flow:</b>       |                                                                                   |
| <b>Alternate Flow:</b>  | N/A                                                                               |

| UC-07: Approve / Reject Entry |                                                                                |
|-------------------------------|--------------------------------------------------------------------------------|
| <b>Primary Actor:</b>         | FCAU Officer                                                                   |
| <b>Description:</b>           | Approve or reject pre-import entry.                                            |
| <b>Preconditions:</b>         | <ul style="list-style-type: none"> <li>• Documents verified</li> </ul>         |
| <b>Postconditions:</b>        | <ul style="list-style-type: none"> <li>• Entry approved or rejected</li> </ul> |
| <b>Main Flow:</b>             |                                                                                |
| <b>Alternate Flow:</b>        | N/A                                                                            |

| UC-08: Perform Border Inspection |                                                                                                                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>            | Food & Drug Inspector (FDI)                                                                                                                          |
| <b>Description:</b>              | Conduct inspection at entry point.                                                                                                                   |
| <b>Preconditions:</b>            | <ul style="list-style-type: none"> <li>• Entry approved</li> </ul>                                                                                   |
| <b>Postconditions:</b>           | <ul style="list-style-type: none"> <li>• Inspection results recorded</li> </ul>                                                                      |
| <b>Main Flow:</b>                | <ul style="list-style-type: none"> <li>• Inspector performs inspection</li> <li>• Conducts label verification</li> <li>• Records findings</li> </ul> |
| <b>Alternate Flow:</b>           | N/A                                                                                                                                                  |

| UC-09: Label Verification |                                                                             |
|---------------------------|-----------------------------------------------------------------------------|
| Primary Actor:            | FDI                                                                         |
| Description:              | Verify compliance of product labeling.                                      |
| Preconditions:            | <ul style="list-style-type: none"> <li>Inspection initiated</li> </ul>      |
| Postconditions:           | <ul style="list-style-type: none"> <li>Label compliance recorded</li> </ul> |
| Main Flow:                |                                                                             |
| Alternate Flow:           | N/A                                                                         |

| UC-10: Select Sampling |                                                                                 |
|------------------------|---------------------------------------------------------------------------------|
| Primary Actor:         | FDI                                                                             |
| Description:           | Select samples for lab testing.                                                 |
| Preconditions:         | <ul style="list-style-type: none"> <li>Inspection ongoing</li> </ul>            |
| Postconditions:        | <ul style="list-style-type: none"> <li>Samples selected and recorded</li> </ul> |
| Main Flow:             |                                                                                 |
| Alternate Flow:        | N/A                                                                             |

| UC-11: Send Sample to Lab |                                                                             |
|---------------------------|-----------------------------------------------------------------------------|
| Primary Actor:            | FDI                                                                         |
| Description:              | Dispatch selected samples for testing.                                      |
| Preconditions:            |                                                                             |
| Postconditions:           | <ul style="list-style-type: none"> <li>Sample sent to laboratory</li> </ul> |
| Main Flow:                |                                                                             |
| Alternate Flow:           | N/A                                                                         |

| UC-12: Set Compliance Deadline |                                                                     |
|--------------------------------|---------------------------------------------------------------------|
| Primary Actor:                 | FDI                                                                 |
| Description:                   | Define deadline for corrective actions.                             |
| Preconditions:                 |                                                                     |
| Postconditions:                | <ul style="list-style-type: none"> <li>Deadline recorded</li> </ul> |
| Main Flow:                     |                                                                     |
| Alternate Flow:                | N/A                                                                 |

| UC-13: Approve / Reject Conditional Release |                                                                        |
|---------------------------------------------|------------------------------------------------------------------------|
| Primary Actor:                              | FDI                                                                    |
| Description:                                | Decide conditional release of goods.                                   |
| Preconditions:                              | <ul style="list-style-type: none"> <li>Inspection completed</li> </ul> |

|                        |                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------|
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>Conditional release decision recorded</li> </ul> |
| <b>Main Flow:</b>      |                                                                                         |
| <b>Alternate Flow:</b> | N/A                                                                                     |

| UC-14: Re-Inspect / Supplementary Labels |                                                                               |
|------------------------------------------|-------------------------------------------------------------------------------|
| <b>Primary Actor:</b>                    | FDI                                                                           |
| <b>Description:</b>                      | Perform follow-up inspection for compliance.                                  |
| <b>Preconditions:</b>                    | <ul style="list-style-type: none"> <li>Conditional release granted</li> </ul> |
| <b>Postconditions:</b>                   | <ul style="list-style-type: none"> <li>Compliance verified</li> </ul>         |
| <b>Main Flow:</b>                        |                                                                               |
| <b>Alternate Flow:</b>                   | N/A                                                                           |

| UC-15: Laboratory Processing |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>        | Laboratory                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Description:</b>          | Lab processing                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Preconditions:</b>        | Samples to receive                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Postconditions:</b>       | Upload the lab report                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Main Flow:</b>            | <p><b>UC-15A: Receive Sample</b></p> <ul style="list-style-type: none"> <li>Sample received and logged</li> </ul> <p><b>UC-15B: Conduct Testing</b></p> <ul style="list-style-type: none"> <li>Perform analysis</li> </ul> <p><b>UC-15C: Generate Lab Results</b></p> <ul style="list-style-type: none"> <li>Results generated</li> </ul> <p><b>UC-15D: Upload Lab Report</b></p> <ul style="list-style-type: none"> <li>Report uploaded to system</li> </ul> |
| <b>Alternate Flow:</b>       | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| UC-16: Release / Re-Test / Destroy Decision |                                                                                                                                                                   |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>                       | System / Authority                                                                                                                                                |
| <b>Description:</b>                         | Determine final disposition based on results.                                                                                                                     |
| <b>Preconditions:</b>                       | <ul style="list-style-type: none"> <li>Inspection and/or lab results available</li> </ul>                                                                         |
| <b>Postconditions:</b>                      | <ul style="list-style-type: none"> <li>Decision recorded: <ul style="list-style-type: none"> <li>Release</li> <li>Re-test</li> <li>Destroy</li> </ul> </li> </ul> |
| <b>Main Flow:</b>                           |                                                                                                                                                                   |

|                        |     |
|------------------------|-----|
| <b>Alternate Flow:</b> | N/A |
|------------------------|-----|

| UC-17: Review Final Decision |                                                                      |
|------------------------------|----------------------------------------------------------------------|
| <b>Primary Actor:</b>        | Authority                                                            |
| <b>Description:</b>          | Review final outcome before release.                                 |
| <b>Preconditions:</b>        | <ul style="list-style-type: none"> <li>Decision prepared</li> </ul>  |
| <b>Postconditions:</b>       | <ul style="list-style-type: none"> <li>Decision confirmed</li> </ul> |
| <b>Main Flow:</b>            |                                                                      |
| <b>Alternate Flow:</b>       | N/A                                                                  |

| UC-18: Approve Final Release |                                                                                |
|------------------------------|--------------------------------------------------------------------------------|
| <b>Primary Actor:</b>        | Authority                                                                      |
| <b>Description:</b>          | Approve goods for release.                                                     |
| <b>Preconditions:</b>        |                                                                                |
| <b>Postconditions:</b>       | <ul style="list-style-type: none"> <li>Goods approved for clearance</li> </ul> |
| <b>Main Flow:</b>            |                                                                                |
| <b>Alternate Flow:</b>       | N/A                                                                            |

| UC-19: Update ASYCUDA System |                                                                           |
|------------------------------|---------------------------------------------------------------------------|
| <b>Primary Actor:</b>        | System                                                                    |
| <b>Description:</b>          | Update customs system with clearance status.                              |
| <b>Preconditions:</b>        | <ul style="list-style-type: none"> <li>Final approval granted</li> </ul>  |
| <b>Postconditions:</b>       | <ul style="list-style-type: none"> <li>External system updated</li> </ul> |
| <b>Main Flow:</b>            |                                                                           |
| <b>Alternate Flow:</b>       | N/A                                                                       |

| UC-20: Release Goods   |                                                                      |
|------------------------|----------------------------------------------------------------------|
| <b>Primary Actor:</b>  | System                                                               |
| <b>Description:</b>    | Complete goods release process.                                      |
| <b>Preconditions:</b>  | <ul style="list-style-type: none"> <li>Approval completed</li> </ul> |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>Goods released</li> </ul>     |
| <b>Main Flow:</b>      |                                                                      |
| <b>Alternate Flow:</b> | N/A                                                                  |

| UC-21: Receive Clearance Status |                                   |
|---------------------------------|-----------------------------------|
| <b>Primary Actor:</b>           | Customs (ASYCUDA – External)      |
| <b>Description:</b>             | Receive updated clearance status. |

|                        |                                                                                                  |
|------------------------|--------------------------------------------------------------------------------------------------|
| <b>Preconditions:</b>  |                                                                                                  |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>• Clearance status reflected in customs system</li> </ul> |
| <b>Main Flow:</b>      |                                                                                                  |
| <b>Alternate Flow:</b> | N/A                                                                                              |

|                                     |                                                                                  |
|-------------------------------------|----------------------------------------------------------------------------------|
| <b>UC-22: Request Re-Inspection</b> |                                                                                  |
| <b>Primary Actor:</b>               | Importer / CHA                                                                   |
| <b>Description:</b>                 | Request re-inspection if required.                                               |
| <b>Preconditions:</b>               | <ul style="list-style-type: none"> <li>• Initial inspection completed</li> </ul> |
| <b>Postconditions:</b>              | <ul style="list-style-type: none"> <li>• Re-inspection scheduled</li> </ul>      |
| <b>Main Flow:</b>                   |                                                                                  |
| <b>Alternate Flow:</b>              | N/A                                                                              |



## 2.6 Export Control Module

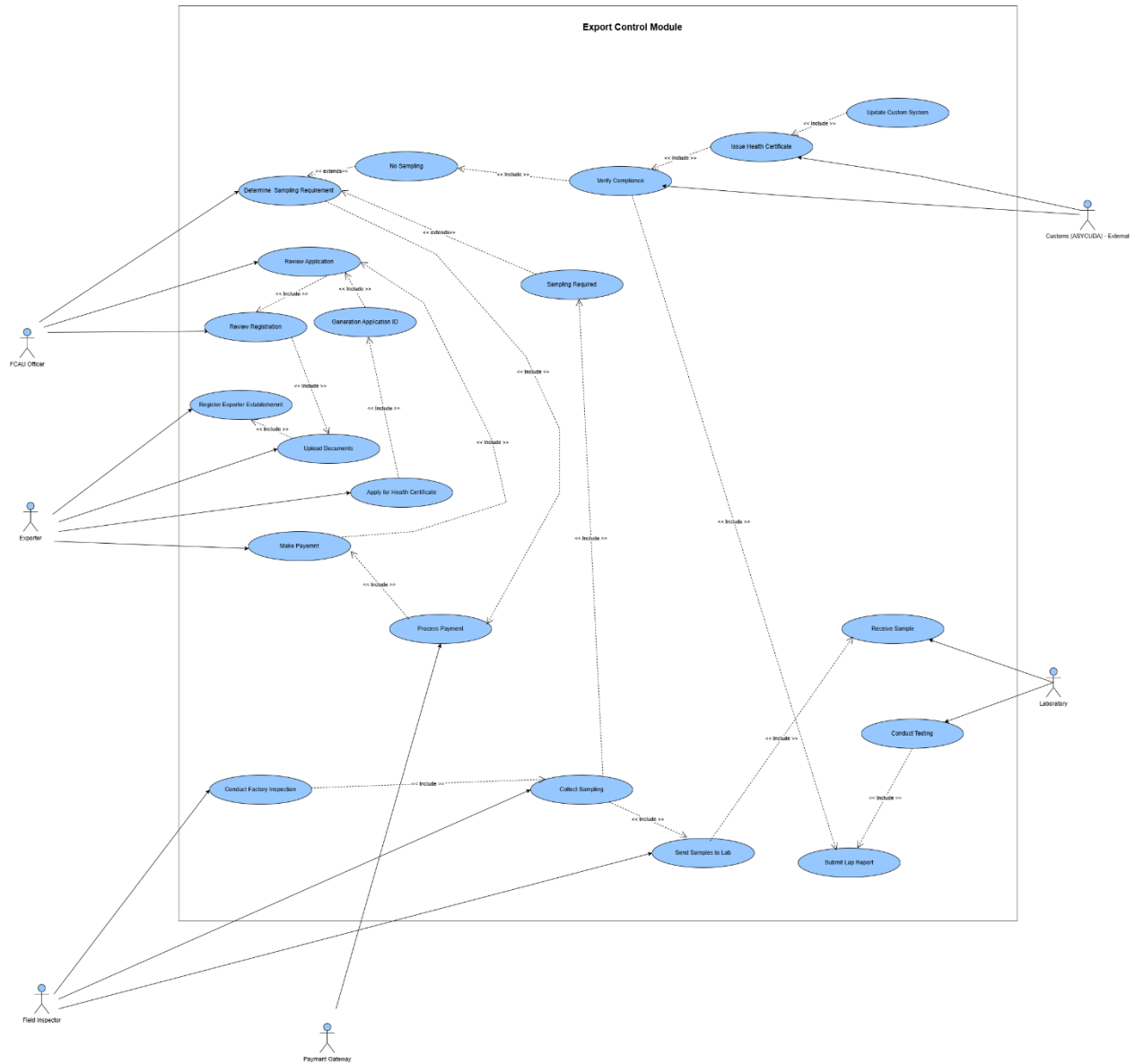


Figure 2.5

## Use Case Narratives

| UC-01: Register Exporter Establishment |                                                                                                                                                                                                                         |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                         | Exporter                                                                                                                                                                                                                |
| Description:                           | Register an exporter and their establishment in the system.                                                                                                                                                             |
| Preconditions:                         | <ul style="list-style-type: none"> <li>• Exporter is not registered</li> </ul>                                                                                                                                          |
| Postconditions:                        | <ul style="list-style-type: none"> <li>• Exporter establishment registered and available for review</li> </ul>                                                                                                          |
| Main Flow:                             | <ul style="list-style-type: none"> <li>• Exporter submits registration details</li> <li>• Uploads required documents</li> <li>• System stores registration data</li> <li>• FCAU Officer reviews registration</li> </ul> |
| Alternate Flow:                        | N/A                                                                                                                                                                                                                     |

| UC-02: Upload Documents |                                                                                                |
|-------------------------|------------------------------------------------------------------------------------------------|
| Primary Actor:          | Exporter                                                                                       |
| Description:            | Upload required documents for registration or certification.                                   |
| Preconditions:          | <ul style="list-style-type: none"> <li>• Registration or application initiated</li> </ul>      |
| Postconditions:         | <ul style="list-style-type: none"> <li>• Documents stored and linked to application</li> </ul> |
| Main Flow:              |                                                                                                |
| Alternate Flow:         | N/A                                                                                            |

| UC-03: Review Registration |                                                                                       |
|----------------------------|---------------------------------------------------------------------------------------|
| Primary Actor:             | FCAU Officer                                                                          |
| Description:               | Review exporter registration details.                                                 |
| Preconditions:             | <ul style="list-style-type: none"> <li>• Registration submitted</li> </ul>            |
| Postconditions:            | <ul style="list-style-type: none"> <li>• Registration approved or rejected</li> </ul> |
| Main Flow:                 |                                                                                       |
| Alternate Flow:            | N/A                                                                                   |

| UC-04: Apply for Health Certificate |                                                                                                                             |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                      | Exporter                                                                                                                    |
| Description:                        | Apply for export health certification.                                                                                      |
| Preconditions:                      | <ul style="list-style-type: none"> <li>• Exporter registered</li> </ul>                                                     |
| Postconditions:                     | <ul style="list-style-type: none"> <li>• Application submitted</li> </ul>                                                   |
| Main Flow:                          | <ul style="list-style-type: none"> <li>• Exporter submits application</li> <li>• System generates application ID</li> </ul> |

|                 |                                                                               |
|-----------------|-------------------------------------------------------------------------------|
|                 | <ul style="list-style-type: none"> <li>Application sent for review</li> </ul> |
| Alternate Flow: | N/A                                                                           |

| UC-05: Generate Application ID |                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------|
| Primary Actor:                 | System                                                                           |
| Description:                   | Generate a unique identifier for the application.                                |
| Preconditions:                 |                                                                                  |
| Postconditions:                | <ul style="list-style-type: none"> <li>Unique application ID assigned</li> </ul> |
| Main Flow:                     |                                                                                  |
| Alternate Flow:                | N/A                                                                              |

| UC-06: Make Payment |                                                                         |
|---------------------|-------------------------------------------------------------------------|
| Primary Actor:      | Exporter                                                                |
| Description:        | Pay certification or inspection fees.                                   |
| Preconditions:      | <ul style="list-style-type: none"> <li>Application submitted</li> </ul> |
| Postconditions:     | <ul style="list-style-type: none"> <li>Payment initiated</li> </ul>     |
| Main Flow:          |                                                                         |
| Alternate Flow:     | N/A                                                                     |

| UC-07: Process Payment |                                                                               |
|------------------------|-------------------------------------------------------------------------------|
| Primary Actor:         | Payment Gateway (External)                                                    |
| Description:           | Process payment transaction.                                                  |
| Preconditions:         |                                                                               |
| Postconditions:        | <ul style="list-style-type: none"> <li>Payment confirmed or failed</li> </ul> |
| Main Flow:             |                                                                               |
| Alternate Flow:        | N/A                                                                           |

| UC-08: Review Application |                                                                                               |
|---------------------------|-----------------------------------------------------------------------------------------------|
| Primary Actor:            | FCAU Officer                                                                                  |
| Description:              | Review health certificate application.                                                        |
| Preconditions:            | <ul style="list-style-type: none"> <li>Application submitted and payment completed</li> </ul> |
| Postconditions:           | <ul style="list-style-type: none"> <li>Application proceeds to sampling decision</li> </ul>   |
| Main Flow:                |                                                                                               |
| Alternate Flow:           | N/A                                                                                           |

| UC-09: Determine Sampling Requirement |                       |
|---------------------------------------|-----------------------|
| Primary Actor:                        | FCAU Officer / System |

|                        |                                                                                                                                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description:</b>    | Decide whether sampling is required.                                                                                                                                                                             |
| <b>Preconditions:</b>  | <ul style="list-style-type: none"> <li>● Application reviewed</li> </ul>                                                                                                                                         |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>● Sampling decision recorded</li> </ul>                                                                                                                                   |
| <b>Main Flow:</b>      | <ul style="list-style-type: none"> <li>● System evaluates risk criteria</li> <li>● Determines: <ul style="list-style-type: none"> <li>◦ No sampling required</li> <li>◦ Sampling required</li> </ul> </li> </ul> |
| <b>Alternate Flow:</b> | N/A                                                                                                                                                                                                              |

|                           |                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>UC-10: No Sampling</b> |                                                                                                           |
| <b>Primary Actor:</b>     | System                                                                                                    |
| <b>Description:</b>       | Skip sampling and proceed to compliance verification.                                                     |
| <b>Preconditions:</b>     |                                                                                                           |
| <b>Postconditions:</b>    | <ul style="list-style-type: none"> <li>● Application moves directly to compliance verification</li> </ul> |
| <b>Main Flow:</b>         |                                                                                                           |
| <b>Alternate Flow:</b>    | N/A                                                                                                       |

|                                 |                                                                        |
|---------------------------------|------------------------------------------------------------------------|
| <b>UC-11: Sampling Required</b> |                                                                        |
| <b>Primary Actor:</b>           | FCAU Officer / Field Inspector                                         |
| <b>Description:</b>             | Initiate sampling process.                                             |
| <b>Preconditions:</b>           |                                                                        |
| <b>Postconditions:</b>          | <ul style="list-style-type: none"> <li>● Sampling scheduled</li> </ul> |
| <b>Main Flow:</b>               |                                                                        |
| <b>Alternate Flow:</b>          | N/A                                                                    |

|                                          |                                                                                  |
|------------------------------------------|----------------------------------------------------------------------------------|
| <b>UC-12: Conduct Factory Inspection</b> |                                                                                  |
| <b>Primary Actor:</b>                    | Field Inspector                                                                  |
| <b>Description:</b>                      | Inspect exporter's facility.                                                     |
| <b>Preconditions:</b>                    | <ul style="list-style-type: none"> <li>● Sampling required</li> </ul>            |
| <b>Postconditions:</b>                   | <ul style="list-style-type: none"> <li>● Inspection findings recorded</li> </ul> |
| <b>Main Flow:</b>                        |                                                                                  |
| <b>Alternate Flow:</b>                   | N/A                                                                              |

|                                |                          |
|--------------------------------|--------------------------|
| <b>UC-13: Collect Sampling</b> |                          |
| <b>Primary Actor:</b>          | Field Inspector          |
| <b>Description:</b>            | Collect product samples. |

|                        |                                                                                |
|------------------------|--------------------------------------------------------------------------------|
| <b>Preconditions:</b>  |                                                                                |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>Samples collected and logged</li> </ul> |
| <b>Main Flow:</b>      |                                                                                |
| <b>Alternate Flow:</b> | N/A                                                                            |

|                                   |                                                                              |
|-----------------------------------|------------------------------------------------------------------------------|
| <b>UC-14: Send Samples to Lab</b> |                                                                              |
| <b>Primary Actor:</b>             | Field Inspector                                                              |
| <b>Description:</b>               | Dispatch samples for laboratory testing.                                     |
| <b>Preconditions:</b>             |                                                                              |
| <b>Postconditions:</b>            | <ul style="list-style-type: none"> <li>Samples sent to laboratory</li> </ul> |
| <b>Main Flow:</b>                 |                                                                              |
| <b>Alternate Flow:</b>            | N/A                                                                          |

|                                     |                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UC-15: Laboratory Processing</b> |                                                                                                                                                                                                                                                                                                                                            |
| <b>Primary Actor:</b>               | Laboratory                                                                                                                                                                                                                                                                                                                                 |
| <b>Description:</b>                 | Laboratory Processing                                                                                                                                                                                                                                                                                                                      |
| <b>Preconditions:</b>               | Sample received                                                                                                                                                                                                                                                                                                                            |
| <b>Postconditions:</b>              | Submit Lab report                                                                                                                                                                                                                                                                                                                          |
| <b>Main Flow:</b>                   | <p><b>UC-15A: Receive Sample</b></p> <ul style="list-style-type: none"> <li>Sample received and recorded</li> </ul> <p><b>UC-15B: Conduct Testing</b></p> <ul style="list-style-type: none"> <li>Perform analysis</li> </ul> <p><b>UC-15C: Submit Lab Report</b></p> <ul style="list-style-type: none"> <li>Upload test results</li> </ul> |
| <b>Alternate Flow:</b>              | N/A                                                                                                                                                                                                                                                                                                                                        |

|                                 |                                                                                 |
|---------------------------------|---------------------------------------------------------------------------------|
| <b>UC-16: Verify Compliance</b> |                                                                                 |
| <b>Primary Actor:</b>           | FCAU Officer / System                                                           |
| <b>Description:</b>             | Verify compliance based on inspection and lab results.                          |
| <b>Preconditions:</b>           | <ul style="list-style-type: none"> <li>All required checks completed</li> </ul> |
| <b>Postconditions:</b>          | <ul style="list-style-type: none"> <li>Compliance status determined</li> </ul>  |
| <b>Main Flow:</b>               |                                                                                 |
| <b>Alternate Flow:</b>          | N/A                                                                             |

|                                        |                                                                       |
|----------------------------------------|-----------------------------------------------------------------------|
| <b>UC-17: Issue Health Certificate</b> |                                                                       |
| <b>Primary Actor:</b>                  | FCAU Officer                                                          |
| <b>Description:</b>                    | Issue certificate for compliant exports.                              |
| <b>Preconditions:</b>                  | <ul style="list-style-type: none"> <li>Compliance verified</li> </ul> |

|                        |                                                                             |
|------------------------|-----------------------------------------------------------------------------|
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>Health certificate issued</li> </ul> |
| <b>Main Flow:</b>      |                                                                             |
| <b>Alternate Flow:</b> | N/A                                                                         |

| UC-18: Update Customs System |                                                                           |
|------------------------------|---------------------------------------------------------------------------|
| <b>Primary Actor:</b>        | System                                                                    |
| <b>Description:</b>          | Update customs (ASYCUDA) with export clearance.                           |
| <b>Preconditions:</b>        | <ul style="list-style-type: none"> <li>Certificate issued</li> </ul>      |
| <b>Postconditions:</b>       | <ul style="list-style-type: none"> <li>External system updated</li> </ul> |
| <b>Main Flow:</b>            |                                                                           |
| <b>Alternate Flow:</b>       | N/A                                                                       |

## 2.7 Registration of Common Salt Manufacturing and Transports Management Module

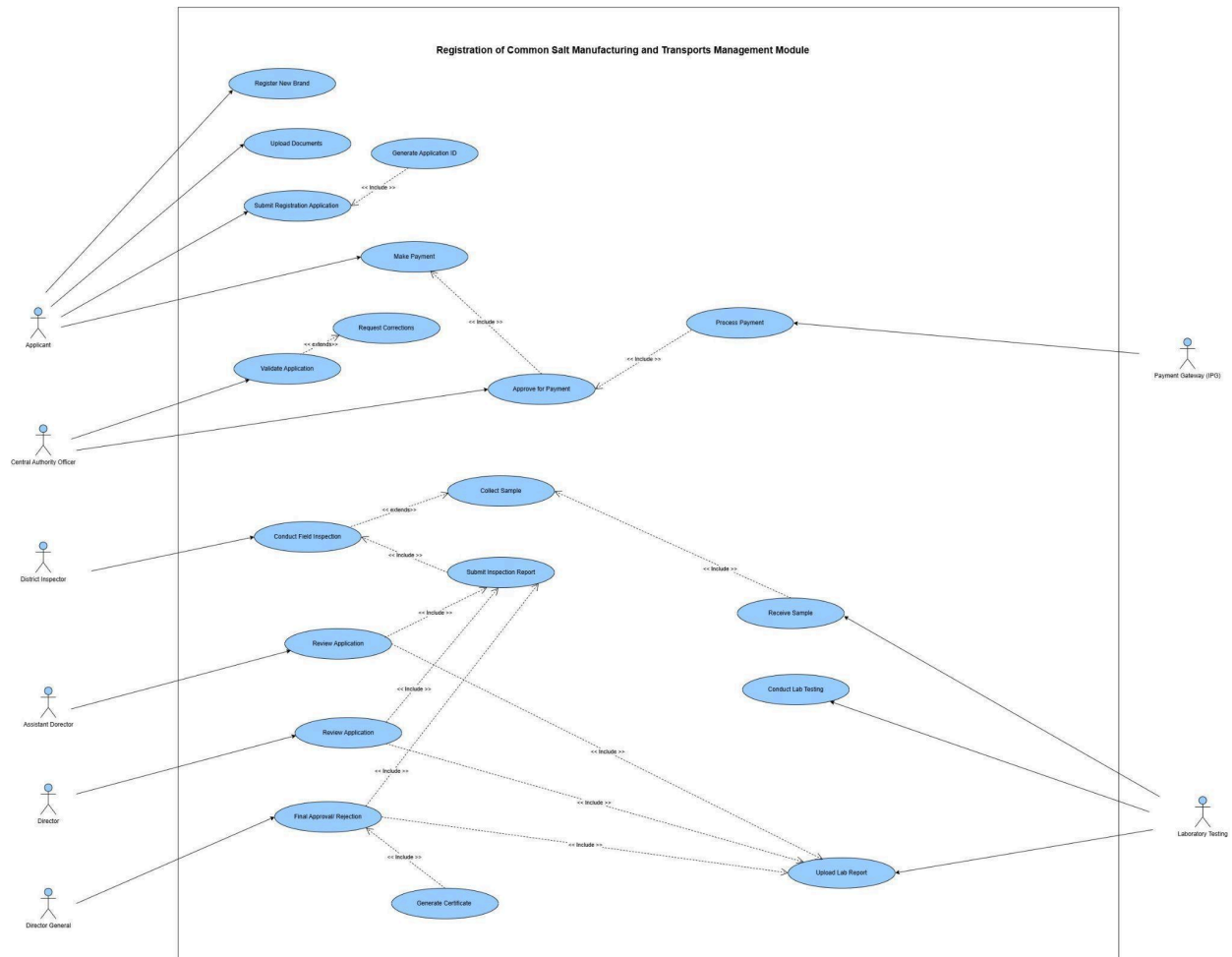


Figure 2.6

## Use Case Narratives

| UC-01: Register New Brand |                                                                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:            | Salt Factory/ Saltern                                                                                                                                       |
| Description:              | Register a Salt brand in the system.                                                                                                                        |
| Preconditions:            | <ul style="list-style-type: none"> <li>• Salt Factory/ Saltern is registered in the system</li> </ul>                                                       |
| Postconditions:           | <ul style="list-style-type: none"> <li>• Brand registration initiated</li> </ul>                                                                            |
| Main Flow:                | <ul style="list-style-type: none"> <li>• Selects “Register New Brand”</li> <li>• Enters brand details</li> <li>• System stores brand information</li> </ul> |
| Alternate Flow:           | N/A                                                                                                                                                         |

| UC-02: Submit Registration Application |                                                                                                                                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                         | Applicant                                                                                                                                                                                              |
| Description:                           | Submit application for registration of salt manufacturing or transport operations.                                                                                                                     |
| Preconditions:                         | <ul style="list-style-type: none"> <li>• Applicant has access to the system</li> </ul>                                                                                                                 |
| Postconditions:                        | <ul style="list-style-type: none"> <li>• Application submitted with unique ID</li> </ul>                                                                                                               |
| Main Flow:                             | <ul style="list-style-type: none"> <li>• Applicant fills application form</li> <li>• Uploads required documents</li> <li>• System generates application ID</li> <li>• Application submitted</li> </ul> |
| Alternate Flow:                        | N/A                                                                                                                                                                                                    |

| UC-03: Upload Documents |                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------|
| Primary Actor:          | Applicant                                                                         |
| Description:            | Upload supporting documents for registration.                                     |
| Preconditions:          | <ul style="list-style-type: none"> <li>• Application initiated</li> </ul>         |
| Postconditions:         | <ul style="list-style-type: none"> <li>• Documents stored successfully</li> </ul> |
| Main Flow:              |                                                                                   |
| Alternate Flow:         | N/A                                                                               |

| UC-04: Generate Application ID |                                             |
|--------------------------------|---------------------------------------------|
| Primary Actor:                 | System                                      |
| Description:                   | Generate unique identifier for application. |
| Preconditions:                 |                                             |



|                        |                                                                                     |
|------------------------|-------------------------------------------------------------------------------------|
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>Unique ID assigned to application</li> </ul> |
| <b>Main Flow:</b>      |                                                                                     |
| <b>Alternate Flow:</b> | N/A                                                                                 |

| UC-05: Validate Application |                                                                                                                                                                   |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>       | Central Authority Officer                                                                                                                                         |
| <b>Description:</b>         | Validate submitted application details.                                                                                                                           |
| <b>Preconditions:</b>       | <ul style="list-style-type: none"> <li>Application submitted</li> </ul>                                                                                           |
| <b>Postconditions:</b>      | <ul style="list-style-type: none"> <li>Application validated or flagged for corrections</li> </ul>                                                                |
| <b>Main Flow:</b>           | <ul style="list-style-type: none"> <li>Officer reviews application</li> <li>Checks completeness and accuracy</li> <li>Approves or requests corrections</li> </ul> |
| <b>Alternate Flow:</b>      | N/A                                                                                                                                                               |

| UC-06: Request Corrections |                                                                                            |
|----------------------------|--------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>      | Central Authority Officer                                                                  |
| <b>Description:</b>        | Request applicant to correct application.                                                  |
| <b>Preconditions:</b>      |                                                                                            |
| <b>Postconditions:</b>     | <ul style="list-style-type: none"> <li>Applicant notified to update application</li> </ul> |
| <b>Main Flow:</b>          |                                                                                            |
| <b>Alternate Flow:</b>     | N/A                                                                                        |

| UC-07: Approve for Payment |                                                                         |
|----------------------------|-------------------------------------------------------------------------|
| <b>Primary Actor:</b>      | Central Authority Officer                                               |
| <b>Description:</b>        | Approve application for payment processing.                             |
| <b>Preconditions:</b>      | <ul style="list-style-type: none"> <li>Application validated</li> </ul> |
| <b>Postconditions:</b>     | <ul style="list-style-type: none"> <li>Payment enabled</li> </ul>       |
| <b>Main Flow:</b>          |                                                                         |
| <b>Alternate Flow:</b>     | N/A                                                                     |

| UC-08: Make Payment    |                                                                                    |
|------------------------|------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>  | Applicant                                                                          |
| <b>Description:</b>    | Pay registration fees.                                                             |
| <b>Preconditions:</b>  | <ul style="list-style-type: none"> <li>Application approved for payment</li> </ul> |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>Payment initiated</li> </ul>                |
| <b>Main Flow:</b>      |                                                                                    |
| <b>Alternate Flow:</b> | N/A                                                                                |

| UC-09: Process Payment |                                                                                 |
|------------------------|---------------------------------------------------------------------------------|
| Primary Actor:         | Payment Gateway (External)                                                      |
| Description:           | Process payment transaction.                                                    |
| Preconditions:         |                                                                                 |
| Postconditions:        | <ul style="list-style-type: none"> <li>Payment confirmed or rejected</li> </ul> |
| Main Flow:             |                                                                                 |
| Alternate Flow:        | N/A                                                                             |

| UC-10: Conduct Field Inspection |                                                                        |
|---------------------------------|------------------------------------------------------------------------|
| Primary Actor:                  | District Inspector                                                     |
| Description:                    | Conduct on-site inspection of facility.                                |
| Preconditions:                  | <ul style="list-style-type: none"> <li>Payment completed</li> </ul>    |
| Postconditions:                 | <ul style="list-style-type: none"> <li>Inspection conducted</li> </ul> |
| Main Flow:                      |                                                                        |
| Alternate Flow:                 | N/A                                                                    |

| UC-11: Collect Sample |                                                                          |
|-----------------------|--------------------------------------------------------------------------|
| Primary Actor:        | District Inspector                                                       |
| Description:          | Collect salt samples during inspection.                                  |
| Preconditions:        | <ul style="list-style-type: none"> <li>Inspection in progress</li> </ul> |
| Postconditions:       | <ul style="list-style-type: none"> <li>Samples collected</li> </ul>      |
| Main Flow:            |                                                                          |
| Alternate Flow:       | N/A                                                                      |

| UC-12: Submit Inspection Report |                                                                               |
|---------------------------------|-------------------------------------------------------------------------------|
| Primary Actor:                  | District Inspector                                                            |
| Description:                    | Submit findings from field inspection.                                        |
| Preconditions:                  | <ul style="list-style-type: none"> <li>Inspection completed</li> </ul>        |
| Postconditions:                 | <ul style="list-style-type: none"> <li>Report submitted for review</li> </ul> |
| Main Flow:                      |                                                                               |
| Alternate Flow:                 | N/A                                                                           |

| UC-13: Laboratory Processing |                    |
|------------------------------|--------------------|
| Primary Actor:               | Laboratory Testing |
| Description:                 |                    |
| Preconditions:               | Sample received    |

|                        |                                                                                                                                                                                                                                                                                                                                          |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Postconditions:</b> | Lab report submitted                                                                                                                                                                                                                                                                                                                     |
| <b>Main Flow:</b>      | <b>UC-12A: Receive Sample</b> <ul style="list-style-type: none"> <li>Sample received and logged</li> </ul> <b>UC-12B: Conduct Lab Testing</b> <ul style="list-style-type: none"> <li>Perform chemical/quality analysis</li> </ul> <b>UC-12C: Upload Lab Report</b> <ul style="list-style-type: none"> <li>Upload test results</li> </ul> |
| <b>Alternate Flow:</b> | N/A                                                                                                                                                                                                                                                                                                                                      |

| <b>UC-14: Review Application (Assistant Director)</b> |                                                                               |
|-------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Primary Actor:</b>                                 | Assistant Director                                                            |
| <b>Description:</b>                                   | Review application and inspection report.                                     |
| <b>Preconditions:</b>                                 | <ul style="list-style-type: none"> <li>Inspection report available</li> </ul> |
| <b>Postconditions:</b>                                | <ul style="list-style-type: none"> <li>Recommendation provided</li> </ul>     |
| <b>Main Flow:</b>                                     |                                                                               |
| <b>Alternate Flow:</b>                                | N/A                                                                           |

| <b>UC-15: Review Application (Director)</b> |                                                                                       |
|---------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>                       | Director                                                                              |
| <b>Description:</b>                         | Review application and make decision.                                                 |
| <b>Preconditions:</b>                       | <ul style="list-style-type: none"> <li>Assistant Director review completed</li> </ul> |
| <b>Postconditions:</b>                      | <ul style="list-style-type: none"> <li>Application approved or rejected</li> </ul>    |
| <b>Main Flow:</b>                           |                                                                                       |
| <b>Alternate Flow:</b>                      | N/A                                                                                   |

| <b>UC-16: Final Approval / Rejection</b> |                                                                               |
|------------------------------------------|-------------------------------------------------------------------------------|
| <b>Primary Actor:</b>                    | Director General                                                              |
| <b>Description:</b>                      | Provide final approval or rejection.                                          |
| <b>Preconditions:</b>                    | <ul style="list-style-type: none"> <li>Director decision available</li> </ul> |
| <b>Postconditions:</b>                   | <ul style="list-style-type: none"> <li>Final decision recorded</li> </ul>     |
| <b>Main Flow:</b>                        |                                                                               |
| <b>Alternate Flow:</b>                   | N/A                                                                           |

| <b>UC-17: Generate Certificate</b> |                                                                        |
|------------------------------------|------------------------------------------------------------------------|
| <b>Primary Actor:</b>              | System                                                                 |
| <b>Description:</b>                | Generate registration certificate.                                     |
| <b>Preconditions:</b>              | <ul style="list-style-type: none"> <li>Application approved</li> </ul> |

|                        |                                   |
|------------------------|-----------------------------------|
| <b>Postconditions:</b> | ● Certificate issued to applicant |
| <b>Main Flow:</b>      |                                   |
| <b>Alternate Flow:</b> | N/A                               |

## 2.8 Bottled Water Manufacturing and Registration Management Module

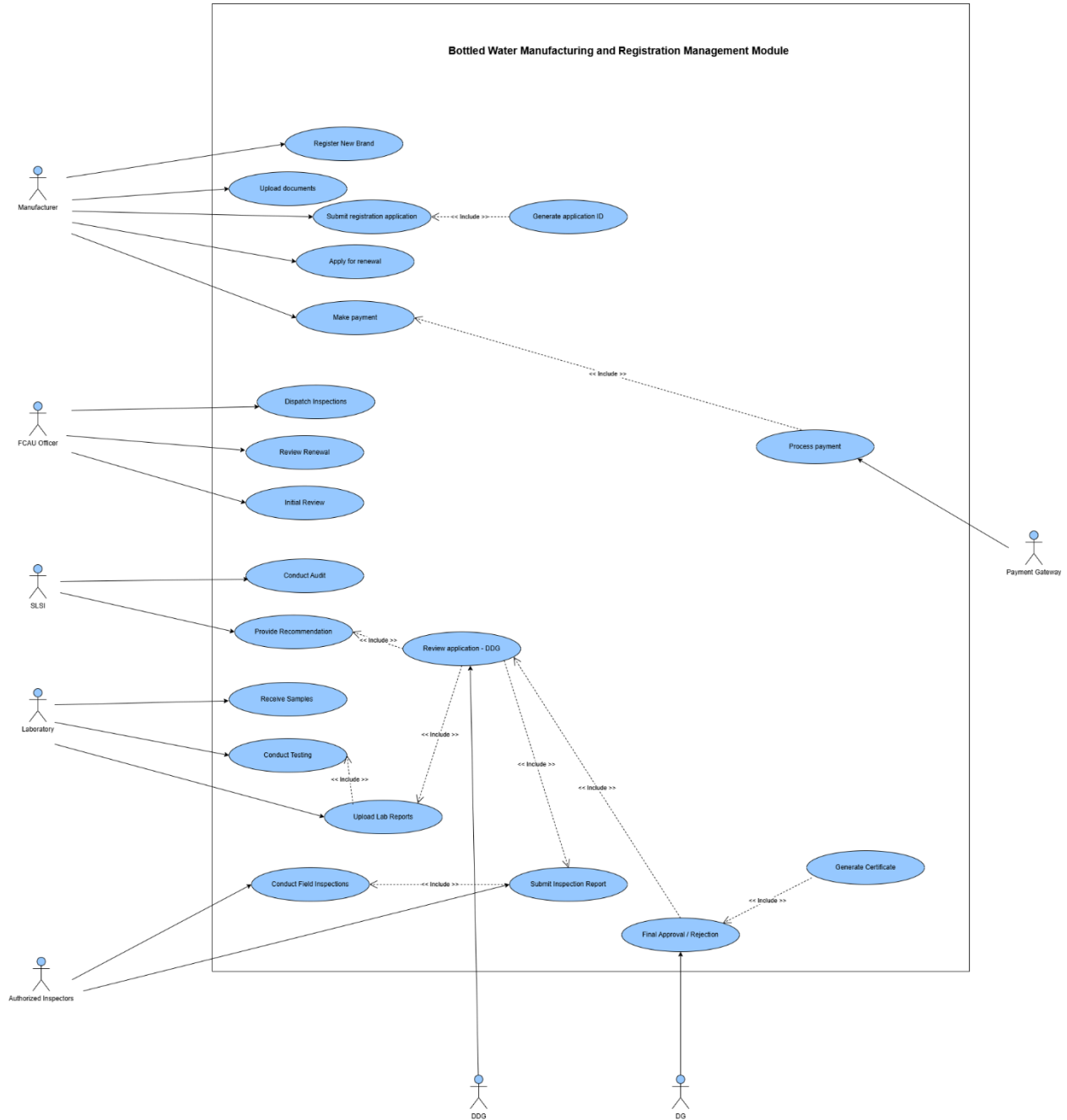


Figure 2.7

## Use Case Narratives

| UC-01: Register New Brand |                                                                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:            | Manufacturer                                                                                                                                                       |
| Description:              | Register a new bottled water brand in the system.                                                                                                                  |
| Preconditions:            | <ul style="list-style-type: none"> <li>Manufacturer is registered in the system</li> </ul>                                                                         |
| Postconditions:           | <ul style="list-style-type: none"> <li>Brand registration initiated</li> </ul>                                                                                     |
| Main Flow:                | <ul style="list-style-type: none"> <li>Manufacturer selects “Register New Brand”</li> <li>Enters brand details</li> <li>System stores brand information</li> </ul> |
| Alternate Flow:           | N/A                                                                                                                                                                |

| UC-02: Upload Documents |                                                                                 |
|-------------------------|---------------------------------------------------------------------------------|
| Primary Actor:          | Manufacturer                                                                    |
| Description:            | Upload required documents for registration or renewal.                          |
| Preconditions:          | <ul style="list-style-type: none"> <li>Application initiated</li> </ul>         |
| Postconditions:         | <ul style="list-style-type: none"> <li>Documents uploaded and stored</li> </ul> |
| Main Flow:              |                                                                                 |
| Alternate Flow:         | N/A                                                                             |

| UC-03: Submit Registration Application |                                                                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Actor:                         | Manufacturer                                                                                                                                                             |
| Description:                           | Submit application for bottled water registration.                                                                                                                       |
| Preconditions:                         | <ul style="list-style-type: none"> <li>Documents uploaded</li> </ul>                                                                                                     |
| Postconditions:                        | <ul style="list-style-type: none"> <li>Application submitted with ID</li> </ul>                                                                                          |
| Main Flow:                             | <ul style="list-style-type: none"> <li>Manufacturer fills application form</li> <li>System generates application ID</li> <li>Application submitted for review</li> </ul> |
| Alternate Flow:                        | N/A                                                                                                                                                                      |

| UC-04: Generate Application ID |                                                                          |
|--------------------------------|--------------------------------------------------------------------------|
| Primary Actor:                 | System                                                                   |
| Description:                   | Generate unique identifier for application.                              |
| Preconditions:                 |                                                                          |
| Postconditions:                | <ul style="list-style-type: none"> <li>Application ID created</li> </ul> |

|                        |     |
|------------------------|-----|
| <b>Main Flow:</b>      |     |
| <b>Alternate Flow:</b> | N/A |

| UC-05: Apply for Renewal |                                                                                 |
|--------------------------|---------------------------------------------------------------------------------|
| <b>Primary Actor:</b>    | Manufacturer                                                                    |
| <b>Description:</b>      | Apply to renew existing registration.                                           |
| <b>Preconditions:</b>    | <ul style="list-style-type: none"> <li>Existing valid registration</li> </ul>   |
| <b>Postconditions:</b>   | <ul style="list-style-type: none"> <li>Renewal application submitted</li> </ul> |
| <b>Main Flow:</b>        |                                                                                 |
| <b>Alternate Flow:</b>   | N/A                                                                             |

| UC-06: Make Payment    |                                                                         |
|------------------------|-------------------------------------------------------------------------|
| <b>Primary Actor:</b>  | Manufacturer                                                            |
| <b>Description:</b>    | Pay registration or renewal fees.                                       |
| <b>Preconditions:</b>  | <ul style="list-style-type: none"> <li>Application submitted</li> </ul> |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>Payment initiated</li> </ul>     |
| <b>Main Flow:</b>      |                                                                         |
| <b>Alternate Flow:</b> | N/A                                                                     |

| UC-07: Process Payment |                                                                               |
|------------------------|-------------------------------------------------------------------------------|
| <b>Primary Actor:</b>  | Payment Gateway (External)                                                    |
| <b>Description:</b>    | Process payment transaction.                                                  |
| <b>Preconditions:</b>  |                                                                               |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>Payment confirmed or failed</li> </ul> |
| <b>Main Flow:</b>      |                                                                               |
| <b>Alternate Flow:</b> | N/A                                                                           |

| UC-08: Initial Review  |                                                                                                                   |
|------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>  | FCAU Officer                                                                                                      |
| <b>Description:</b>    | Conduct preliminary review of application.                                                                        |
| <b>Preconditions:</b>  | <ul style="list-style-type: none"> <li>Application submitted</li> </ul>                                           |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>Application moves to inspection stage or requires clarification</li> </ul> |
| <b>Main Flow:</b>      |                                                                                                                   |
| <b>Alternate Flow:</b> | N/A                                                                                                               |

| UC-09: Review Renewal |  |
|-----------------------|--|
|-----------------------|--|

|                        |                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>  | FCAU Officer                                                                              |
| <b>Description:</b>    | Review renewal applications.                                                              |
| <b>Preconditions:</b>  | <ul style="list-style-type: none"> <li>Renewal request submitted</li> </ul>               |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>Renewal proceeds for further processing</li> </ul> |
| <b>Main Flow:</b>      |                                                                                           |
| <b>Alternate Flow:</b> | N/A                                                                                       |

#### UC-10: Dispatch Inspections

|                        |                                                                            |
|------------------------|----------------------------------------------------------------------------|
| <b>Primary Actor:</b>  | FCAU Officer                                                               |
| <b>Description:</b>    | Assign inspections to authorized inspectors.                               |
| <b>Preconditions:</b>  | <ul style="list-style-type: none"> <li>Application under review</li> </ul> |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>Inspection scheduled</li> </ul>     |
| <b>Main Flow:</b>      |                                                                            |
| <b>Alternate Flow:</b> | N/A                                                                        |

#### UC-11: Conduct Field Inspections

|                        |                                                                                |
|------------------------|--------------------------------------------------------------------------------|
| <b>Primary Actor:</b>  | Authorized Inspectors                                                          |
| <b>Description:</b>    | Perform on-site inspection of manufacturing facility.                          |
| <b>Preconditions:</b>  | <ul style="list-style-type: none"> <li>Inspection assigned</li> </ul>          |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>Inspection findings recorded</li> </ul> |
| <b>Main Flow:</b>      |                                                                                |
| <b>Alternate Flow:</b> | N/A                                                                            |

#### UC-12: Submit Inspection Report

|                        |                                                                              |
|------------------------|------------------------------------------------------------------------------|
| <b>Primary Actor:</b>  | Authorized Inspectors                                                        |
| <b>Description:</b>    | Submit inspection findings.                                                  |
| <b>Preconditions:</b>  | <ul style="list-style-type: none"> <li>Inspection completed</li> </ul>       |
| <b>Postconditions:</b> | <ul style="list-style-type: none"> <li>Report submitted to system</li> </ul> |
| <b>Main Flow:</b>      |                                                                              |
| <b>Alternate Flow:</b> | N/A                                                                          |

#### UC-13: Laboratory Processing

|                        |                                |
|------------------------|--------------------------------|
| <b>Primary Actor:</b>  | Laboratory                     |
| <b>Description:</b>    | Lab processing                 |
| <b>Preconditions:</b>  | Receive sample                 |
| <b>Postconditions:</b> | Submit Report                  |
| <b>Main Flow:</b>      | <b>UC-13A: Receive Samples</b> |



|                 |                                                                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <ul style="list-style-type: none"> <li>Samples received</li> </ul> <b>UC-13B: Conduct Testing</b> <ul style="list-style-type: none"> <li>Perform quality testing</li> </ul> <b>UC-13C: Upload Lab Reports</b> <ul style="list-style-type: none"> <li>Upload test results</li> </ul> |
| Alternate Flow: | N/A                                                                                                                                                                                                                                                                                 |

| UC-14: Conduct Audit |                                                                            |
|----------------------|----------------------------------------------------------------------------|
| Primary Actor:       | SLSI                                                                       |
| Description:         | Conduct audit on manufacturing processes.                                  |
| Preconditions:       | <ul style="list-style-type: none"> <li>Application under review</li> </ul> |
| Postconditions:      | <ul style="list-style-type: none"> <li>Audit completed</li> </ul>          |
| Main Flow:           |                                                                            |
| Alternate Flow:      | N/A                                                                        |

| UC-15: Provide Recommendation |                                                                            |
|-------------------------------|----------------------------------------------------------------------------|
| Primary Actor:                | SLSI                                                                       |
| Description:                  | Provide recommendation based on audit.                                     |
| Preconditions:                | <ul style="list-style-type: none"> <li>Audit completed</li> </ul>          |
| Postconditions:               | <ul style="list-style-type: none"> <li>Recommendation submitted</li> </ul> |
| Main Flow:                    |                                                                            |
| Alternate Flow:               | N/A                                                                        |

| UC-16: Review Application (DDG) |                                                                                             |
|---------------------------------|---------------------------------------------------------------------------------------------|
| Primary Actor:                  | DDG                                                                                         |
| Description:                    | Review application, inspection, and lab reports.                                            |
| Preconditions:                  | <ul style="list-style-type: none"> <li>All reports and recommendations available</li> </ul> |
| Postconditions:                 | <ul style="list-style-type: none"> <li>Application forwarded for final decision</li> </ul>  |
| Main Flow:                      |                                                                                             |
| Alternate Flow:                 | N/A                                                                                         |

| UC-17: Final Approval / Rejection |                                                                                    |
|-----------------------------------|------------------------------------------------------------------------------------|
| Primary Actor:                    | DG                                                                                 |
| Description:                      | Make final decision on application.                                                |
| Preconditions:                    | <ul style="list-style-type: none"> <li>DDG review completed</li> </ul>             |
| Postconditions:                   | <ul style="list-style-type: none"> <li>Application approved or rejected</li> </ul> |
| Main Flow:                        |                                                                                    |

|                        |     |
|------------------------|-----|
| <b>Alternate Flow:</b> | N/A |
|------------------------|-----|

| <b>UC-18: Generate Certificate</b> |                                                                        |
|------------------------------------|------------------------------------------------------------------------|
| <b>Primary Actor:</b>              | System                                                                 |
| <b>Description:</b>                | Generate registration certificate.                                     |
| <b>Preconditions:</b>              | <ul style="list-style-type: none"><li>• Application approved</li></ul> |
| <b>Postconditions:</b>             | <ul style="list-style-type: none"><li>• Certificate issued</li></ul>   |
| <b>Main Flow:</b>                  |                                                                        |
| <b>Alternate Flow:</b>             | N/A                                                                    |

### 3. Sequence Modeling

Sequence modeling is a dynamic design technique used to represent how system components interact over time to execute a specific functionality. Within the context of this Software Design Document (SDD), sequence modeling provides a time-ordered view of interactions between actors, system components, and microservices, illustrating how individual use cases are realized step by step.

Unlike static models, sequence models focus on behavioral flow, showing how messages are exchanged between different entities in a chronological sequence. These entities may include users (actors), application services, APIs, databases, and external systems. By capturing these interactions, sequence modeling helps in understanding not just *what* the system does, but *how* it performs operations internally.

Sequence diagrams, which are the primary artifact of sequence modeling, typically include:

- **Actors and Lifelines** – Represent users and system components involved in the interaction
- **Messages** – Show communication between components (synchronous or asynchronous)
- **Activation Bars** – Indicate processing time within a component
- **Control Structures** – Such as loops, conditions, and parallel processing

In a microservices-based architecture, sequence modeling plays a critical role in visualizing service orchestration and communication patterns. It helps define:

- API interactions between services
- Event-driven communication (e.g., message queues, publish–subscribe patterns)
- Data flow across distributed services
- Error handling and fallback mechanisms

Sequence models are especially valuable for complex workflows such as application submission, inspection processes, lab coordination, enforcement actions, and import/export handling. They ensure clarity in system behavior, reduce ambiguity in implementation, and support developers in designing efficient and scalable interactions.

Overall, sequence modeling acts as a bridge between use case narratives and technical implementation, enabling a precise and shared understanding of runtime system behavior among architects, developers, and stakeholders.

### 3.1 Premises Registration and Lifecycle Management Module

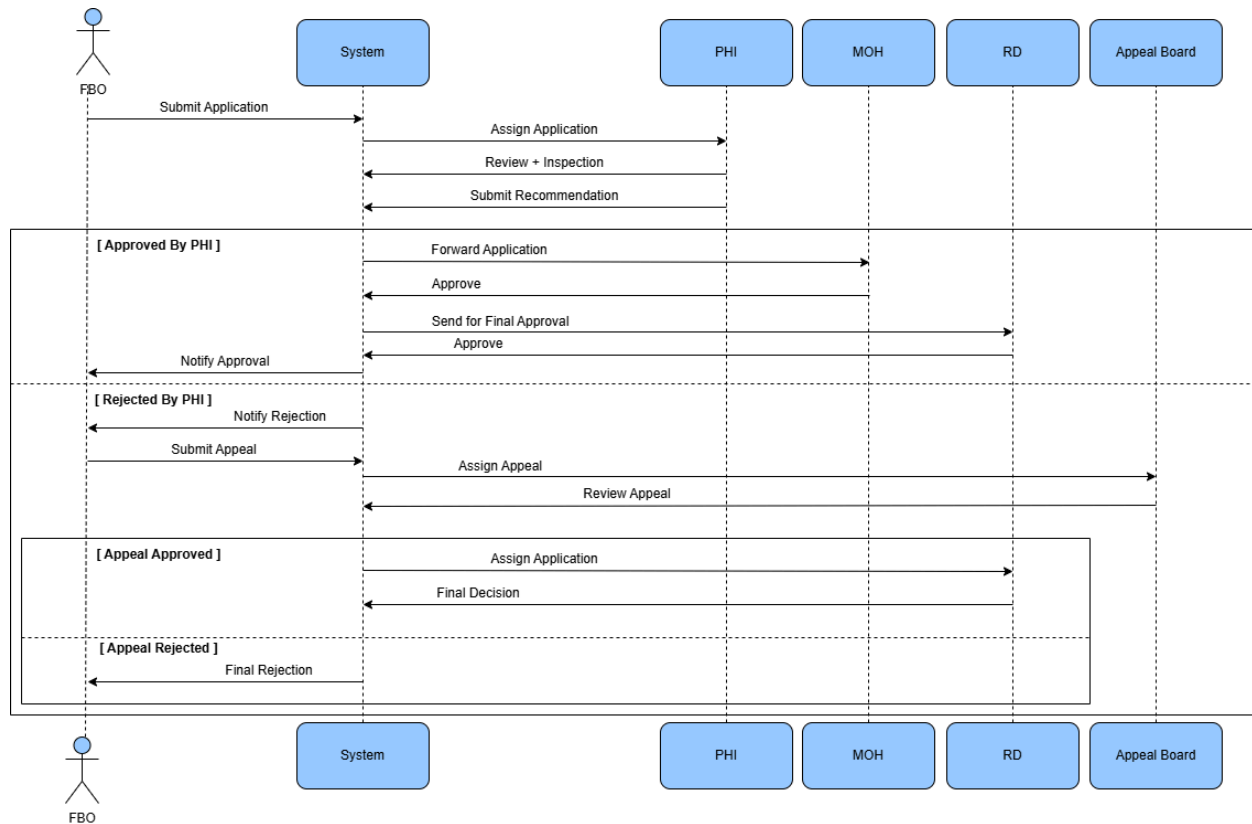


Figure 3.0

## 3.2 Inspection and Grading Module

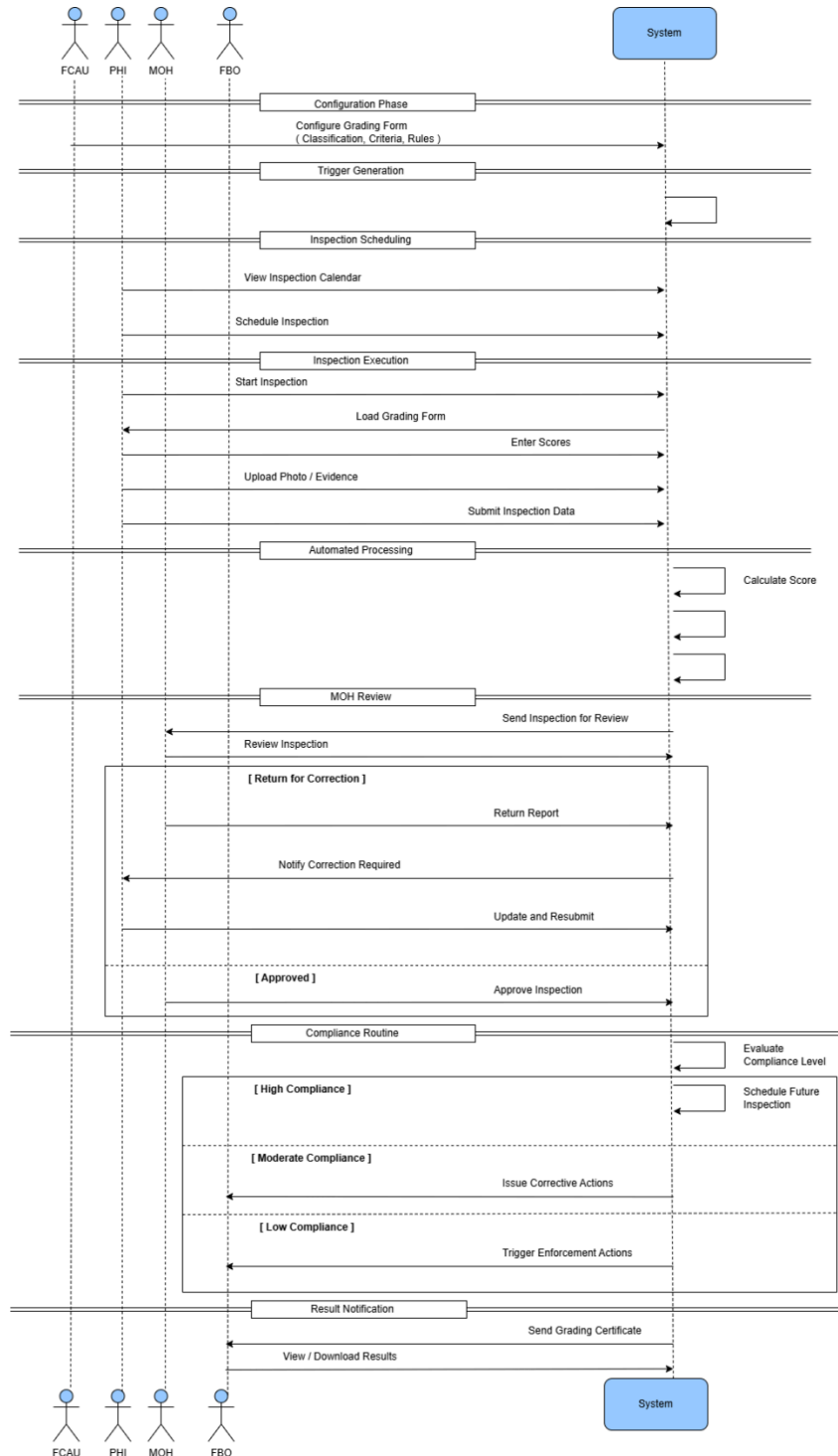


Figure 3.1

### 3.3 Laboratory Coordination Module

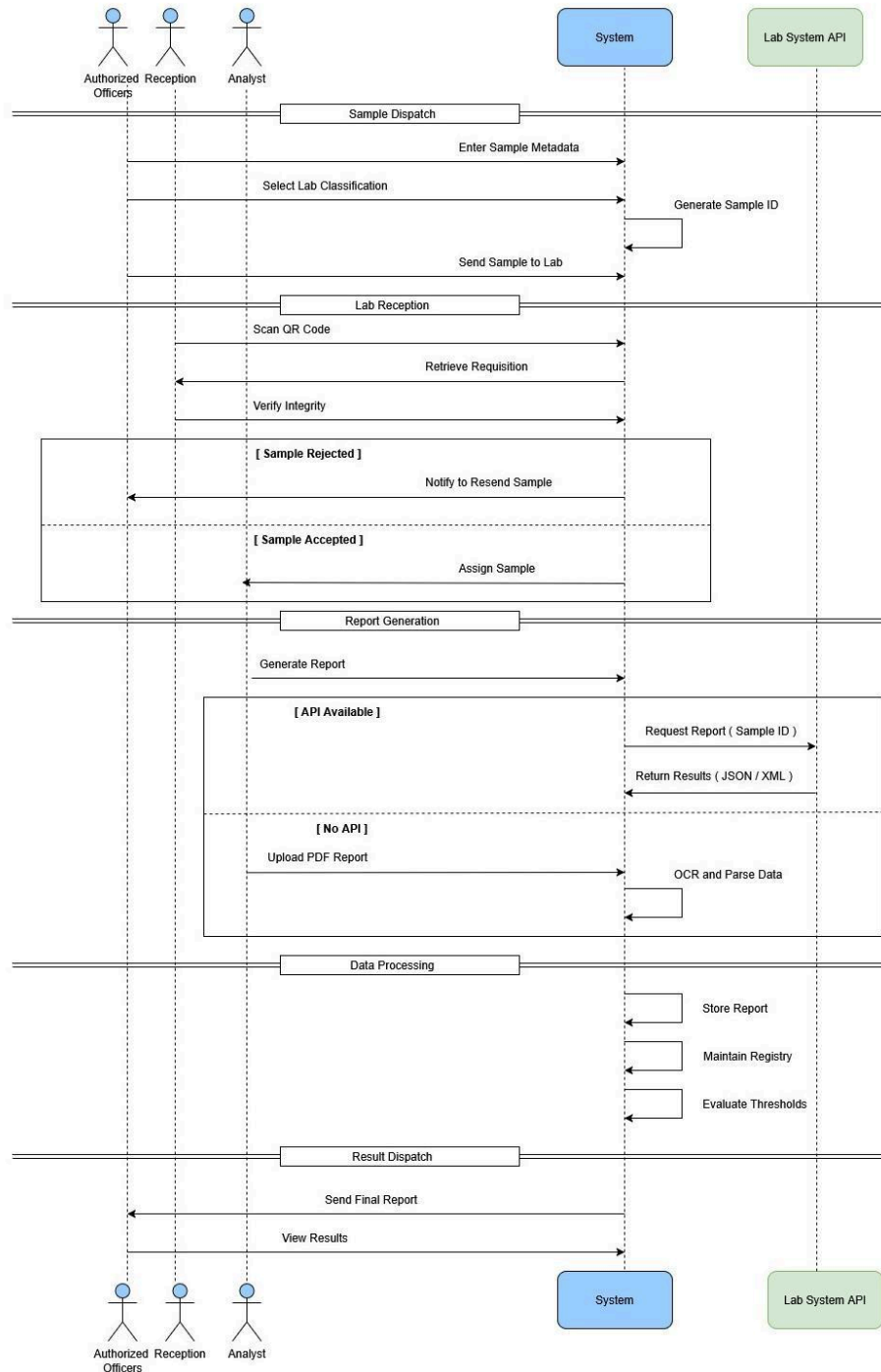


Figure 3.2

### 3.4 Legal, Prosecution and Enforcement Management Module

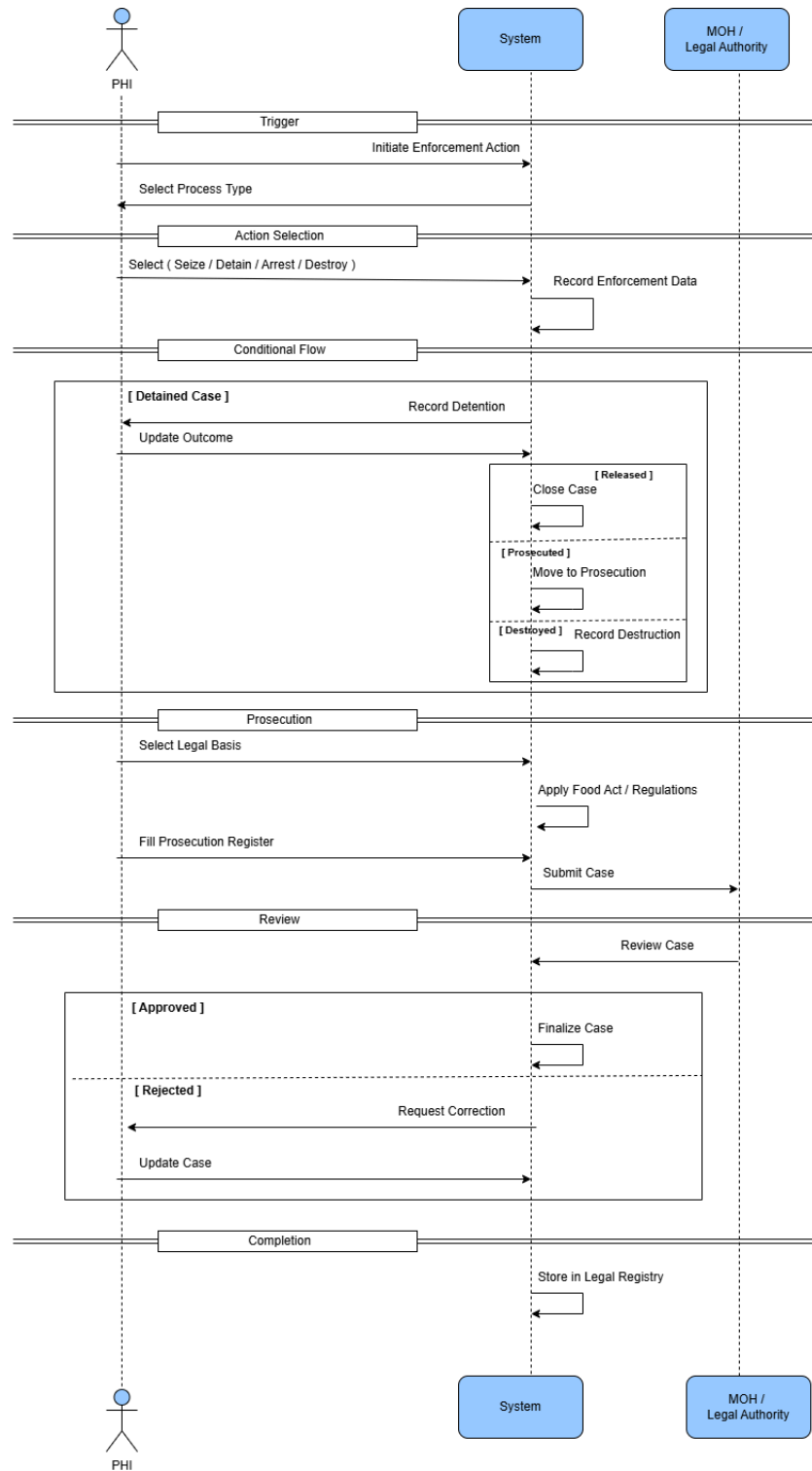


Figure 3.3

## 3.5 Import Control Module

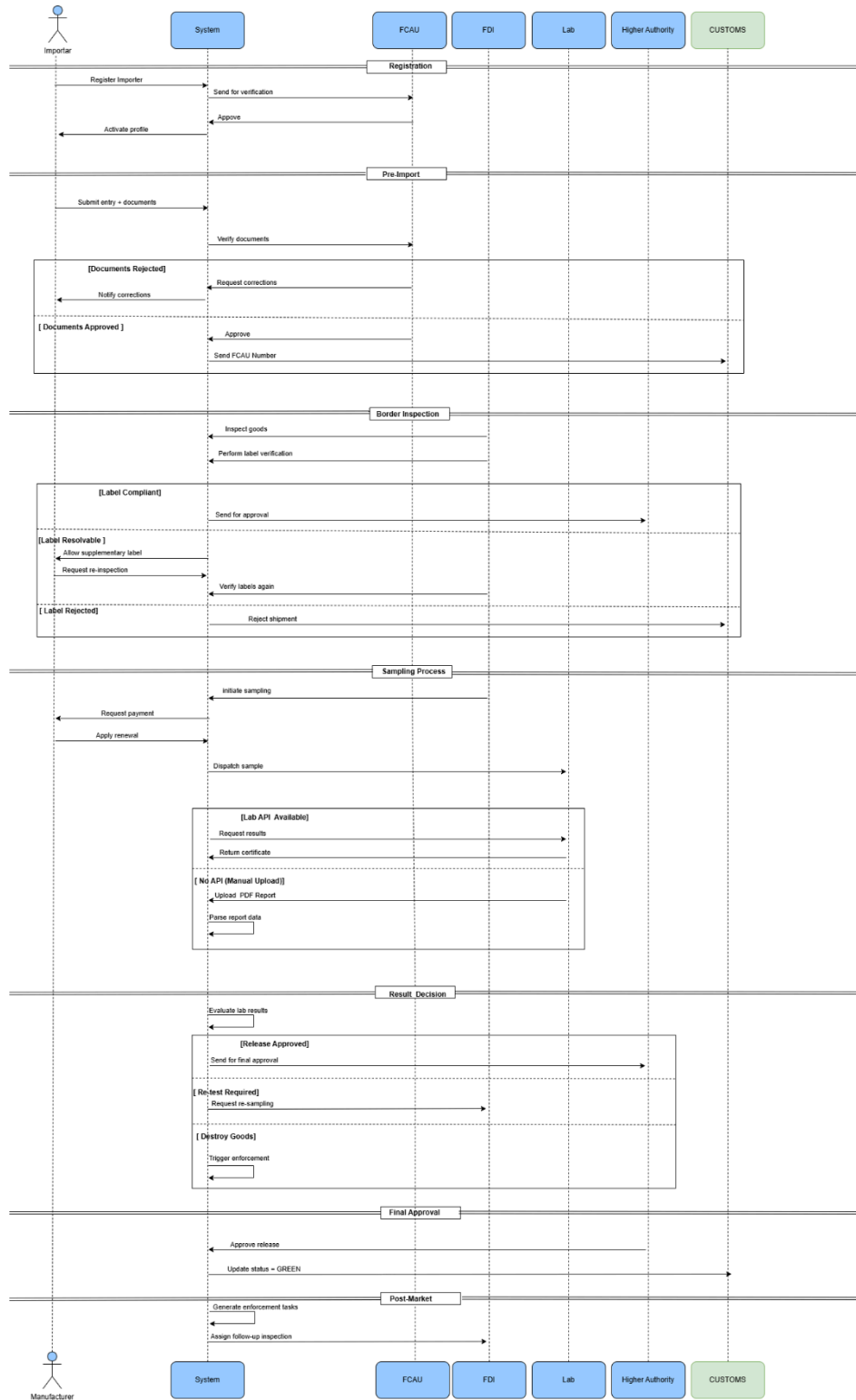


Figure 3.4



### 3.6 Export Control Module

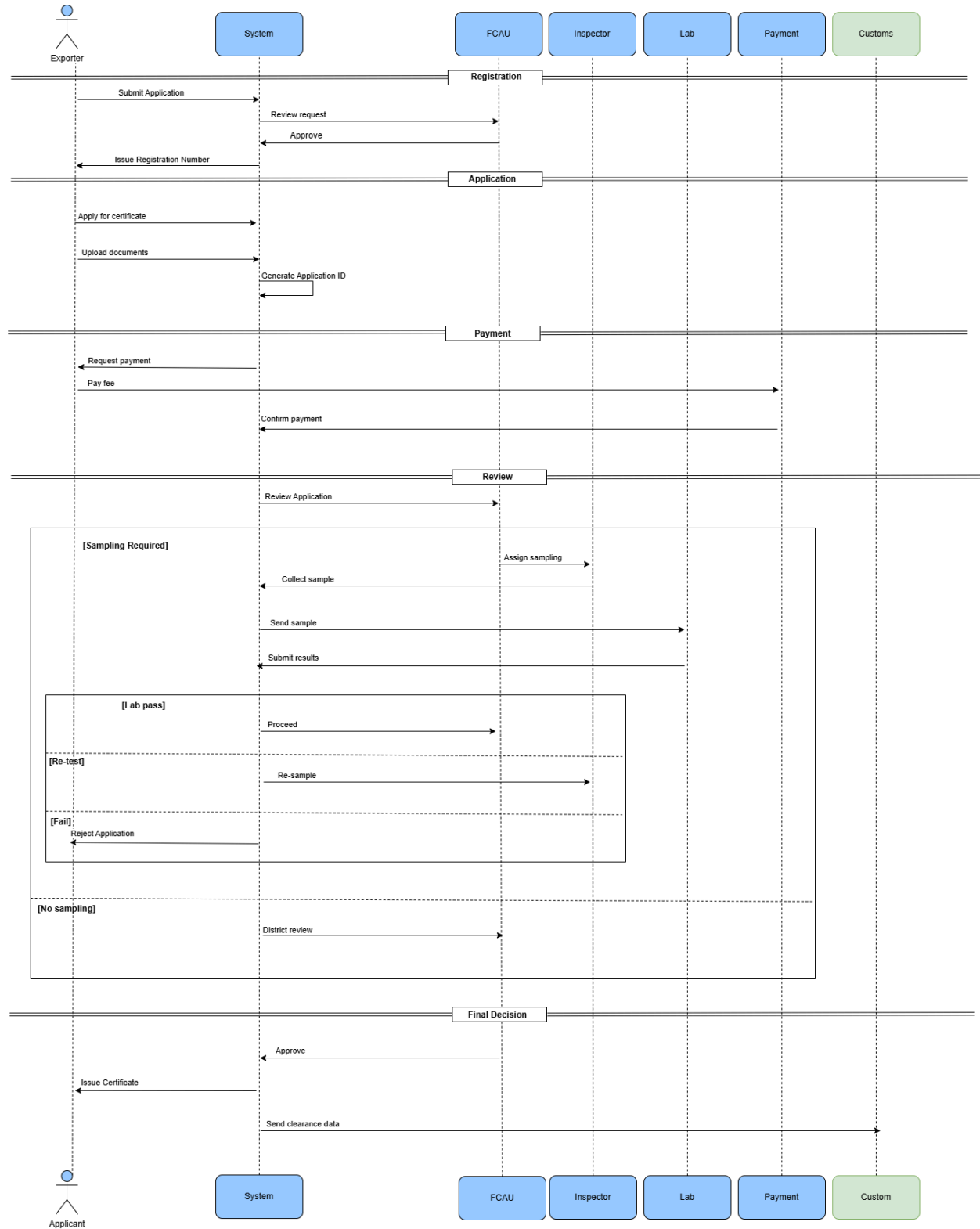


Figure 3.5

## 3.7 Registration of Common Salt Manufacturing and Transports Management Module

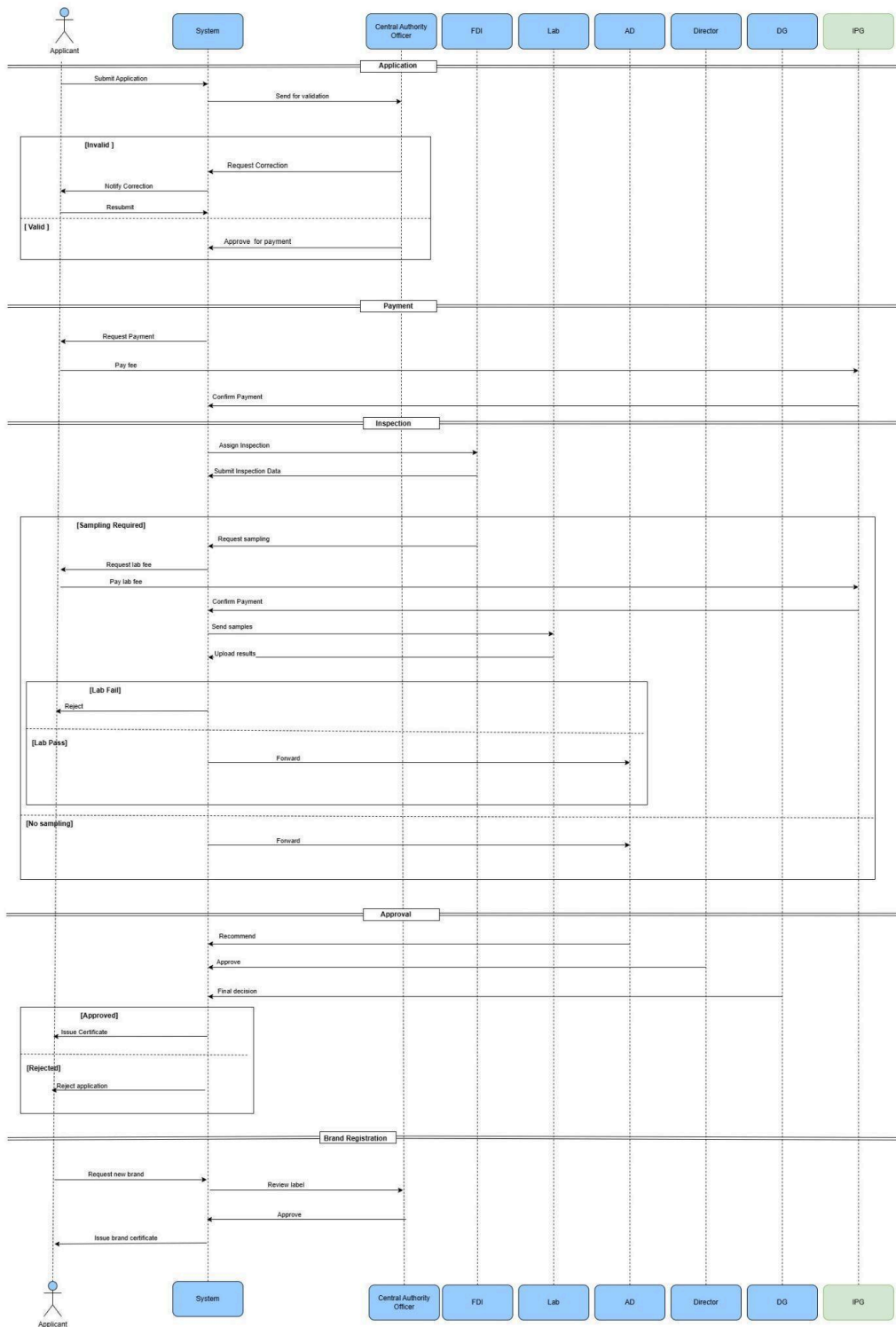


Figure 3.6

### 3.8 Bottled Water Manufacturing and Registration Management Module

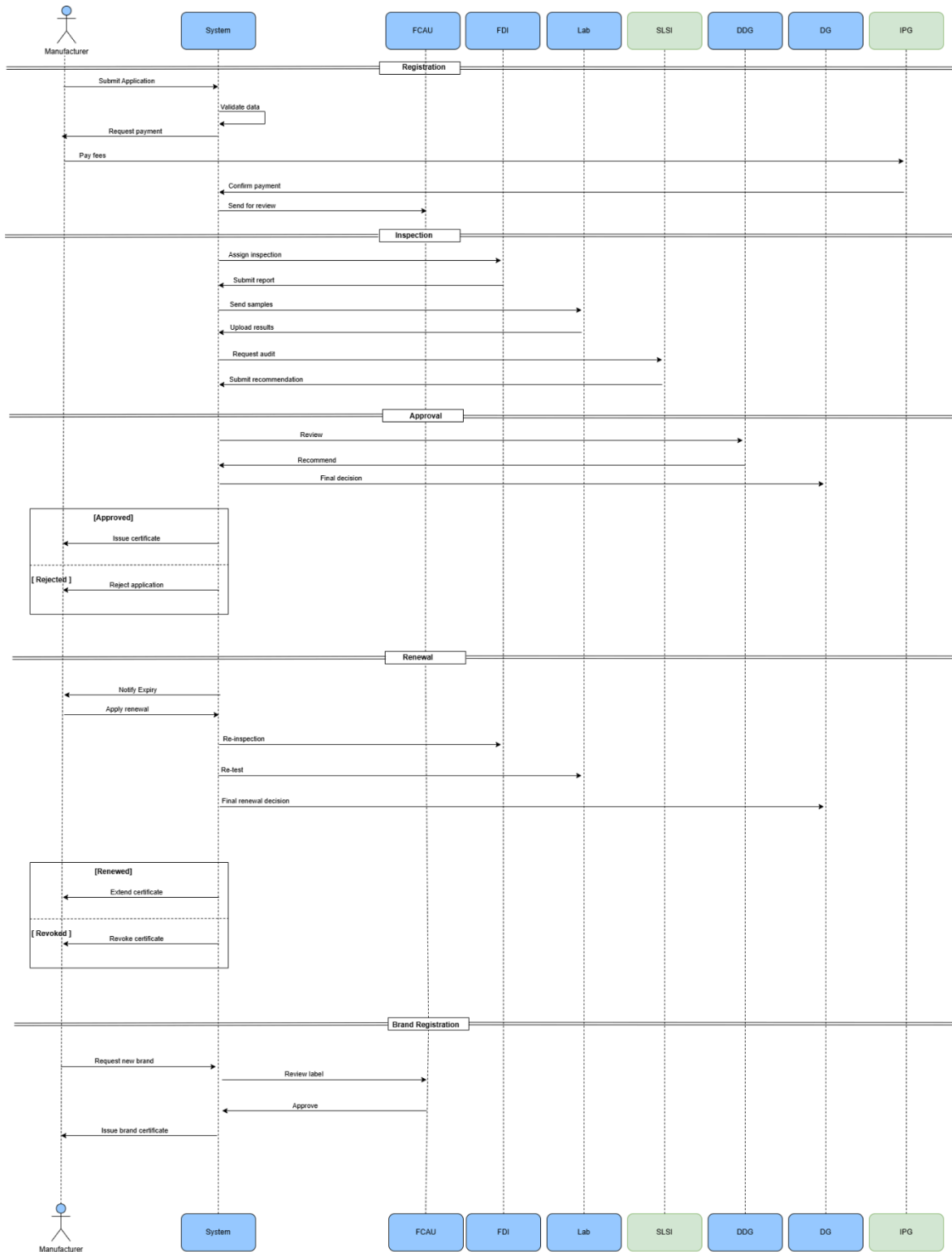


Figure 3.7

## 4. Activity Modeling

Activity modeling is a behavioral modeling technique used to represent the flow of control and activities within a system, capturing how processes are executed from start to end. In this Software Design Document (SDD), activity modeling provides a clear visualization of business processes, system workflows, and decision logic across different modules.

Unlike sequence modeling, which focuses on interactions over time, activity modeling emphasizes the overall process flow, including parallel activities, decision points, and transitions between different states. It is particularly useful for illustrating complex workflows that involve multiple actors, conditional paths, and iterative steps.

Activity diagrams, the primary artifact of activity modeling typically include:

- **Initial and Final Nodes** – Indicating the start and end of a process
- **Activities/Actions** – Representing individual steps or tasks
- **Control Flows** – Showing the sequence of execution
- **Decision Nodes** – Capturing branching logic based on conditions
- **Merge Nodes** – Combining alternative flows
- **Fork and Join Nodes** – Representing parallel processing

In the context of a microservices based architecture, activity modeling helps to:

- Visualize end-to-end business workflows across multiple services
- Identify service boundaries and responsibilities
- Highlight parallel processing opportunities for performance optimization
- Define decision logic and exception handling paths
- Support orchestration and choreography of distributed services

Activity models are especially valuable for processes such as registration workflows, inspection and grading procedures, lab coordination, enforcement handling, and import/export operations.

Overall, activity modeling serves as a bridge between business process understanding and system design, ensuring that workflows are logically structured, efficient, and aligned with system requirements. It enhances communication among stakeholders and provides a solid foundation for developing robust and scalable system implementations.

## 4.1 Premises Registration and Lifecycle Management Module

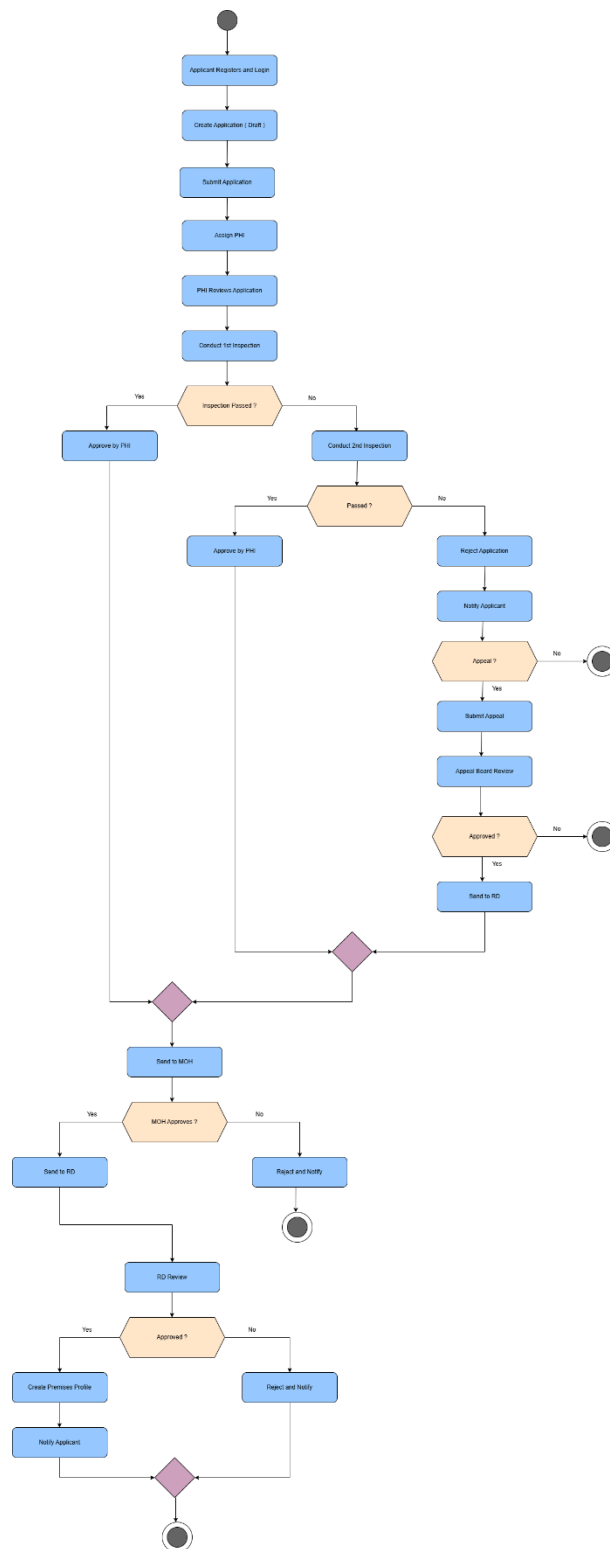


Figure 4.0

## 4.2 Inspection and Grading Module

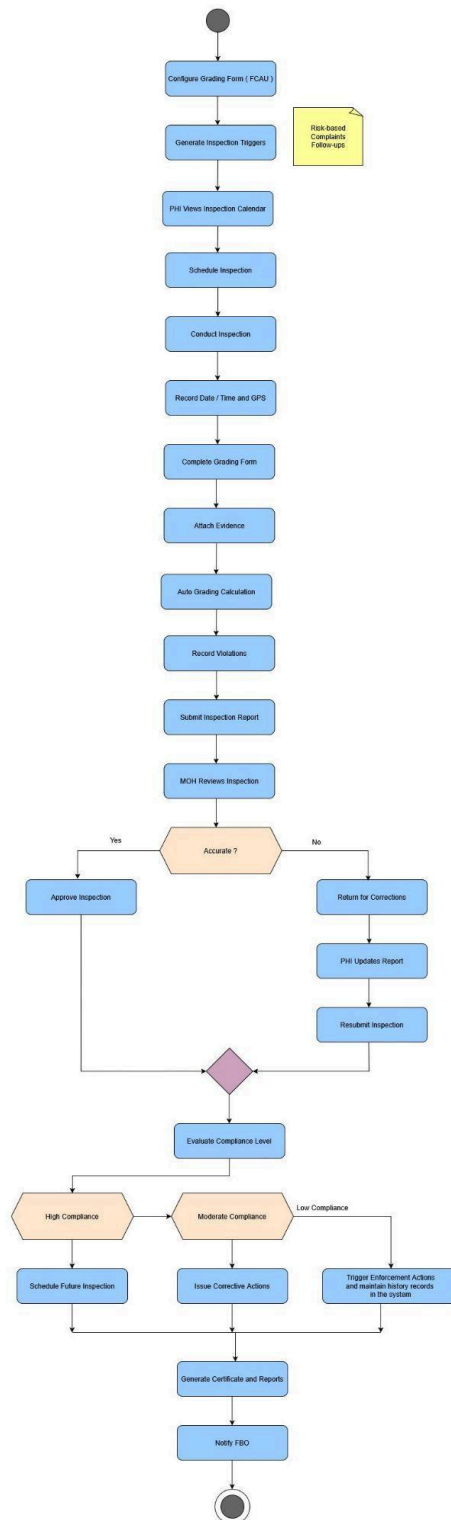


Figure 4.1

## 4.3 Laboratory Coordination Module

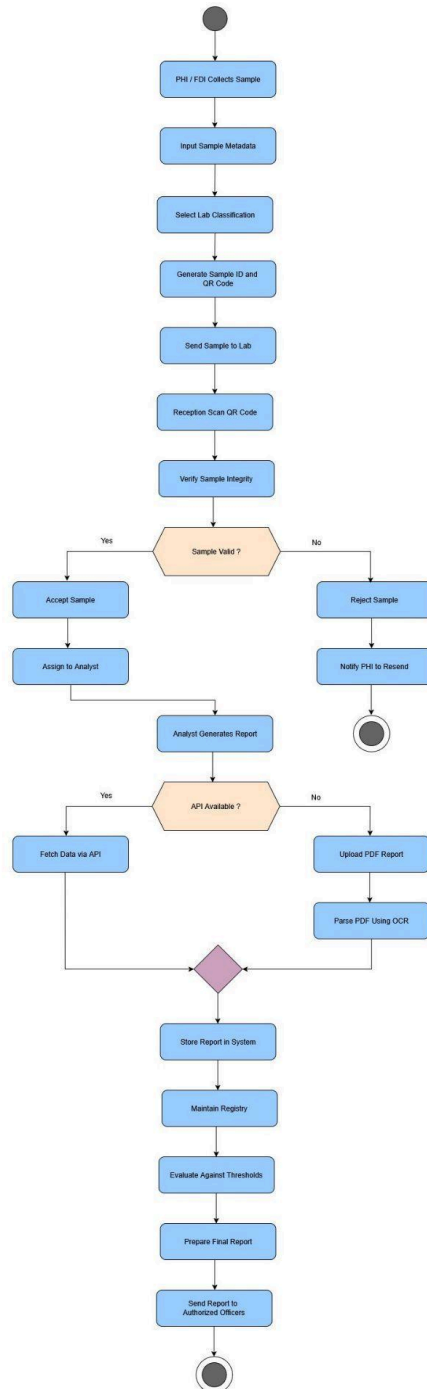


Figure 4.2

## 4.4 Legal, Prosecution and Enforcement Management Module

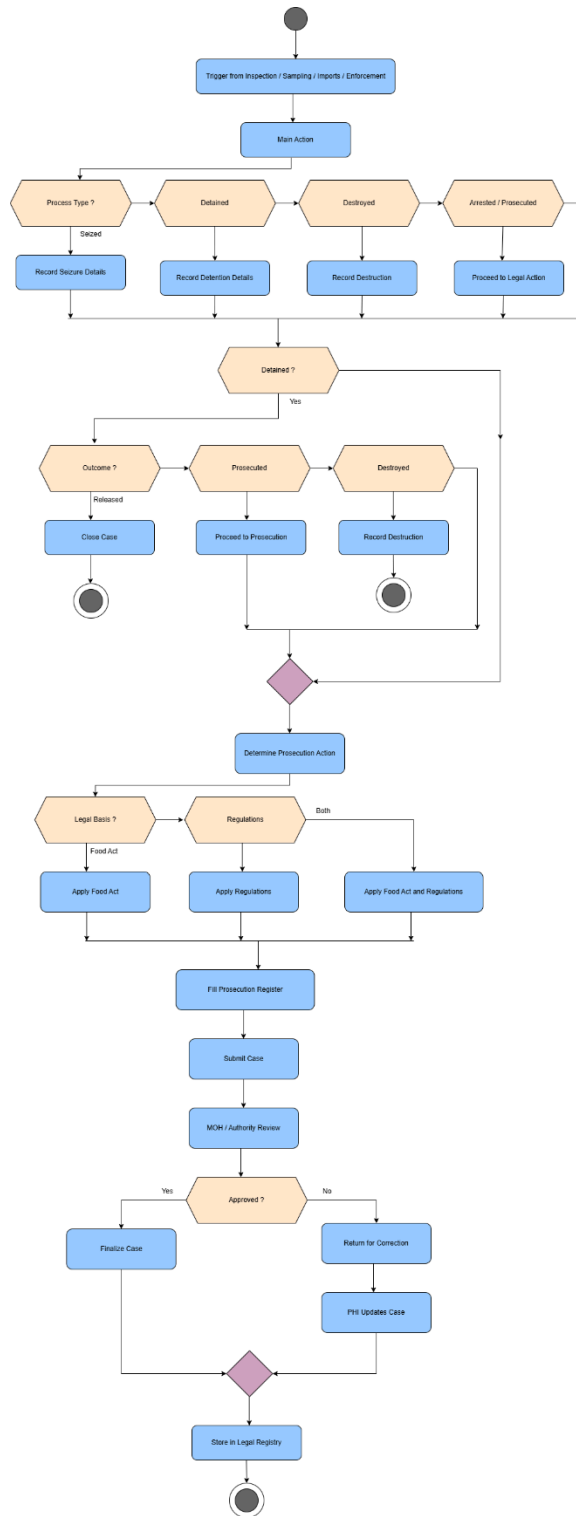


Figure 4.3



## 4.5 Import Control Module

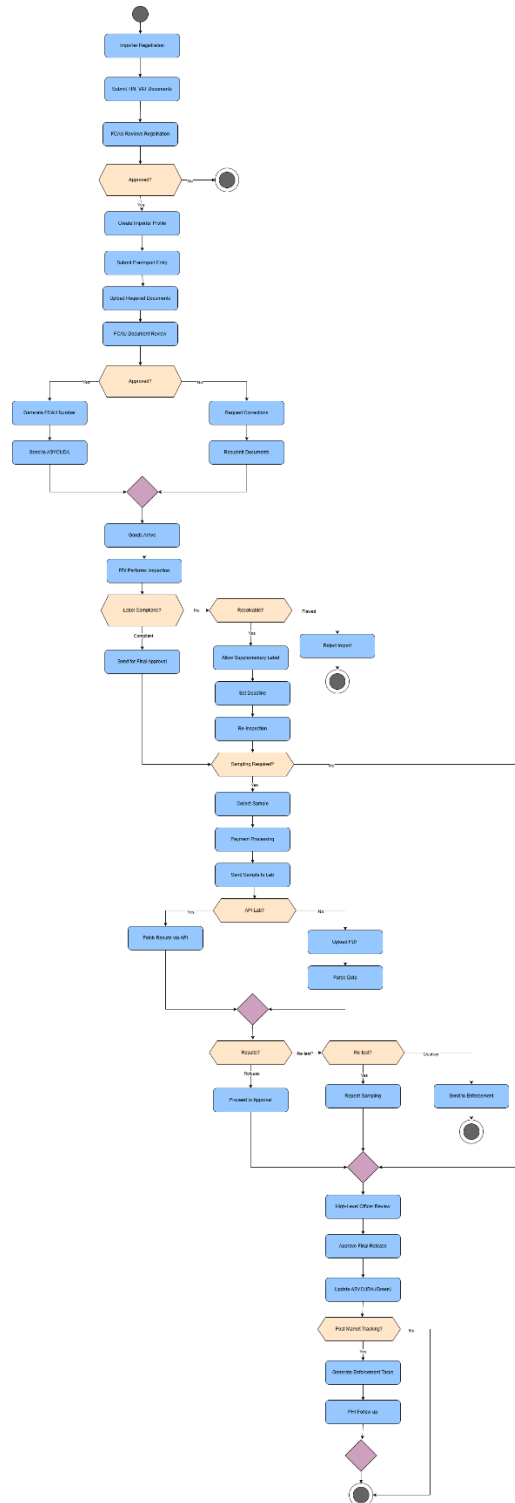


Figure 4.4

## 4.6 Export Control Module

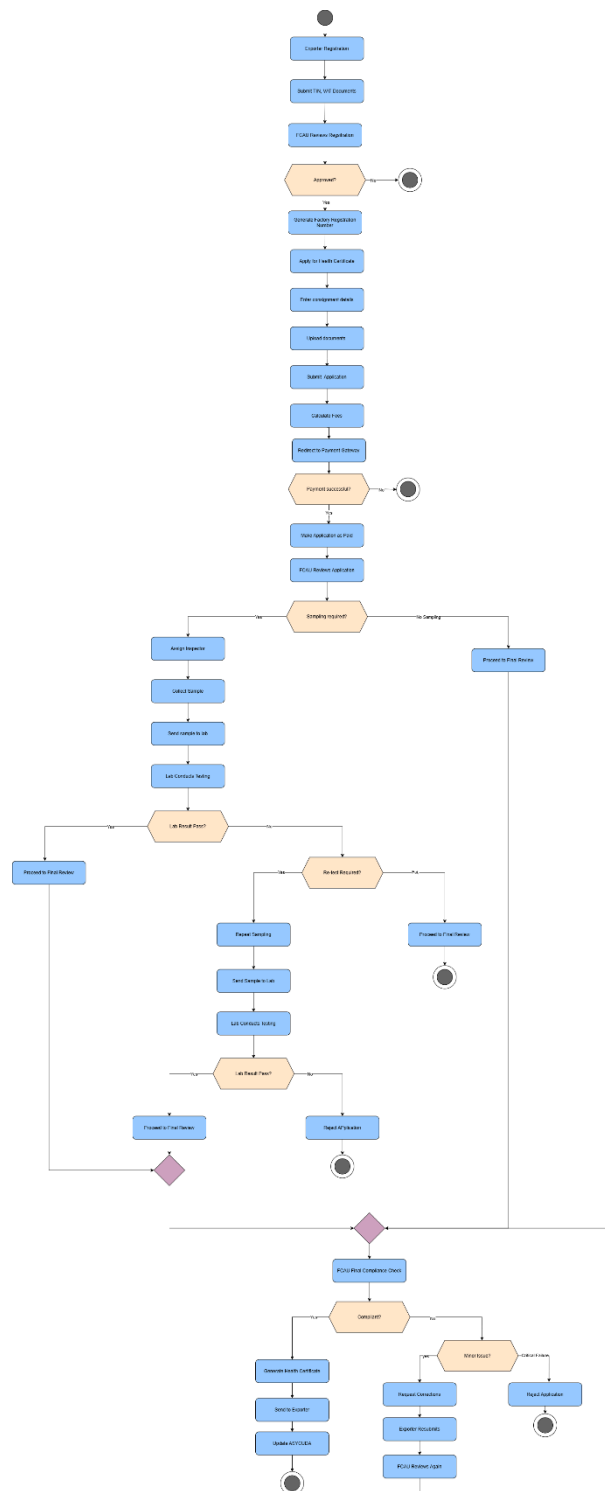


Figure 4.5

#### 4.7 Registration of Common Salt Manufacturing and Transports Management Module

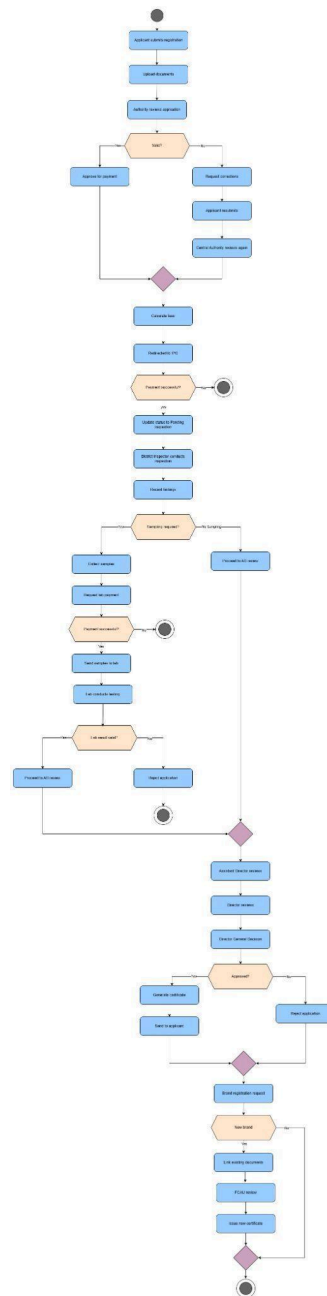


Figure 4.6

## 4.8 Bottled Water Manufacturing and Registration Management Module

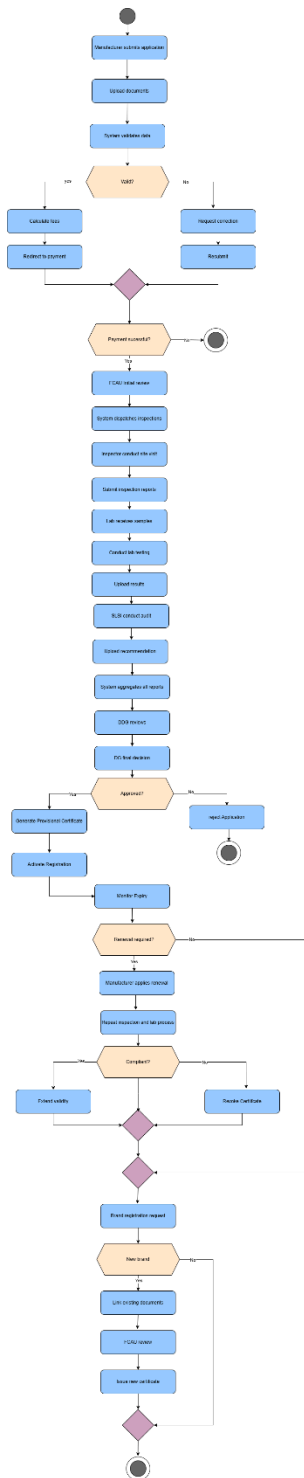


Figure 4.7